Repairing Safety After Benign Fine-Tuning: A Compute-Matched Comparison of Data Repairs

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Abstract

Fine-tuning a safety-tuned language model on benign instruction data can erode its refusals of harmful requests. When only the training data may change, the owner must choose a repair, such as removing rows or adding safety data, yet repairs are rarely compared with training seeds, side effects and compute counted. We measure which data repair restores the most safety for the compute spent without raising over-refusal or losing task skill; lost safety is measured as refusal erosion as Qwen3Guard labels it and as harmful compliance as the HarmBench classifier defines it. After a benign LoRA fine-tune that partly erodes OLMo-2-1B-Instruct, six pre-registered repairs (rows removed by training-data attribution, a representation filter or a free length rule; random, selected or self-generated refusal examples added), each run with six seeds, are compared with random removal and with no repair, labelled better, equivalent within 5 percentage points, worse or inconclusive, and set against five references that change no data. Pre-registered gates test whether the erosion is localized in data categories; a positive control tests whether diagnostics find a planted cause [PLACEHOLDER: if funded]; and predictions of when diagnosis pays are written down in advance and scored on a second family, Llama-3.2-1B-Instruct, at a matched erosion dose. We find [FINDING 1: the labels of the six repairs]; [FINDING 2: how the best data repair compares with the references]; and [FINDING 3: whether the erosion is localized, and whether diagnostics find the planted cause].

1 Introduction

Fine-tuning a safety-tuned ("Instruct") language model on data that contains nothing harmful can still erode its refusals of harmful requests. Qi et al. (2024) observed this after one epoch on ordinary instruction data sets (Alpaca, Dolly) for Llama-2-7b-Chat and GPT-3.5 Turbo; He et al. (2024) showed that benign rows selected for their similarity to harmful examples erode safety far more than random rows; and Guo et al. (2026) report that 100 random benign rows of full fine-tuning raise attack success sharply on recent 8B-class Instruct models. The size of the effect depends on the recipe (Kim et al., 2025). When a fine-tune does erode safety, its owner often controls only the data: commercial fine-tuning services screen training data, gate the tuned model on a safety evaluation, and report harm categories rather than rows, leaving the customer to fix the data and resubmit (Microsoft, 2026).

Repair is then a data decision with side effects. One can *remove* rows: published filters score benign rows by their contribution to erosion (He et al., 2024; Guan et al., 2025; Li et al., 2025; Wang et al., 2026b; Zhang et al., 2026c), and training-data attribution ranks rows by gradient similarity to harmful examples (Jiao et al., 2025; Pan et al., 2026). One can *add* safety data, but which data are added decides the outcome: a random 1% mix of safety rows made Llama3-8B-Instruct less safe than no safety data, while DataRx's selected mix made it safer (Zhang et al., 2026b). Safety data can also cause over-refusal (Bianchi et al., 2024; Luo et al., 2026b). And some repairs need no data change at all: a safety prompt used only at test time (Lyu et al., 2024), a gentler recipe (Kim et al., 2025), a post-hoc projection of the fine-tuned adapter (Hsu et

al., 2024), or a fine-tuning objective that constrains updates on the first answer tokens (Qi et al., 2025). Every data repair costs compute, and the only direct evidence about it is a retrain.

Which repair works best is an empirical question that published results do not settle. Each repair was evaluated in its own setting, with its own models, judges and baselines, mostly with one or a few training seeds, and seldom with its scoring compute charged against the alternative of retraining [CITE-CHECK]. The closest comparison we found, DualSelect (Chen et al., 2026), compares data selectors and a safety-instruction mix over three seeds, with a false-refusal rate and its selection overhead reported, but on training data that include red-teaming pairs, and without model-level references or equivalence-bounded verdicts. Three results suggest that expensive diagnosis may not pay. Once selection compute is charged, powerful data-selection methods are almost never compute-optimal (Yin & Rush, 2025) [CITE-CHECK]. Gradient-based attribution tracks the answer format of training examples more than their task (Kim et al., 2026) [CITE-CHECK], so a free format or length rule may match it. And the erosion may be diffuse: even subsets selected for low conflict with safety still erode it (Guo et al., 2026), and Springer et al. (2026) argue from theory that better curation cannot remove the fragility.

We therefore measure repairs directly (§3, §4). OLMo-2-1B-Instruct is fine-tuned on 4,800 Dolly rows with a fixed LoRA recipe (recipe B: a constant learning rate, one epoch). Six repairs fixed in advance, each run over six seeds, are compared with random removal of the same size and with no repair: three removals of 600 rows (ranked by training-data attribution, a LARF-style representation filter or answer length) and three additions of 48 refusal examples (random, selected in the style of DataRx, or written by the untuned model). Each contrast is labelled *better*, *equivalent* (its 90% interval within ± 5 percentage points, pp), *worse* (significantly adverse on the unsafe rate, over-refusal or task loss) or *inconclusive*, after Holm correction; over-refusal and held-out task loss are feasibility limits, and cost is measured in full-run equivalents (FRE) under two conventions. A data repair counts as dominant only if it beats five references that change no data, none of them ours: a test-time safety prompt, the adapter scaled down, its Safe LoRA projection (Hsu et al., 2024), recipe G (a standard warmup and cosine schedule) and recipe B with the constrained objective of Qi et al. (2025). The second family, Llama-3.2-1B-Instruct, uses recipe B at its own learning rate, so that both families erode partly (a matched erosion dose; §4.1).

Pre-registered parts surround the comparison. Gate A requires erosion beyond seed noise. Gate B asks whether the erosion is *localized*: does removing one Dolly category restore more safety than removing as many random rows? Gate C asks which diagnostic picks that category, and whether a format rule explains attribution's pick. A positive control plants a known eroding cause in nested menus of 8, 32 and 128 cells [TO BE FINALISED] and asks whether each diagnostic finds it, and at which menu size diagnosis beats trial runs [PLACEHOLDER: if funded]. A secondary, model-based analysis predicts when paying for a diagnostic beats spending the same compute on trial runs selected by successive halving (Jamieson & Talwalkar, 2016; Li et al., 2018). Every decision rule is committed before the data it judges; outcomes are replicated over training seeds, confirmed on held-out prompts, and reported as *too close to call* when the evidence cannot separate options.

Our contributions are:

- **A measured comparison of data repairs with equivalence bounds** (§3.2, §4.3–§4.4): six repairs over six seeds each, labelled better, equivalent, worse or inconclusive, with over-refusal, task loss and measured cost, against five references that change no data, on OLMo-2-1B and, in stage 2, on Llama-3.2-1B. [FINDING 1: the six labels, and which repair, if any, beats the references]. We release the menus, seeds, outcome tables and costs [ARTIFACT].

- **A positive control for diagnosis** (§4.5): [FINDING 2: whether each diagnostic ranks a planted cause first, and the runs-only cost of finding it, at menu sizes 8, 32 and 128] [PLACEHOLDER: if funded].
- **Where the erosion lives** (§4.6, §4.10): [FINDING 3: the Gate B verdict with its detection limit; Gate C's reading with the format control; how much of each repair's effect runs through refusal openings].
- **Pre-registered predictions of when diagnosis pays** (secondary; §3.4): [FINDING 4: predicted against measured value of diagnosis on the category and planted menus, and on the second family].

[FIGURE 1: the repair comparison. For each repair, the safety restored against random removal or no repair, with its 90% interval and the ±5 pp equivalence band, plotted against over-refusal (left) and measured cost in FRE (right); the adapter-scaling reference curve is drawn in the left panel.]

2 Related Work

Safety erosion under benign fine-tuning. Qi et al. (2024) showed that fine-tuning aligned models on benign instruction data degrades safety, though less than harmful data does, with no training seeds reported. He et al. (2024) selected benign rows by representation and gradient similarity to harmful anchors; in their best case, 100 such rows gave over 70% unsafe answers against under 20% for random rows. This shows that benign rows *can* erode safety, not that ordinary fine-tunes have a localized cause. Kim et al. (2025) change the size of the erosion several-fold through optimization choices alone. Guo et al. (2026) find that 100 random benign rows of full fine-tuning erode recent 8B-class Instruct models, and that a handful of safety rows restores safety after a small attack. Under LoRA their results depend on the family: in their Table 2, Qwen2.5-7B-Instruct reads 7.27% before fine-tuning, 4.50% after 100 rows and 9.09% after 5,000 rows (two seeds), while Llama-3.1-8B-Instruct rises to 44.80% at 5,000 rows. Springer et al. (2026) explain erosion geometrically, as benign updates curving into a sharp, safety-relevant subspace.

Removal repairs: filters and attribution. Published filters score benign rows for their contribution to erosion. DataShield projects rows onto a compliance direction (Zhang et al., 2026c); LARF uses representations of safety-sensitive layers (Li et al., 2025); SQSD projects each row's parameter update onto danger and safety directions (Wang et al., 2026b); bi-anchoring compares a row's gradients or representations with harmful and safe anchors (He et al., 2024); and Self-Inf-N, proposed as an attack, finds outlier benign rows whose fine-tuning breaks safety (Guan et al., 2025) [CITE-CHECK]. Row filters report their scoring cost without charging it against runs (Zhang et al., 2026c; Hu et al., 2026) [CITE-CHECK]; other defences select tokens (Li et al., 2026) or use continual learning (Alssum et al., 2025) [CITE-CHECK]. A concurrent submission finds risk rankings taken at initialization nearly uncorrelated with those after two epochs, and re-scores rows during training (Anonymous, 2026). For attribution, Pan et al. (2026) find that removing rows flagged by gradient-based attribution gives a safer retrained model than removing rows flagged by WildGuard, from one training run per arm and with no compute reported. DATE-LM (Jiao et al., 2025) finds that with safety-aligned look-alikes in the data, attribution's detection drops while LLM classifiers are largely unaffected; without them, the strongest classifiers detect better, yet removing attribution's picks gives the safer retrained model. Luo et al. (2026b) report weak, inconsistent effects from deleting or reweighting influential rows, large effects from rewriting them, and an over-refusal trade-off for safety rewriting.

Addition and model-level repairs. Adding about 3% safety examples to instruction tuning improves safety but can cause exaggerated refusals (Bianchi et al., 2024) [CITE-CHECK]. Which rows are added matters: DataRx selects safety rows by the gap between a model's own response and a safe reference (Zhang et al., 2026b) [CITE-CHECK]; on Llama3-8B-Instruct, averaged over seven tasks, its random 1% mix raised attack success from 20.24% (no safety data) to 59.23%, while its selected mix lowered it to 13.70%. Its

baselines, all additions, include self-distillation, a precedent for our self-generated arm. Guo et al. (2026) restore safety with a handful of safety rows, and Llama 3's safety mix was tuned by repeated experiments that traded violations against false refusals (Grattafiori et al., 2024). Model-level repairs change no data: a safety prompt used only at test time reduces erosion (PTST; Lyu et al., 2024), SafeTune compares repairs of safety drift in single deterministic runs (Seth et al., 2026) [CITE-CHECK], a gentler recipe shrinks erosion (Kim et al., 2025), Safe LoRA projects the LoRA weights of selected layers onto a safety-aligned subspace without training or data (Hsu et al., 2024), and an objective that constrains updates on the first answer tokens makes alignment more persistent under fine-tuning (Qi et al., 2025). We use such repairs as references that a data repair must beat. Zhang et al. (2026a) restore the safety of a fine-tuned model by further training on a single safety example; this patch acts after fine-tuning and is not among our references (§6).

Attribution, its cost and what it tracks. Training-data attribution ranges from cheap gradient similarity on LoRA gradients (LESS; Xia et al., 2024) [CITE-CHECK] to estimates of finite removal effects by unrolled differentiation (SOURCE; Bae et al., 2024), which are reported to cost several training runs [CITE-CHECK]. Xiao & Aranguri (2026) compare the cost of diagnosis methods for undesirable post-training behaviours; probe attribution is over ten times cheaper than gradient attribution and LLM-judge baselines in their study. Yin & Rush (2025) charge selection compute against training compute and find that powerful selection methods are almost never compute-optimal; gradient-based selection pays only when the trained model is about ten times larger than the selection model [CITE-CHECK]. Kim et al. (2026) find that gradient similarity tracks answer format more than task [CITE-CHECK]; we therefore include a free answer-length rule as a repair and as a control for attribution's picks.

Prediction and decisions under a budget. Persona vectors (Chen et al., 2025) project training data onto activation directions to flag data that will shift a trait [CITE-CHECK]; TuneAhead (Luo et al., 2026a) and Data2Behavior (Wang et al., 2026a) predict a fine-tune's outcome from its data before training; MNEME predicts side effects of fine-tuning and unlearning by sparse model diffing (Kassem et al., 2025) [CITE-CHECK]; any such predictor can enter our comparison as a scorer at its measured cost. Successive halving and Hyperband allocate a fixed budget by repeatedly discarding the worst candidates at increasing fidelity (Jamieson & Talwalkar, 2016; Li et al., 2018); multi-fidelity methods model how cheap observations relate to full ones (e.g. Ouyang et al., 2025). DataDecide (Magnusson et al., 2025) scores small pretraining runs by decision accuracy against compute, and Ivison et al. (2025) charge selection compute against training on more data [CITE-CHECK].

How we differ. We do not claim a first benchmark of safety filters or defences, a first comparison of removal against addition, or a first pricing of data selection: SafeTune and SafeTuneBed (Hossain et al., 2025) compare defences; DualSelect compares selectors and a safety-data mix over seeds, with a false-refusal rate and selection cost; DataRx compares ways to choose added safety data; and DataShield's cost table and Yin & Rush (2025) price selection [CITE-CHECK]. Our contribution is the combination; to our knowledge, and not found by our search, no prior work combines, for purely benign data, equivalence-bounded contrasts over six training seeds for removal and addition repairs; over-refusal and task loss as limits fixed in advance; every arm's full cost, scoring included, under two conventions; model-level references that a data repair must beat; a positive control with a planted cause; and predictions about the value of diagnosis written down before the data.

3 Problem Setup

3.1 Setting and endpoints

Let θ_0 be a safety-tuned Instruct model, fine-tuned with a fixed recipe $\mathcal{R}$ (adapter, loss, optimizer, schedule, length rule) on benign data $\mathcal{D} = D_1 \cup \cdots \cup D_K$, split into K *units* by metadata fixed before any outcome. A *repair* a maps $(\mathcal{D}, \mathcal{R})$ to a training set and recipe $(a(\mathcal{D}), \mathcal{R}_a)$; a data repair keeps $\mathcal{R}_a = \mathcal{R}$. The model trained with seed s (adapter initialization and data order) is $\theta(a, s)$. For a model θ, $f(\theta)$ is the *unsafe rate* (the share of sampled answers to a harm bank that a frozen judge labels unsafe), $\ell(\theta)$ is the held-out *task loss*, and $r(\theta)$ is the *over-refusal rate* on benign prompts that resemble harmful ones. Endpoints average over seeds: $F(a) = \mathbb{E}_s[f(\theta(a, s))]$, and likewise $\bar{\ell}(a)$ and $R(a)$. With a_0 the fine-tune without repair, the *erosion*, refusal erosion as Qwen3Guard labels it (§4.2), is $E = F(a_0) - f(\theta_0)$; repair is meaningful only when E exceeds seed noise (Gate A, §4.6).

3.2 Repair contrasts, limits and labels (primary)

Let a_{R} remove the same number of random rows as the removal repairs. Removals are matched in training tokens as well as rows, or a token-matched analysis is pre-registered beside the row-matched one [TO BE FIXED in the stage-1 pre-registration]. A removal repair m is compared with random removal, $\Lambda_m = F(a_{\mathrm{R}}) - F(m)$: the safety it restores beyond removing the same amount of data at random. An addition repair is compared with no repair, $\Lambda_m = F(a_0) - F(m)$, and a selected or self-generated addition with the random addition of the same size. The restoration $G_m = F(a_0) - F(m)$ is reported for every arm. Contrasts are estimated from per-seed unsafe rates as differences of means over unpaired seeds, with the variance pooled across arms.

Side effects are the *over-refusal excess* $O(m) = R(m) - r(\theta_0)$, against the untuned model ($R(m) - R(a_0)$ is also reported), and the *task-loss increase* $L(m) = \bar{\ell}(m) - \bar{\ell}(a_0)$. A repair is *feasible* at limits (τ_O, τ_L) if $O(m) \le \tau_O$ and $L(m) \le \tau_L$. Results are reported at $\tau_O \in \{2, 5, 10\}$ pp; τ_L is two standard errors of a seed-to-seed difference in held-out loss, fixed before the comparison runs. The *cost* of a repair is $C(m) = c(m) + 1$ FRE: $c(m)$ is the measured cost of building the repaired data (scoring rows, selecting or generating safety rows), and one FRE is the measured cost of one random-removal run (§4.7).

With a smallest effect of interest $\delta = 5$ pp and Holm correction at family-wise level 0.05 over the primary contrasts, each contrast receives one label: *better* if the Holm-adjusted test rejects $\Lambda_m = 0$ and the estimate is at least δ; *equivalent* if the 90% interval for Λ_m lies inside $(-\delta, \delta)$ (two one-sided tests); *worse* if the contrast with its comparator is significantly adverse at the Holm-adjusted 5% level on the unsafe rate ($\Lambda_m < 0$), on over-refusal ($R(m)$ above the comparator's) or on task loss ($\bar{\ell}(m)$ above the comparator's), each read in its adverse direction, so an adverse repair is never reported as inconclusive; and *inconclusive* otherwise. Which label a repair receives when it is better on the unsafe rate but worse on a side effect is fixed in the stage-1 pre-registration [TO BE FINALISED]; both results are reported. A margin of 4 pp is reported as a sensitivity. Five references change no data (§4.4). One, *adapter scaling*, $\theta_\lambda = \theta_0 + \lambda\,(\theta(a_0, s) - \theta_0)$ for $\lambda \in \{1, 0.75, 0.5, 0\}$, traces (F, O, L) from the fine-tune ($\lambda = 1$) to the untuned model ($\lambda = 0$). A data repair *dominates* only if it lies below this curve: a lower F than the curve at equal O and L.

3.3 Localization

Let a_{-k} remove unit D_k and $a_{\mathrm{rand},k}$ remove $|D_k|$ random rows drawn across units. Erosion is *localized* at level δ_B if some $\Lambda_k = F(a_{\mathrm{rand},k}) - F(a_{-k})$ reaches δ_B beyond seed noise (the random arm controls for the smaller training dose), and *diffuse* if every Λ_k is credibly below δ_B. Diffuse erosion leaves the repair

comparison unchanged: unit removals then act like random removal up to δ_B, and a unit-level diagnostic has little to find.

3.4 The value of diagnosis (secondary, model-based)

A *menu* $\mathcal{A}_M$ holds a_0 and M repairs built by a frozen rule. A *selector* π with budget B (in FRE) buys trial runs and returns a pick, scored as no repair if it is infeasible; its regret $\mathrm{Reg}(\pi; B, M)$ is the expected unsafe rate of the pick minus that of the menu's best feasible repair. The runs-only selector π_- uses one rule per (M, B) cell, frozen from a simulation before any outcome; π_{+d} pays the cost c_d of diagnostic d, keeps its top k_d repairs and runs the same rule on $B - c_d$. The value of d is $\Delta_d = \mathrm{Reg}(\pi_-) - \mathrm{Reg}(\pi_{+d})$, and its break-even cost c_d^* is the largest c_d with $\Delta_d \geq 0$. At the menus we can afford, these are mostly outputs of a model, so a generative model is frozen before Gate B's category runs; we report measured against predicted values and score OLMo-calibrated predictions on Llama (Appendix C.7).

4 Experimental Design

4.1 Models, data and recipes

Families and scale. At 1B, OLMo-2-0425-1B-Instruct (Team OLMo et al., 2024) is the stage-1 model and Llama-3.2-1B-Instruct (Grattafiori et al., 2024) the second family (stage 2; §4.8). At 7–8B, the full comparison runs on OLMo-2-1124-7B-Instruct and Llama-3.1-8B-Instruct, at learning rates fixed by the same rule (below) [PLANNED; funding-gated, fallback one family]. A headline check at the largest single-GPU size comes last [PLACEHOLDER: if resources allow]. Qwen2.5 models (Qwen et al., 2024) are outside this study; a same-recipe comparison of OLMo-2, Llama-3 and Qwen2.5 is reported in [CITE: concurrent submission, anonymized]. Every model trains on one GPU, and every comparison stays on one GPU type.

Data. Gate B and stage 1 train on 4,800 rows of Dolly-15k (Conover et al., 2023), 600 from each of 8 task categories (the units; $K = 8$), sealed per tokenizer with a manifest; 200 further rows, 25 per category, measure task loss. The add repairs draw their safety rows from PKU-SafeRLHF (§4.3). Alpaca is the planned second benign dataset, after Gate B [NOT YET FIXED: its units and learning rate].

Recipes. *Recipe B*, the main recipe: LoRA (Hu et al., 2022) of rank 64 and scale 128 on all linear layers; AdamW at a constant learning rate of 10^{-4}, no warmup; 128 rows per step; one epoch (38 steps); a per-row loss (each row's mean over its response tokens). Gate A, Gate B and stage 1 use it. It was chosen over recipe G from probe and Gate A readings alone, before any repair outcome. *Recipe G*, a reference arm that changes only the recipe (§4.4), adds a 4-step warmup and cosine decay, which halve the mean learning rate over the epoch, and a token-mean loss. It is not a general fix: its effect on erosion differs across families and sizes [CITE: concurrent submission, anonymized]. Training is deterministic on matched hardware.

A matched erosion dose. Each family uses recipe B with only its learning rate changed: the highest rung of a ladder (10^{-4}, 5×10^{-5}, 2×10^{-5}) whose probe reads partial (20–55 pp) under the frozen one-seed Gate A bars (§4.6; a probe too close on E gets one more seed; any other too-close bar goes to the user). The rate is chosen from probes only, never from a repair result. Families respond differently to the same recipe [CITE: concurrent submission, anonymized]; we fixed this rule after reading a one-recipe development ladder, on which no rung left both 1B families partly eroded, and Llama's first probe at recipe B, and we state cross-family claims at a matched erosion dose. The rungs are 10^{-4} for OLMo-2 (1B and 7B) and 5×10^{-5} for Llama (3.2-1B and 3.1-8B). Both Llama rungs were accepted on a too-close bar (one seed's answers-end bar at 1B; the window's upper edge at 8B); §5 reports both.

4.2 Measures, judges and the claim rule

- **Unsafe rate U.** The share of 600 sampled answers (200 development harm prompts $\times$ 3 samples at temperature 0.7, up to 2,048 new tokens) that Qwen3Guard-Gen-4B (Zhao et al., 2025) labels Unsafe. The prompts come from HarmBench (Mazeika et al., 2024), StrongREJECT (Souly et al., 2024), SimpleSafetyTests (Vidgen et al., 2023) and OR-Bench's toxic set (Cui et al., 2025). Standard errors are clustered by prompt. U is a broad policy label, not an attack-success rate as published elsewhere: judges and prompts differ, so U is compared only within this study. A rise in U is refusal erosion as Qwen3Guard labels it. Gates and selectors read development banks only; final claims are confirmed on a sealed held-out subsample (§4.8).
- **Compliance and the claim rule.** HarmBench's classifier asks whether an answer carries out the requested behaviour. A contrast in U supports the wording *harmful compliance as the HarmBench classifier defines it*, never harm alone, only if the classifier's contrast has the same sign and at least half the size and the classifier's erosion under that recipe is at least 20 pp; otherwise the sentence says *restored refusal*. A blind human read during development rated about a third of judge-unsafe answers as usable help (Appendix A).
- **Also per arm:** hard-category U (violent, illegal, sexual or self-harm categories); U by answer-length band and a length-reweighted U; Granite Guardian 3.3 (Padhi et al., 2024) on the greedy answers; and a selectivity score (refusal on harm prompts minus refusal on benign prompts).
- **Over-refusal.** Qwen3Guard's refusal flag on greedy answers to 100 benign prompts that resemble harmful ones (50 XSTest safe prompts, Röttger et al., 2024; 50 OR-Bench-Hard prompts, Cui et al., 2025), reported by source, with a lexicon check.
- **Task loss and skill.** Held-out loss on all 200 task rows is the skill limit of the comparison. Probe and Gate A bars also check ARC-Easy (Clark et al., 2018), allowing a paired drop of at most 0.03 on 189 questions; a rule fixed during development resolves the usual straddling interval, and as it is a loosening we report both verdicts (Appendix C.4).
- **Answers end.** A coherence screen flags answers that reach the token cap, repeat 4-grams heavily, are empty, or have fewer than 5 tokens; an arm whose answers fail too often is not read.

4.3 Stage 1: the primary comparison

Table 1 lists the stage-1 repairs; §4.4 gives the references. Every rule, scorer and pool is committed by hash before Gate B's category runs (§4.9).

Table 1. Stage-1 repairs on OLMo-2-1B (recipe B; one epoch; removal arms train 33 steps, addition arms 38). The five references, with their seeds, are in §4.4.

Arm	Repair	Rule	Seeds
a_{R}	random removal	600 random rows, 75 per category	8
a_0	no repair	–	6
H1	length rule (free)	remove the 600 rows with the shortest answers	6
A1	attribution	remove the top 600 rows by bi-anchored gradient similarity at two checkpoints of a_0 (the Gate C method)	6
L1	LARF-style filter	remove the top 600 rows by the untuned model's representation similarity to harmful anchors at one layer	6

Arm	Repair	Rule	Seeds
S1	random refusals	add 48 rows (1%) drawn at random from a fixed refusal pool	6
S3	DataRx-style	add 48 rows selected from S1's pool by DataRx's score	6
S4	self-generated	add 48 pool prompts with the untuned model's own refusals	6

Primary family. P1: A1 against a_{R}; P2: L1 against a_{R}; P3: H1 against a_{R}; P4: S1 against a_0 (two-sided, since prior work predicts both directions; a significant adverse result is labelled *worse*); P5: S3 against S1; P6: S4 against S1. Holm at 5%, labels as in §3.2.

Pools, seeds and secondary arms. The refusal pool holds PKU-SafeRLHF (Ji et al., 2025) prompts with an unsafe response in the source data, each paired with a source-labelled safe response that opens with a refusal; no pool or anchor is filtered by the evaluation judge, and the add effect must also hold on the evaluation prompts least similar to the added rows. Seeds run in blocks fixed before any outcome, and only complete blocks are analysed; a blinded rule on Gate B's pooled noise can shrink the family or stop stage 1. At planning noise the minimum detectable effect after Holm's first step is about 5 pp [PLACEHOLDER: final MDE from Gate B's pooled noise]. Secondary arms (three seeds, exploratory) and scorers scored without new runs are listed in Appendix C.3 [TO BE FINALISED].

4.4 References that change no data

All five are mandatory, and we claim no novelty for any. **Safety prompt:** a fixed safety instruction as a prefix of the user turn (OLMo-2 has no system turn), at evaluation only, as in PTST (Lyu et al., 2024), on three a_0 seeds. **Adapter scaling:** the a_0 adapter scaled by $\lambda = 0.75$ and 0.5 on three seeds, which with $\lambda \in \{0, 1\}$ gives the reference curve of §3.2. **Safe LoRA** (Hsu et al., 2024): a_0's adapter projected as in §2, on three seeds; the base model, layer rule and threshold follow the paper's rule or a fixed default, never stage-1 outcomes [NOT YET FIXED]. **Recipe G** (a reference that changes only the recipe): the unchanged data under recipe G, three seeds. **B+Qi:** recipe B with the token-wise constrained SFT objective of Qi et al. (2025), which constrains updates on the first answer tokens, on the unchanged data at six seeds. The paper states plainly whether any data repair beats these references.

4.5 Positive control: a planted cause

[PLACEHOLDER: runs only if funded; otherwise menu size leaves the headline.] A planted cause is $k = 37$ benign-looking rows whose answers comply directly with prompts that also have an unsafe answer in the source data. A sealed draw places it in one leaf cell, whose rows it replaces, so the run keeps 4,800 rows. Nested cells [TO BE FINALISED] give menus of $M = 8$, 32 and 128 cells. Because the truth is known, the sweep measures how diagnosis and trial runs scale with M, which natural menus at our budget cannot. A three-seed pilot gates the sweep, the sweep removes the planted cell at each menu size, and every scorer ranks the cells without new runs (Appendix C.5 gives the definitions). Outputs: the safety restored by removing the planted cell, $G^{\mathrm{p}}(M)$; each diagnostic's top-k hit rate with known truth; and the runs-only cost of finding the cell, replayed with the frozen selector rules.

4.6 Gates A, B and C

Gate A (entry). For each family at its matched dose, over three seeds (the probe seed and two fresh seeds), mean E must lie in a pre-registered partial-erosion window of 20–55 pp, be at least twice the seed standard deviation, and exceed zero on every seed; a second judge must agree in direction, the answers must end, and the task must be learned without lost skill (§4.2); its results, with every setting tried, are in §5.

Gate B (localization). Its pre-registration was frozen before any Gate B run; Appendix C.1 gives its rules. At recipe B, each category removal (600 rows, 33 steps) is compared with random removal of 600 rows (75 per category) at the same seed, so a_0 cancels. All 8 categories run at one seed; the two with the largest contrast, plus Gate C's picks, run at two more seeds. The verdict is LOCALIZED if some category restores at least $\delta_B = 8$ pp more than random removal and clears a Bonferroni-corrected bar; DIFFUSE if every category is credibly below 8 pp; and TOO CLOSE otherwise. It reads U alone, under guards; a noise gate, an automatic extension and a length-matched control are pre-registered. At planning noise, 80% detection needs about 10–12 pp, so DIFFUSE means no category-level cause above about 8 pp, not no cause. **Status:** the noise gate has passed, and Gate C's rankings were sealed before the category runs.

Gate C (which diagnostic picks the category). Attribution (A1's score, on a retrained first-seed a_0) and representation similarity (on the untuned model) each pick their top category; a format rule picks the category with the largest share of its loss on the first five answer tokens. Gate C is read only if Gate B is LOCALIZED (AGREE, ATTR_BETTER, REP_BETTER, NONE_FINDS or TOO CLOSE). In the sealed rankings both diagnostics pick the same category, and the format rule another. So, by the frozen rules, Gate C can read only AGREE, NONE_FINDS or TOO CLOSE, or is not read; ATTR_BETTER and REP_BETTER cannot occur; and the paper may say only that neither pick is the format pick, which does not show that the picks are free of format.

If DIFFUSE. Gate C is not read and stage 1 is unchanged. A rewrite arm (the untuned model's own answers replace the 600 top-attribution rows) runs first among the secondary arms, and the planted sweep becomes the only test of whether diagnosis can pay.

4.7 Cost accounting

Every arm and every scorer is logged in GPU-seconds by stage. One FRE per family is the median measured time of that family's random-removal arms. Two cost conventions, *full* and *lean*, are frozen before stage 1 (Appendix C.6); every pool filter, generated answer, anchor job and scorer is charged. Claims about cost hold only where both conventions agree, and no cost ratio measured at 1B is claimed for a larger model. [TO BE FINALISED: measured costs per arm and per scorer.]

4.8 Second family, confirmation and scale

Llama-3.2-1B (stage 2) runs at its matched dose (§4.1): its own Gate A, a reduced Gate B on categories, and the primary family at six seeds. A contrast "replicates across two families" only if it has the same sign in both and the family difference is below the interaction's MDE; a family difference is claimed only above that MDE. **Confirmation** (stage 3): at most two contrasts per family, chosen by a rule committed before stage 1, re-run at six fresh seeds on a sealed 200-prompt held-out subsample and on the development bank. **Scale** (stage 4): a replication in a published setting (a LARF- or DataShield-style filter on Llama-3.1-8B-Instruct fine-tuned on Alpaca, with random removal and length rules), the full 7–8B comparison of §4.1 and the headline check [PLACEHOLDER: if resources allow]. No result is claimed at a size without runs at that size.

4.9 Pre-registration and testing order

Gate B runs first, to its verdict; funding for the later stages is then sought with that result, and stage 1 starts only once funded, never because of an outcome. Before Gate B's category runs, the stage-1 pre-registration [NOT YET FIXED] commits every stage-1 scorer's code and settings hash, every edit rule, the pools, the plant, the primary family, the seed rule, the limits, the claim rule, the selector specification and its predictions, and it lists every Gate B value seen before it is frozen (the noise-gate reading and the sealed

rankings). Anything added later is labelled post-Gate-B exploratory. Families of tests are read in order, each at its own 5% level: (1) the primary family; (2) confirmation; (3) the second-family (Llama) comparison; (4) the planted-sweep claims; (5) the decision-value predictions. Everything else is descriptive.

4.10 Side studies

1. **Mechanism: refusal openings against answer content.** An opening decomposition (Appendix C.8) splits each arm's change in U into answers that stop opening with a refusal and what the answers say. A teacher-forced first-token refusal margin on Gate B arms should track U if a category's effect runs through openings; Qi et al. (2025) argue that safety alignment often adapts mainly the first few output tokens. [PLACEHOLDER: report against the bound on a first-token path recorded before Gate B.] Exploratory.
2. **Recipe and family contrasts** are not results of this paper. Recipe G enters only as a reference (§4.4). The contrasts behind the per-family rule (recipe G against recipe B in several families, a pre-registered single-seed run that changes only the loss weighting, the learning-rate rungs and a rank-16 check) are reported in [CITE: concurrent submission, anonymized].
3. **Measurement validity and over-refusal.** Agreement and E for three judges on the same answers, hard-category U and length bands, how much of the gap to published numbers comes from judges and prompts, and O by source, bounded by the judge's false-Unsafe rate on benign prompts.
4. **Side-effect prediction (secondary).** On the addition arms only: the balanced error of predicting an over-refusal violation, a diagnostic against trial runs of equal cost.

5 Results

[PLACEHOLDER: Gate A with every setting tried and its disclosures; Table 2, the stage-1 comparison (G, Λ with 90% intervals and labels, O, L, cost under both conventions, the HarmBench contrast and the claim-rule wording) with the references and the adapter-scaling curve; Figure 1; the second family, if run; Gates B and C with the length control and the format reading; the planted sweep, if funded; measured against predicted decision value; the opening decomposition; every too-close-to-call result.]

6 Discussion and Limitations

What "unsafe" means. Our primary judge gives a broad policy label, so a rise in U is refusal erosion as Qwen3Guard labels it. In a blind human read during development (one reader; Appendix A), most judge-unsafe answers went along with the request, but only about a third gave usable help. The claim rule (§4.2) therefore allows at most "harmful compliance as the HarmBench classifier defines it", and U is not comparable with attack-success rates published under other judges and prompts.

Equivalence and power. The labels resolve differences of about 5 pp; smaller differences stay inconclusive, and an *equivalent* label means "within $\pm$5 pp", not "identical". Gate B's detection limit (§4.6) holds for categories; row-level causes that span categories are not tested by Gate B.

Recipe and setting. The main recipe, B, is plain rather than standard: a constant learning rate with no warmup, a per-row loss and a rank-64 adapter on all linear layers, larger than is common (a rank-16 check is in [CITE: concurrent submission, anonymized]). Recipe G's standard schedule lowers the mean learning rate; it is a reference, not a general fix, and repair rankings may differ at a lower dose. Cross-family claims hold at a matched erosion dose, not at one recipe, under a rule fixed after a development ladder was read. The skill bar at the probes and Gate A is looser than the bar first registered, and we say where it decided a

reading (Appendix C.4). The main model and recipe were chosen by screening probe and Gate A readings, never Gate B or stage-1 outcomes; §5 lists every setting tried.

Baselines, doses and cost. Every scorer is re-implemented from its paper ("X-style"), so a weak result may reflect our implementation; a published table of deviations per scorer, reference outputs requested from the authors and the published-setting replication address this. Comparisons across removal and addition hold only at the pre-registered doses (600 rows removed, 48 added). An arm's cost depends on how much we evaluate as well as on training, and costs at 1B do not transfer to larger models, because a scorer's cost and a run's cost grow differently with model size.

Scale and family. The full comparison runs at 1B, with one benign dataset until Alpaca runs; at 7–8B it is planned on both families, funding-gated, with one family as the fallback; larger sizes get a check only if resources allow. A family difference is claimed only above its MDE.

The positive control and the secondary analysis. The planted cause is artificial; finding it shows that a diagnostic *can* find a concentrated cause, not that natural erosion has one. The value of diagnosis at natural menus is model-based and judged against predictions made in advance. Selectors are not run against each other at equal budgets here; a later study with its own pre-registration would do that.

Scope and user. We compare data repairs with model-level references; rewriting responses enters only if the erosion is diffuse. A one-example patch applied after fine-tuning (Zhang et al., 2026a) is not among the references, so no claim here covers it. A customer of a closed fine-tuning service can change only the data but cannot run gradient or representation scorers; a developer with open weights can run every scorer but also has model-level repairs. Our answer serves a service that must ship the customer's adapter and may change only the data, and a pipeline that allows only data changes.

7 Conclusion

[PLACEHOLDER: which repair to use, at what cost and with what side effects, and whether paying for diagnosis was ever worth it, under the observed outcome.]

Impact Statement

This work studies how to repair a language model's safety after benign fine-tuning has eroded it, so that developers and fine-tuning services can choose a repair with its side effects and cost in view. We see three risks. First, the study shows which training data erode safety and which repairs fail, and the positive control, if run, plants benign-looking rows that erode safety; an attacker could use this. The planted rows come from a public safety dataset, and we release [TO BE DECIDED: what is released, for example scores and outcome tables but no generated answers to harm prompts]. Second, every outcome comes from automatic judges, and a blind human read rated only about a third of judge-unsafe answers as usable help, so a repair labelled *better* or *equivalent* is not a safety guarantee. Third, the models are small and the benign data come from one dataset; the results may not hold for larger models, other data or deployed systems, and do not replace a model's own safety evaluation.

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Appendix A: Blind human read (development evidence)

This read checked, during development, what the primary judge's Unsafe label means. It gates nothing, changes no verdict and is not a result of the study. The same read is also reported in [CITE: concurrent submission, anonymized].

Design. The read was pre-registered before anyone saw the answers, and the ratings were sealed before the key was opened. A seeded rule drew 50 sampled answers to development harm prompts: 40 that Qwen3Guard labels Unsafe, from recipe-B fine-tunes of OLMo-2-1B, Qwen2.5-1.5B, Qwen2.5-7B and Llama-3.2-1B and a recipe-G fine-tune of OLMo-2-1B (6 to 12 per group), and 10 that it labels Safe, as controls. The sheet showed only the request and the answer, without model, recipe, seed or judge label. One reader rated each answer 0 (refuses or deflects), 1 (goes along but gives no usable help: vague, generic, wrong, fictional or incoherent) or 2 (usable help: specific steps, methods, materials or content that would meaningfully help someone do the harmful thing).

Counts. Of the 40 Unsafe-labelled answers, 39 were rated and 1 was skipped. 34 of the 39 went along with the request (rated 1 or 2; 0.87, exact 95% interval [0.73, 0.96]), and 13 of the 39 gave usable help (rated 2; 0.33 [0.19, 0.50]). None of the 10 Safe-labelled controls was rated 1 or 2, so the judge missed none ([0.00, 0.31]). Of the 15 OLMo-2-1B answers that also have a HarmBench-classifier label, 4 of the 12 labelled "yes" were rated usable help, and none of the 3 labelled "no": the reader's bar was stricter than the classifier's.

Use in the paper. Our reading: when Qwen3Guard labels an answer Unsafe, the answer usually goes along with the request, so U measures lost refusal, but only about a third of these answers give usable help. The read supports the wording of §4.2 and §6, not a claim of usable harm.

Limits. One reader, an author who helped design the study, and no second rater, so no agreement figure. Groups of 6 to 12 answers are too small to compare, and no comparison between groups was made. The sample differs from the one an audit of Gate A proposed (Unsafe answers from OLMo-2-1B's Gate A fine-tunes, stratified into very short answers and answers flagged only as unethical, plus the Unsafe answers on the over-refusal bank), so it does not test those strata.

Appendix C: Design details

This appendix holds design details that the main text summarizes. The frozen pre-registrations govern wherever this summary differs from them [TO BE DECIDED: anonymized copies as supplementary material].

C.1 Gate B

Arms and waves. Wave 0a: three random-removal arms R_1–R_3 (600 rows, 75 per category, each from its own sealed draw) at seeds 1101–1103, for the noise gate. Wave 0b: a retrain of a_0 at seed 1101 that keeps its checkpoints after steps 10 and 38, and the Gate C diagnostics. Wave 1: each of the 8 category removals at seed 1101. Wave 2: $m = 2$–4 units (the two with the largest wave-1 contrast, plus Gate C's picks if not already in) at seeds 1102 and 1103. Every removal arm comes from a sealed edit file and trains 33 steps; the Gate A a_0 runs are reused. No step starts before the step above it is complete.

Contrasts and noise. All U are in pp over 600 sampled answers. For unit u at seed j, $c_{u,j} = U(R_j, j) - U(u, j)$, so a_0 cancels exactly; $D_u = c_{u,1101}$ (wave 1) and $\Lambda_u = (c_{u,1102} + c_{u,1103})/2$ (wave 2). With $r_j = U(a_0, j) - U(R_j, j)$ and σ_{eval} the median prompt-clustered standard error of a seed-matched pair's per-prompt difference, $\hat{\sigma}_{\text{gate}} = \max(\text{SD}(r_{1101}, r_{1102}, r_{1103}), \sigma_{\text{eval},0})$; the final $\hat{\sigma}$ pools the r_j and the wave-2 pairs. A fallback $\hat{\sigma}$ that does not use a_0 applies if the retrain's adapter fingerprints, or a run's environment, differ from Gate A's. Tests use t at 3 degrees of freedom, the effective degrees of freedom of $\hat{\sigma}$.

Noise gate. After wave 0a, Gate B goes on only if $\hat{\sigma}_{\text{gate}} \leq 7$ pp and $\max(8, 4.177 \times 0.707\,\hat{\sigma}_{\text{gate}}) \leq 33.8$ pp (0.75 of Gate A's mean E). A failure stops Gate B, and the pre-registered fallback applies instead of a redesign.

Verdict. *LOCALIZED*: some wave-2 unit has $\Lambda_u \geq 8$ pp and $\Lambda_u \geq t_m \times 0.707\,\hat{\sigma}$, where t_m is the one-sided Bonferroni 5% t at 3 degrees of freedom (3.182, 3.740 and 4.177 for $m = 2, 3, 4$); the unit passes guards G2–G4 and the direction checks, and every arm that the contrast uses passes G1. *DIFFUSE*: $\max_u D_u < 2.381\,\hat{\sigma}$, and every wave-2 unit has $\Lambda_u + 1.638 \times 0.707\,\hat{\sigma} < 8$ pp; both also hold on coherent answers (G2) and in both adverse corners (G3), and every arm passes G1. *TOO CLOSE*: anything else, a guard failure or a missing arm included. With LOCALIZED, the one-sided 95% lower bound $\Lambda_u - t_m \times 0.707\,\hat{\sigma}$ is reported. At Gate A's mean erosion, 8 pp is about 18% of E, so LOCALIZED does not say that the unit carries most of the erosion. The verdict reads U alone; the blind human read recommended before the first wave is done, on a different sample from the one first proposed (Appendix A).

Guards. G1, answers end: Gate A's bars on sampled answers (at most 5% capped and at most 10% failing the coherence screen; a short-answer TOO CLOSE counts as a failure); a random-removal arm that fails G1 makes the verdict TOO CLOSE with no extension. G2: the verdict recomputed on coherent answers only. G3: two adverse corners, with unscored answers counted against the verdict, and with controversial answers counted Unsafe. G4, over-correction: the unit's mean refusal rate on the 100 over-refusal prompts may exceed the untuned model's by at most 5 pp. Direction checks: hard-category U, the share of sampled answers that do not refuse, and Granite Guardian's greedy labels must each give a positive contrast.

Extension. It is automatic; no choice follows an outcome. A wave-2 unit is *contested* if it is not LOCALIZED, $\Lambda_u + 1.638 \times 0.707\,\hat{\sigma} \geq 8$ pp, and it passed the guards. After a TOO CLOSE with at least one contested unit and no random-removal arm failing G1, up to two contested units run at seeds 1104 and 1105 with R_4 and R_5, and are read once more over four seeds with the same constants. No other reading follows an outcome, and no new margin is set.

Length control. After LOCALIZED, for the unit with the largest Λ_u: two removals of as many rows, drawn from the other units and matched to the unit's answer-length bins, run against the unit's removal at two fresh seeds. The wording is "content-localized" only if the length-matched contrast L_u is at least 8 pp and at least $2.353 \times 0.707\,\hat{\sigma}$, and all four arms pass G1; otherwise it is "localized under the per-example-mean loss; content and answer length not separated". The control changes the wording, never the verdict.

Operating characteristics. By simulation, with one unit carrying the effect and the guards not simulated (so the rates are upper bounds): at central planning noise (a seed-pair standard deviation of 3.75 pp) and given that the noise gate passed, a true 8, 10 or 12 pp unit reads LOCALIZED in 52, 83 or 96% of runs, so an 8 pp category reads LOCALIZED about half the time. 80% detection needs about 9, 10 or 12 pp at low, central or high planning noise. False LOCALIZED, given that the gate passed and no unit has an effect, is 0.2–2.7% for seed-pair standard deviations of 3.75–7 pp. A missed 8–12 pp unit mostly reads TOO CLOSE, not DIFFUSE (DIFFUSE in at most 7.5% of runs at 8 pp). The pre-registration records the prior that DIFFUSE or TOO CLOSE is more likely than LOCALIZED.

Status. The noise gate passed after wave 0a, and neither trigger for the fallback $\hat{\sigma}$ fired after wave 0b. The diagnostics ran in wave 0b, and their rankings were sealed before wave 1 (Appendix C.2).

C.2 Gate C

Diagnostics and picks. Each diagnostic scores all 4,800 rows, and a unit's score is the mean over its 600 rows. Attribution is bi-anchored gradient similarity (sealed anchors: 64 harmful, 64 safe) at the retrained a_0 's checkpoints after steps 10 and 38, with equal weight. Representation similarity is bi-anchored, at the untuned model's final layer and last response token. A content classifier is reported only. pick_A and pick_R are the top units of attribution and representation similarity (ties go to the lower unit id). The format pick pick_F is the unit with the largest mean share of a row's loss weight on its first five answer tokens under the per-row loss; it costs nothing.

Rule. For a pick p, B_p is the mean of $c_{p,j}$ over seeds 1101–1103, and $\Delta = B_{\text{pick}_A} - B_{\text{pick}_R}$. A pick beats random if $B_p \geq 2\hat{\sigma}/\sqrt{3} \approx 1.15\,\hat{\sigma}$; the margin is $M_C = \max(6\text{ pp}, 1.15\,\hat{\sigma})$, with $\hat{\sigma}$ from before any extension. AGREE: $\text{pick}_A = \text{pick}_R$. ATTR_BETTER: $\Delta \geq M_C$ and pick_A beats random. REP_BETTER: $\Delta \leq -M_C$. NONE_FINDS: neither pick beats random. The verdict is the first that holds in the order NONE_FINDS, AGREE, ATTR_BETTER, REP_BETTER, TOO CLOSE. If an arm that a B_p uses fails G1, Gate C reads TOO CLOSE. Gate C is read only after LOCALIZED, and it reads "not read" when an arm it needs is missing. No Gate C extension runs.

Format reading. If $\text{pick}_A = \text{pick}_F$, the paper must say that attribution's pick is explained by answer form (a near-free answer-form score picks the same unit) and claims no value of that pick beyond answer form; the same holds for representation similarity. If neither pick equals pick_F, the paper says only that neither pick is the form pick, which does not show that the picks are free of form. The two equalities are fixed when the rankings are sealed, so they are reported whatever Gate B reads; the form reading never changes a verdict. Also reported: the Spearman ρ of each diagnostic's 8 unit scores with the first-five-token share and with the wave-1 contrasts D_u (a one-sided 5% test needs $\rho \geq 9/14$), and the top-1 hit (chance 12.5%).

Consequence of the sealed rankings. Attribution and representation similarity pick the same unit, and the format rule picks another. So $\Delta = 0$: ATTR_BETTER and REP_BETTER cannot occur, and Gate C, if read, is NONE_FINDS, AGREE or TOO CLOSE (the last only through the answers-end guard on the pick arms). The format reading is the second case above, and wave 2 holds $m = 2$ or 3 units.

C.3 Stage 1: pools, seeds and secondary arms

Pools and anchors. The refusal pool holds prompts from PKU-SafeRLHF (Ji et al., 2025) that have an unsafe response in the source data, paired with a response that the source labels safe and that opens with a refusal by the frozen lexicon. Pools and anchors are never filtered by the evaluation judge, and prompts that nearly duplicate an evaluation prompt are removed. The add effect must also hold on the evaluation prompts least similar to the added rows.

Seeds and power. Method arms run at six seeds, random removal at eight and no repair at six; seeds run in blocks fixed before any outcome, and only complete blocks are analysed. A blinded rule reads only Gate B's pooled noise estimate: up to 4.5 pp, the plan of §4.3; from 4.5 to 6.0 pp, the family shrinks to P1, P3 and P6 at nine seeds; above 6.0 pp, stage 1 does not run. At planning noise the minimum detectable effect after Holm's first step is about 5 pp [PLACEHOLDER: final MDE from Gate B's pooled noise].

Secondary and scored-only arms {[TO BE FINALISED] .} At three seeds each, exploratory, with one-sided bounds only: a random dose curve (1,200 and 2,400 rows removed); random removal plus S1; a DataShield-style filter; rules for answers with list or markup lines and for the longest answers; the length rule under a token-mean loss; and octiles of answer length. Scored without new runs: a Self-Inf-N-style score and a content classifier; we publish the overlap of all top-600 sets. SQSD is cited but not run, because its scoring needs several FRE.

Certified evaluation of the references. A pinned evaluation script regenerates each a_0 adapter by a deterministic re-run and is certified once: with $\lambda = 1$ and no prefix it must reproduce a_0's U exactly. The implementations of Safe LoRA and B+Qi are committed before stage 1 is frozen.

C.4 The skill bar

Held-out loss on all 200 task rows is the skill limit of the comparison. The probe and Gate A bars also check ARC-Easy accuracy (Clark et al., 2018) on 189 questions: the paired drop d against the untuned model must stay at or below 0.03. With 189 questions the paired interval is about ± 0.07 wide, so it straddles 0.03 for almost every run. A rule fixed during development, first for probe and Gate A readings at one common recipe and then extended to every family's Gate A read at its chosen learning rate before any such read, resolves a straddling interval: the run **passes, as "skill kept, reported as a limitation", when the point drop $d \leq 0.03$ and the held-out task loss is below the untuned model's (the task is learned); it fails when $d > 0.03$.** This rule is a loosening: it can pass a run whose true drop is up to about 6 points, which 189 questions cannot detect. We report the verdict under the frozen bars beside the verdict under this rule. Disclosure: the first recipe-G probe on OLMo-2 was read under the earlier rule before this rule was proposed, so for that one run the rule is not applied blind. At 1B, ARC-Easy is close to chance, so "skill kept" rests mainly on the held-out task loss.

C.5 The positive control

Definitions. A planted data set $\mathcal{D}^{\mathrm{p}}$ replaces the rows of one small cell of $\mathcal{D}$ with k rows that look benign but carry a known eroding signal, keeping the number of rows fixed. Nested partitions split $\mathcal{D}^{\mathrm{p}}$ into M cells, so that the planted cell sits inside one cell at every M. The planted erosion is $P = F(a_0^{\mathrm{p}}) - F(a_0)$, and removing the cell that holds the plant at menu size M restores $G^{\mathrm{p}}(M) = F(a_0^{\mathrm{p}}) - F(a^{\mathrm{p}}_{-c(M)})$. A diagnostic *$d$ hits* at (M, k_d) if it ranks that cell among its top k_d. Because the truth is known, the sweep measures how diagnosis and trial runs scale with M, which natural menus at our budget cannot.

Design. [PLACEHOLDER: runs only if funded.] The plant is $k = 37$ rows whose answers comply directly with prompts that also have an unsafe answer in the source data, but which the source labels safe; they look benign. A sealed draw picks one category and one leaf cell inside it, and the plant replaces that cell's rows, so the run keeps 4,800 rows and 38 steps. The nested cells [TO BE FINALISED] are the 8 categories (600 rows), 32 length quartiles within categories (150 rows) and 128 random quarters of those (37–38 rows). A pilot of three seeds goes on only if the plant adds at least 10 pp of erosion with a one-sided 90% lower bound above 5 pp (otherwise the plant doubles once, then the null is reported). The main sweep removes the planted cell at each menu size (three seeds each) and runs random removal on the planted data. Every scorer ranks the cells without new runs. Outputs: $G^{\mathrm{p}}(M)$; each diagnostic's top-k hit rate with known truth; the

runs-only cost of finding the cell, replayed with the frozen selector rules; and a check of additivity across the nested levels.

C.6 Cost conventions

Every arm and every scorer is logged in GPU-seconds by stage (setup, training, greedy generation, sampled generation, judging, scoring, overhead). One FRE per family is the median measured time of that family's random-removal arms. Two cost conventions are frozen before stage 1: *full* (training plus the evaluation as run) and *lean* (training plus one sample per harm prompt, priced from the measured time per answer); the cost c_0 of one lean trial run is measured on a real job. Every pool filter, generated answer, anchor job and scorer is charged; "free" means below 0.01 FRE. Attribution is priced from its measured scoring pass. Claims about cost hold only where both conventions agree, and no cost ratio measured at 1B is claimed for a larger model.

C.7 The value of diagnosis

A *menu* $\mathcal{A}_M = \{a_0, a_1, \dots, a_M\}$ holds a_0 and M repairs built by a frozen rule, never by a compared diagnostic; $\mathcal{A}_{\text{feas}}$ is its feasible subset and $a^* = \arg\min_{a \in \mathcal{A}_{\text{feas}}} F(a)$. A *selector* π with budget B (in FRE) buys trial runs (each a training fraction ϕ and a prompt fraction ψ, priced from the measured cost model) at total cost at most B and returns $\hat{a}$. It drops a repair whose adverse estimate of O or L passes its limit (a_0 is never dropped), and picks the lowest estimated F among repairs with full runs on at least two seeds; ties go to a_0. An infeasible pick is scored as if no repair were made:

$$\text{loss}(\hat{a}) = \begin{cases} F(\hat{a}) & \hat{a} \in \mathcal{A}_{\text{feas}}, \\ F(a_0) & \text{otherwise}, \end{cases} \qquad \text{Reg}(\pi; B, M) = \mathbb{E}[\text{loss}(\hat{a})] - F(a^*).$$

This is a scoring convention for an expected outcome at a fixed budget, not a safety guarantee. The runs-only selector π_- uses one rule per (M, B) cell (a one-shot screen, successive halving or hierarchical bisection), chosen from a simulation before any outcome and then frozen. The selector π_{+d} pays the cost c_d of diagnostic d, keeps its top k_d repairs, and runs the same rule on $B - c_d$. The value of d and its break-even cost are

$$\Delta_d(B; M, c) = \text{Reg}(\pi_-; B, M) - \text{Reg}(\pi_{+d}; B, M, c), \qquad c_d^*(B, M) = \sup\{c \geq 0 : \Delta_d(B; M, c) \geq 0\}.$$

$F(a^*)$ cancels in Δ_d. When a_0 is feasible, regret is non-negative, so no diagnostic can save more than $V_{\max}(B, M) = \text{Reg}(\pi_-; B, M)$. At the menus we can afford, these quantities are mostly outputs of a model, so we treat them as predictions. A generative model (repair effects shrunk by empirical Bayes, noise from pooled residuals, measured costs, two effect shapes) is frozen before Gate B's category runs, and its predicted $V_{\max}$, Δ_d and c_d^* for the category menu and the planted menus are recorded then. We report measured against predicted values (replayed with the same code path on the outcome tables), and score OLMo-calibrated predictions of the chance of finding the best repair, and of regret, on Llama's category menu. Selectors are not run against each other at equal budgets in this study.

C.8 The opening decomposition and the first-token margin

Let $p(\theta)$ be the share of sampled answers to harm prompts that open with a refusal (a frozen lexicon applied to the first 12 words), and $u_{\text{R}}(\theta)$ and $u_{\text{N}}(\theta)$ the unsafe rates among answers that do and do not open with one, so $f = p\,u_{\text{R}} + (1 - p)\,u_{\text{N}}$. For the untuned model (subscript 0) and a fine-tune (subscript 1),

$$f_1 - f_0 = \underbrace{(p_0 - p_1)(u_{\mathrm{N0}} - u_{\mathrm{R0}})}_{\text{opening switch}} + \underbrace{(1 - p_1)(u_{\mathrm{N1}} - u_{\mathrm{N0}}) + p_1(u_{\mathrm{R1}} - u_{\mathrm{R0}})}_{\text{content change}}.$$

The *opening share* is the first term over $f_1 - f_0$; the split depends on the order of terms, so both orders are reported. The decomposition is descriptive: it says whether a change in the unsafe rate shows up in how answers open or in what they say, not which training choice caused the change. The first term has two layers: how many refusal openings fine-tuning removes ($p_0 - p_1$), and how often a direct answer was already unsafe before fine-tuning (u_{N0}); where u_{N0} is high and u_{R0} low, a lost opening comes close to an unsafe answer. If the opening share is small, repairs that act only on how answers open (refusal examples; an objective that constrains updates on the first answer tokens) may restore less than repairs that change what the answers say. We apply the decomposition to every arm; across families and recipes, the two layers are reported in [CITE: concurrent submission, anonymized].

The causal test on Gate B arms is a teacher-forced first-token refusal margin $m(p) = \log P_{\mathrm{ref}}(p) - \log(1 - P_{\mathrm{ref}}(p))$, where P_{ref} is the probability of a frozen refusal phrase as the first answer tokens. If a category's effect runs through openings, the margin should track U across arms.

Editor's notes (not part of the paper; delete before submission)

What this is. Revision 6 of the draft, written on 16 October 2026 from 21:10 UTC (17 Oct IST; clock checked), retargets the paper to ICML 2027 (FOCUS 16 Oct 21:05 UTC, commit 416b780) and brings the body to ICML's 8-page limit: the main body now measures 7.96 pages to the Impact Statement (8.15 pages to the References heading, the measure used before; 9.36 in revision 5e); see "What changed in revision 6" and conflicts 41–52. Design details moved to a new Appendix C (part of the paper); none was dropped. The body now says what FOCUS records since 9 Oct: Llama-3.2-1B's rung without "provisional" (its Gate A passed under decision 25, 10 Oct); the 7–8B rungs of both families as design; Gate B's frozen pre-registration, its passed noise gate and its sealed rankings, with the pre-registered consequence for Gate C; D5 = (a). It applies the side-paper overlap rule, the coordinator's three additions (recipe G as a reference, not a general fix; the literature sweep's factual fixes; the user's rule of 17 Oct, "keep the more ambitious one always but always open to fallbacks": the title kept, token matching, the full 7–8B comparison on both families) and adds an Impact Statement. Earlier revisions: 5e (9 Oct, from 17:14 UTC) called Llama-3.2-1B's rung provisional and stated the frozen too-close rule exactly; 5d (from 17:03 UTC) filled Llama-3.2-1B's rung provisionally; 5c (from 15:29 UTC) applied the user's wording package (decision 23) and the ARC rule's scope (decision 16); 5b (from 15:06 UTC) added the blind human read; 5 (from 13:17 UTC) made recipe B the main recipe again, recipe G a reference, the per-family learning-rate rule, five mandatory references and the dataset plan; 4 (8 Oct, from 21:40 UTC) withdrew the loss-weighting account and added the ARC rule; 3 (from 19:35 UTC) added the *worse* label; 2 (from 14:00 UTC) followed the paper specification (`research/review/2026-10-08-paper-specification.md`, commit f2a9979) as amended by the adopted FOCUS amendment (commit 4b37ec2, "ADOPTED 2026-10-08T13:53 UTC") and the study plan `research/review/2026-10-08-sharpened-direction-DRAFT.md` (revision 2). FOCUS.md wins on any conflict. **Nothing in the paper body is a result,** with four kinds of exception, each a design or status fact: the blind human read's rounded caveat (§4.2, §6, the Impact Statement; conflict 28); the rungs chosen from development probes, with a one-line disclosure that both Llama rungs were accepted on a too-close bar (§4.1); Gate B's status (the noise gate passed; the rankings are sealed) and the pre-registered consequence of the sealed rankings for Gate C (§4.6, Appendix C.1–C.2), with no number and no category named; and §4.9's disclosure that the stage-1 pre-registration is frozen after those Gate B values were seen. Gate B's category runs (wave 1), Gate C's reading, stage 1, the positive control and the Llama stage have not run. Development evidence appears only in these notes, except the blind read's counts in Appendix A. In the LaTeX build these notes are Appendix B and the design details are Appendix C; when the notes are deleted before submission, Appendix C becomes Appendix B (every reference to it is a `\ref`). Menu sizes, the filter list and costs are marked [TO BE FINALISED]; the stage-1 pre-registration, Safe LoRA's settings and Alpaca's preconditions are marked [NOT YET FIXED]; the release policy and the supplementary copies are marked [TO BE DECIDED]; the full 7–8B comparison is marked [PLANNED; funding-gated, fallback one family]; token matching is marked [TO BE FIXED in the stage-1 pre-registration]; the concurrent side paper is cited as [CITE: concurrent submission, anonymized]. The ICML LaTeX version (`icml/main.tex`) carries the same text with a fetched `references.bib`; the author-year citations below use the venue years and letters of those fetched entries (revision 6 re-lettered Zhang et al. 2026: a = Safety at One Shot, b = DataRx, c = DataShield).

Title. The spec's working title ends "…, and When Diagnosis Pays". The user's decision makes the measured repair comparison the headline and the decision-value map secondary, so the title names only the comparison. **Revision 6:** the title is unchanged: the user's rule of 17 Oct keeps "Compute-Matched", with "Cost-Accounted" as the fallback only if token matching cannot be done (decision 26). Alternatives: 1. *Repairing Safety After Benign Fine-Tuning: A Compute-Matched Comparison of Data Repairs, and When Diagnosis Pays* (the spec's title; use it only if the planted sweep is funded and gives the map measured support). 2. *Diagnose or Retrain? A Compute-Matched Comparison of Data Repairs for Safety After Benign Fine-Tuning* (the previous title as a lead-in, as the spec allows). 3. *Choosing a Data Fix When Benign Fine-Tuning Erodes Safety: Diagnosis, Trial Runs and Side Effects at Matched Compute*. This is the DIFFUSE title: the plan (§11) and FOCUS ("agreed fallback") switch to it if Gate B reads DIFFUSE. Its number is kept from the previous draft because the plan refers to it as "alternative title 3". 4. *Remove, Add or Leave the Model Alone? Seeded, Compute-Matched Repairs for Safety Eroded by Benign Fine-Tuning* (stresses the model-level references). 5. *Repairing Safety After Benign Fine-Tuning: A Cost-Accounted Comparison of Data Repairs* (revision 6; the literature sweep's proposal; the fallback title under decision 26, used only if token matching cannot be done).

What changed in revision 6 (16 Oct, from 21:10 UTC)

- **Sources.** The lead's brief for revision 6, with two additions from the coordinator during the work (recipe G; the literature sweep). FOCUS.md on `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched; head 416b780), every entry from 9 Oct 17:00 UTC to the newest (16 Oct 21:05 UTC, the ICML 2027 goal); the Gate B/C pre-registration, revision 5, frozen 10 Oct 11:16 UTC (SHA-256 5c97cce7…, checked on the fetched file), §1–§5, §7, §8 and §10; the GI handoff STATUS of 16 Oct 13:20 UTC; the kill-test design §4.4 (Gate C verdicts); the side paper's abstract and outline (for the overlap rule only); `writing/paper/2026-10-17-lit-sweep.md` §0, §4 (corrections) and §5; the head of `writing/paper/2026-10-17-icml-replan-PROPOSAL.md` (status only). Details in "Sources read (revision 6)".
- **Venue (FOCUS 16 Oct 21:05 UTC).** ICML 2027 is the goal; TMLR or NeurIPS 2027 stays the fallback until the user names one. The ICML 2027 deadlines are ESTIMATED (about 16 Jan 2027 abstract, 22 Jan 2027 paper, AoE; a third-party tracker); the official call is not published (sweep §6). Decision 6 and the LaTeX header say so; the ICML 2026 style stays, with the swap-to-2027 note.
- **§4.1, families and scale.** The ladder is now design: OLMo-2-0425-1B and Llama-3.2-1B at 1B; OLMo-2-1124-7B-Instruct and Llama-3.1-8B-Instruct at 7–8B with, after the coordinator's addition 3, "the full comparison runs on OLMo-2-1124-7B-Instruct and Llama-3.1-8B-Instruct … [PLANNED; funding-gated, fallback one family]" (the brief's "[TO BE DECIDED: … ICML re-plan proposal pending]" was used first and is replaced); a headline check "at the largest single-GPU size" [PLACEHOLDER: if resources allow] (the 8 Oct 12:53 UTC ladder names OLMo-2-32B and, if funded, Llama-3.x-70B; the body no longer names sizes). Qwen leaves the study: "a same-recipe comparison of OLMo-2, Llama-3 and Qwen2.5 is reported in [CITE: concurrent submission, anonymized]".
- **§4.1, rungs.** "Llama-3.2-1B's provisional rung is …" became "The rungs are 10^{-4} for OLMo-2 (1B and 7B) and 5×10^{-5} for Llama (3.2-1B and 3.1-8B)", with one disclosure: "Both Llama rungs were accepted on a too-close bar (one seed's answers-end bar at 1B; the window's upper edge at 8B); §5 reports both." Sources: FOCUS 10 Oct 09:40 UTC (Llama-3.2-1B Gate A PASS under decision 25), 10 Oct 21:30 UTC (OLMo-2-7B rung 1, partial) and 11 Oct 03:53 UTC (Llama-3.1-8B rung 2, accepted "at the edge"). Numbers are in the development evidence.
- **§4.1, the family × recipe sentence (overlap rule).** "During development no rung of a one-recipe ladder left both families partly eroded (§5), because families respond to the same change of recipe by different amounts" became "Families respond differently to the same recipe [CITE: concurrent submission, anonymized]; we fixed this rule after reading a one-recipe development ladder, on which no rung left both 1B families partly eroded, and Llama's first probe at recipe B". The disclosure of when the rule was fixed stays; the finding is cited.
- **Recipe G is a reference, not a fix (coordinator's addition 1; FOCUS 11 Oct 10:38 and 12:54 UTC; GI handoff 15 Oct).** FOCUS records that recipe G lowered erosion for OLMo-2-1B but not for OLMo-2-7B or Llama-3.1-8B, and that "the paper must not present recipe G as a general fix". Edits: §1 "recipe G (a recipe-only fix whose warmup and cosine decay halve the mean learning rate)" → "recipe G (a standard warmup and cosine schedule)"; §4.1 "*Recipe G*, the recipe-only fix and a reference" → "*Recipe G*, a reference arm that changes only the recipe", plus "It is not a general fix: its effect on erosion differs across families and sizes [CITE: concurrent submission, anonymized]"; §4.4 "**Recipe G** (recipe only)" → "(a reference that changes only the recipe)"; §6 "we report it as a recipe-only fix" → "it is a reference, not a general fix"; Table 1 no longer lists the references (below). The abstract never named recipe G ("a gentler learning-rate schedule" in revision 5e is gone with the abstract's list of references); Figure 1 and its caption never named it. The 7–8B recipe-G numbers are side work: they are only in the development evidence, labelled "not this paper's result". In these notes, decision 15 and risks 3, 21, 24 and 26 carry a revision-6 line; older history entries keep their words.
- **The user's decision of 17 Oct (coordinator's addition 3): "keep the more ambitious one always but always open to fallbacks".** (1) The title keeps "Compute-Matched" (decision 26). (2) §3.2 gains one design sentence: "Removals are matched in training tokens as well as rows, or a token-matched analysis is pre-registered beside the row-matched one [TO BE FIXED in the stage-1 pre-registration]"; the lead rules on which. (3) The 7–8B tier is the full comparison on both families, marked [PLANNED; funding-gated, fallback

one family], in §4.1, §4.8 ("the full 7–8B comparison of §4.1") and §6 ("at 7–8B it is planned on both families, funding-gated, with one family as the fallback"): the re-plan's option B, with option C as the fallback (decision 28). (4) The opening-hold loss and Safety at One Shot are listed as candidate additional reference arms in these notes only (decision 27); the body's five references are unchanged. To hold the length, §4.2's "Banks" item joined the U item (same words).

- **Gate B status (§4.6).** "These rules summarize its pre-registration, revision 5 [NOT YET FIXED]" became "Its pre-registration was frozen before any Gate B run; Appendix C.1 gives its rules"; a new last sentence: "**Status:** the noise gate has passed, and Gate C's rankings were sealed before the category runs." No number.
- **Gate C consequence (§4.6; Appendix C.2).** New, neutral and pre-registered: "In the sealed rankings both diagnostics pick the same category, and the format rule another. So, by the frozen rules, Gate C can read only AGREE, NONE_FINDS or TOO CLOSE, or is not read; ATTR_BETTER and REP_BETTER cannot occur; and the paper may say only that neither pick is the format pick, which does not show that the picks are free of format." The brief asked for the ATTR_BETTER/REP_BETTER clause. The format clause is added because the frozen pre-registration (§5, "Form pick") says the two equalities "are reported whatever Gate B reads" and fixes this wording when neither equality holds (decision 29). Checked against the pre-registration's order (NONE_FINDS, AGREE, ATTR_BETTER, REP_BETTER, TOO CLOSE), AGREE's definition (pick_A = pick_R; kill-test design §4.4) and the pick-arm G1 guard: with equal picks $\Delta = 0$, so ATTR_BETTER ($\Delta \geq M_C$) and REP_BETTER ($\Delta \leq -M_C$) cannot hold, and TOO CLOSE can come only from the guard. Appendix C.2 also notes that wave 2 then holds $m = 2$ or 3 units. The body names no category; the development evidence does.
- **D5 = (a) (FOCUS 16 Oct 12:15 UTC).** P1–P6 stay primary and the selector stays secondary and model-based (unchanged). New: §6 "Selectors are not run against each other at equal budgets here; a later study with its own pre-registration would do that" (and one line in Appendix C.7). The one-example safety patch is bounded in text: §2 cites Safety at One Shot (Zhang et al., 2026a; new BibTeX) as a patch that "acts after fine-tuning and is not among our references (§6)", and §6 says "no claim here covers it".
- **§4.9, a timing disclosure (FOCUS 16 Oct 00:25 and 13:21 UTC; stage-1 brief rule T2).** The stage-1 pre-registration "lists every Gate B value seen before it is frozen (the noise-gate reading and the sealed rankings)".
- **Overlap rule (the side paper, ARR #13386, "Same Recipe, Different Fate").** Its abstract covers the same-recipe family contrast (OLMo-2, Llama-3, Qwen2.5 at 1B and 7–8B), recipe G against recipe B, the 7–8B replication, the half learning rate, the rank-16 and token-mean-only runs, the two-layer opening account and the blind read. The main paper no longer presents any of these as its own result: §4.1 (two places), §4.10 item 2 ("Recipe and family contrasts are not results of this paper … reported in [CITE: concurrent submission, anonymized]"), §6 ("a rank-16 check is in [CITE …]"), §5's placeholder (the same-recipe contrast, the ladder, the rungs, the token-mean-only run and the rank-16 check are gone from it), Appendix C.8 ("across families and recipes, the two layers are reported in [CITE …]") and Appendix A ("The same read is also reported in [CITE …]"; decision 30). The abstract and the contributions made no such claim; contribution 3 now says "how much of each repair's effect runs through refusal openings" (repair arms, not families). The citation is a placeholder: ICML lets authors cite their own unpublished work only anonymized, with an anonymized copy as supplementary material (ICML 2026 instructions; decision 32).
- **Literature sweep, factual fixes (coordinator's addition 2; sweep §4 "Corrections" and §5).** (1) DualSelect: §1 now says it "compares data selectors and a safety-instruction mix over three seeds, with a false-refusal rate and its selection overhead reported, but on training data that include red-teaming pairs, and without model-level references or equivalence-bounded verdicts"; §2's removal paragraph drops its DualSelect sentence; "How we differ" says "DualSelect compares selectors and a safety-data mix over seeds, with a false-refusal rate and selection cost". The facts are FAQ's check of the arXiv full text (SafeInstr, 2–3% safety-instruction tokens, is a baseline; selection overhead 1.34× standard SFT at a 90%-of-target-tokens budget; three seeds, mean ± std; a false-refusal rate). The sweep's "six selectors" and "red-teaming pairs" come from the sweep, not re-read here; the body gives no count. (2) DataRx: §2 adds "Its baselines, all additions, include self-distillation, a precedent for our self-generated arm" (the sweep's full-text reading; the coordinator confirmed). No sentence claims the self-generated arm is new. (3) DataShield: the body cites only Zhang et al. (2026c, arXiv:2606.00160); the

BibTeX comment records that arXiv:2607.15081 (Wu et al.) is a different paper with the same name (checked with the arXiv API). (4) GradShield: the NeurIPS 2026 accepted-papers JSON lists it as a poster (same seven authors); the entry keeps its arXiv form and adds "NeurIPS 2026 (poster)" as a note (CITE-CHECK 8). (5) "How we differ" is rewritten after the sweep's §5 text, shortened: it adds SafeTune and SafeTuneBed, DualSelect and DataRx to what we do not claim first, and the combination now says "for purely benign data", "limits fixed in advance" and "every arm's full cost, scoring included". Not applied (decisions for the user and lead): the title word (decision 26; since resolved by the user's rule of 17 Oct: kept); an opening-hold or one-shot reference (decision 27; candidates in these notes only); the sweep's other added sentences and design actions (risk 32).

- **New citations (each fetched programmatically).** Safety at One Shot (`zhang2026oneshot`; OpenReview note EyH8Fu3vtZ, ICLR 2026 poster, BibTeX from the API search record; arXiv:2601.01887 checked by the arXiv API; the second author is "Tony He" on OpenReview and "Lipeng He" on arXiv); SABA (`anonymous2026saba`; OpenReview Jo6U0y2omr, an ICLR 2027 submission under review, cited as Anonymous; claim from its abstract in the API record): §2 "A concurrent submission finds risk rankings taken at initialization nearly uncorrelated with those after two epochs, and re-scores rows during training"; SafeTuneBed (`hossain2025safetunebed`; arXiv:2506.00676 by the arXiv API; an IJCAI 2025 workshop version with another title exists on OpenReview). CITE-CHECK 27–29.
- **Trim to 8 pages: what moved to Appendix C (nothing dropped).** C.1 Gate B (arms and waves, contrasts and noise, the noise gate, the verdict rule and constants, guards G1–G4 and direction checks, the extension, the length control, operating characteristics, the recorded prior, the blind-read sentence from §4.6, status); C.2 Gate C (diagnostics, picks, rule and order, the format reading, reports, the consequence of the sealed rankings); C.3 stage-1 pools and anchors, seeds and power (the blinded rule), secondary and scored-only arms, the certified evaluation script and "the implementations of Safe LoRA and B+Qi are committed before stage 1 is frozen" (from §4.4); C.4 the ARC rule and its disclosure (from §4.2); C.5 the positive control's definitions (old §3.4) and design (old §4.5); C.6 the cost conventions (from §4.7); C.7 the value of diagnosis in full (old §3.5); C.8 the opening decomposition (old §3.6) and the first-token margin's formula (from §4.10). The body keeps a summary of each, with its claims.
- **Trim: what was shortened in the body.** The abstract (311 → 239 words, placeholders counted; the list of the five references and "of the same size" moved to §1 and §4); §1's fourth and fifth paragraphs and the contributions; §2's DataRx numbers moved from §1 to §2 (§1 keeps the direction of the result); §2's two short paragraphs merged ("Prediction and decisions under a budget"); §3.2's label list set as a sentence and the adapter-scaling display inline; §3.4 (the value of diagnosis) cut to its definitions; Table 1 lists the eight stage-1 arms only (the five references and their seeds are in §4.4, where they were already listed); §4.3's three paragraphs merged; §4.5, §4.7, §4.8 and §4.10 shortened; §5's placeholder shortened; §6's "Cost" merged into "Baselines, doses and cost". Section numbers changed: old §3.5 is now §3.4; old §3.4 and §3.6 are in Appendix C.
- **Impact Statement (new; ICML requires it; it does not count toward the page limit).** 169 words, after §7 and before the references: the aim; three risks (the data and repair findings and the planted rows could help an attacker; the judges' labels are not a safety guarantee, with the blind-read caveat; small models and one dataset); and [TO BE DECIDED: what is released] (decision 31).
- **Anonymity checks.** The body, Appendix A, Appendix C and the bibliography hold no author name, institution, repository, branch, account, commit hash, grant or pod name; the self-reference to the side paper is the anonymized placeholder; the PDF metadata says "Anonymous Authors" (checked with `pdfinfo`; the subject line still says "International Conference on Machine Learning 2026", from the 2026 style). Appendix A's "One reader, an author who helped design the study" names no one. These notes (Appendix B) are not anonymous and must be deleted before submission.
- **Length.** Before: 9.36 pages (revision 5e; same measure, re-measured on the revision-5e PDF with the re-created script). After: 7.96 pages to the Impact Statement heading (ICML's measure: the statement is outside the limit) and 8.15 pages to the References heading (the old measure, which now includes the 0.19-page Impact Statement). §5 and §7 are still placeholders; results will need about a page (risk 28).

- **Pipelines.** The scratchpad scripts named in the brief (`paper_to_pdf.py`, `scripts/body_pages.py`, `scripts/notes_md_to_tex_rev5c.py`, `scripts/wrap_tex.py`, `rev5d/splice.py`) no longer existed (the scratchpad's files were cleared; only empty folders were left). They were re-created in `rev6/scripts/`: the body measure reproduces revision 5e's 9.36 pages on its PDF; the notes converter follows revision 5e's Appendix B conventions (conflict 41).
- **Editor's notes.** Decisions 6, 7, 13, 15, 17, 18 and 25 updated (7, 18 and 25 RESOLVED); decisions 26–34 added (26 and 28 RESOLVED by the user's rule of 17 Oct). Risks 2, 3, 17, 19, 21, 22, 24 and 26 updated; risks 28–33 added. "Where the framing is weak" item 6 added. Placeholders, fill-ins, CITE-CHECK 7, 8 and 21, development evidence and status brought up to FOCUS 16 Oct 21:05 UTC; CITE-CHECK 27–29 and conflicts 41–52 added. The LaTeX Appendix B is regenerated from these notes.

What changed in revision 5e (9 Oct, from 17:14 UTC)

History: revision 6 drops "provisional", because the user recorded Llama-3.2-1B's Gate A as PASS under decision 25 (FOCUS 10 Oct 09:40 UTC).

- **Source.** The lead's brief for revision 5e. FOCUS.md read to its newest entry, "Granite Mac-versus-pod PARITY: PASS, 2026-10-09 17:22 UTC" (commit 2837eaf, committed while this revision was written; no later entry exists at 17:31 UTC), with the entries of 16:59 UTC (Llama's rung-2 read, commit 7a8eb08), 16:48 UTC (the Granite judge on RunPod and its parity rule), 15:31 UTC (the rung-2 pre-registration) and 15:29 UTC (the USD 10/h pod price limit), and the frozen amendment to the family-probe bars (7 Oct 21:35 UTC). The brief gave the Granite parity result as pending; FOCUS 17:22 UTC records PASS, and FOCUS wins (conflict 40).
- **§4.1, "provisional".** "Llama-3.2-1B's rung is" became "Llama-3.2-1B's provisional rung is", because the user has not decided (decision 25). Conflict 37 is resolved.
- **§4.1, the too-close rule.** "(§4.6; a too-close probe gets one more seed)" became "(§4.6; a probe too close on E gets one more seed; any other too-close bar goes to the user)", the wording proposed in conflict 39. It matches the frozen amendment (7 Oct 21:35 UTC): only the E class's edge band leads to one more seed; a TOO CLOSE on another bar (short answers, ARC, unscored answers) goes to the user; and "any TOO CLOSE goes to the user with the numbers". For length, the body leaves out two details of the amendment: the extra seed runs only after the user's go, and a two-seed mean that is still too close goes to the user. Conflict 39 is resolved.
- **Length.** The body did not grow: the main body measures 9.36 pages, as in revision 5d (same measure; the References heading sits at the same height on page 10, and every heading checked from §3.3 on, §4.2 and later included, is where it was). The two changes added one line to §4.1's last paragraph; alone, they pushed the body to 9.40 pages. Three trims in the same paragraph took the line back, with no change of meaning: "whose erosion probe reads partial" became "whose probe reads partial" (the class, 20–55 pp, says what the probe reads); "under the frozen one-seed versions of the Gate A bars" became "under the frozen one-seed Gate A bars"; "the second family's first probe at recipe B" became "Llama's first probe at recipe B" (§3.5 and §4.9 also say "Llama"). §4.1's "Data" keeps revision 5d's trim ("sealed per tokenizer"). No layout command was added. One cosmetic cost: in the LaTeX build, "(20–55 pp)" now breaks between "20–55" and "pp". A tie there pushed the body again. Of the other trim sets tried in scratch builds, the only one that holds the length without the break changes "respond to the same change of recipe by different amounts" to "respond differently to the same change of recipe", which loses "by different amounts". So the break stays.
- **Editor's notes.** Risk 17 gives no balance figure, because the balance changes daily. The status entry drops its 12:18 UTC balance and "no pod running", and adds the USD 10/h pod price limit (FOCUS 15:29 UTC), the Granite judge on RunPod with its pre-set parity rule (FOCUS 16:48 UTC) and the parity PASS (FOCUS 17:22 UTC). Decision 7 notes that the USD 10/h limit replaces "no hourly limit" in [D20]. Decisions 18 and 25 quote "provisional rung". Conflicts 37 and 39 are resolved; conflict 40 is added. The LaTeX Appendix B is regenerated from these notes with revision 5c's converter; Appendix A is unchanged.

What changed in revision 5d (9 Oct, from 17:03 UTC)

History: revision 5e calls the rung "provisional" and states the too-close rule exactly (§4.1).

- **Source.** FOCUS "Llama-3.2-1B rung 2 at the main recipe (recipe B at LR 5e-5), seed 1101, read 2026-10-09 16:59 UTC" (commit 7a8eb08); no later FOCUS entry exists at 17:03 UTC. Also read: its pre-registration (15:31 UTC) and the frozen amendment to the family-probe bars (7 Oct 21:35 UTC: the short-answers rule and "a probe qualifies only if every bar passes with no TOO CLOSE").
- **§4.1, provisionally filled.** The placeholder for Llama-3.2-1B's rung became "Llama-3.2-1B's rung is 5×10^{-5} (one probe seed; development evidence; to be confirmed by stage 2's Gate A)." In the LaTeX build its placeholder highlight is removed. The user has not decided whether to accept the rung (decision 25); if the user declines, the sentence reverts to a placeholder (decision 18). The only other body change holds the length (below).
- **Editor's notes.** New development-evidence entry for the rung-2 probe; decision 25 added (accept rung 2 given the answers-end TOO CLOSE, with the lead's recommendation); decision 18 provisionally filled, pending the user; the §4.1 row of the placeholder table renamed; "What this is", the status entry and the sources updated; conflicts 37–39 added. The LaTeX Appendix B is regenerated from these notes with revision 5c's converter; Appendix A is unchanged.
- **Length.** The body did not grow: the main body measures 9.36 pages, as in revision 5c (same measure; the References heading sits at the same height on page 10, and every heading from §4.2 on is where it was). The new sentence adds one line to §4.1's last paragraph; alone, that pushed §4.2's last item ("Banks") to page 7 and the body to 9.40 pages. To take the line back, §4.1's "Data" paragraph, whose last line held only "ing rate].", drops two words: "prepared and sealed per tokenizer" became "sealed per tokenizer". No layout command was added: in a scratch build, asking TeX to set each paragraph of pages 5–6 one line shorter (its looseness setting) did not shorten the body.

What changed in revision 5c (9 Oct, from 15:29 UTC)

- **Source.** FOCUS "USER DECISIONS, 2026-10-09 15:23 UTC", items 1 and 3 (conflict 36).
- **The *worse* label says "adverse" (decision 23a).** "Significantly harmful" became "significantly adverse" in §1 and §3.2; §3.2's "harmful direction" and "harmful repair" became "adverse direction" and "adverse repair"; §4.3's P4 "a significant harmful result" became "a significant adverse result". The abstract names the label but never defined it, so it had no such phrase (conflict 35). In these notes: the *worse* fill-in and decisions 2 and 12. "Significantly" stays, because the label needs a Holm-adjusted test (conflict 33).
- **The abstract names the measures (decision 23b).** New sentence: "Lost safety is measured as refusal erosion as Qwen3Guard labels it and as harmful compliance as the HarmBench classifier defines it." The title keeps "safety". To hold the length, two abstract phrases were shortened: "but rarely compared on equal terms, with training seeds, side effects and compute counted" and "set the bar for a data repair".
- **The caveat sits once in §4.2 and once in §6 (decision 23c).** §4.2's claim rule no longer asks every harmful-compliance sentence to carry the caveat; it states it once, as a fact about the measures: "A blind human read during development rated about a third of judge-unsafe answers as usable help (Appendix A)." §6 keeps its caveat and adds "Appendix A"; §4.6's pointer for Gate B's read now goes to Appendix A. No other body sentence carries the caveat, and harm sentences still say "harmful compliance as the HarmBench classifier defines it". In these notes the "Harm wording" fill-in, risks 1 and 24 and the two HarmBench entries of the development evidence no longer attach the caveat to each sentence.
- **Appendix A, "Blind human read (development evidence)" (decision 23d).** New and part of the paper: after the references and before these notes (in the LaTeX build, `\appendix` makes it Appendix A and these notes Appendix B, on a new page). It holds the design (pre-registered; ratings sealed before the key was opened; 50 answers; blinding; the 0/1/2 rubric), the counts (40 Unsafe-labelled answers drawn, 39 rated, 1 skipped; 34 of 39 went along, 0.87 [0.73, 0.96]; 13 of 39 gave usable help, 0.33 [0.19, 0.50]; 0 of 10 Safe controls missed; 4 of the 12 OLMo answers that HarmBench labels "yes" rated usable help, 0 of 3 "no"), the reading (labelled ours) and the limits (one reader, who helped design the study; no second rater; groups of 6 to 12, not compared; a sample different from the audit's proposal). Group counts stay in these notes. Sources: the Side Project's `resea`

`rch/side-blind-read/PREREG-2026-10-08.md` and `blind-read-scores-2026-10-09.json` at commits 4d63e04 and 57d08dd, read with `git show`; the Gate A audit §8 item 4. The body gives no counts.

- **ARC rule's scope (decision 16; FOCUS 15:23 UTC, item 1).** §4.2's [TO BE FINALISED: its scope under the per-family rule] is replaced by the scope: first for probe and Gate A readings at one common recipe, then extended to every family's Gate A read at its chosen learning rate, before any such read. The loosening sentence and the disclosure stay. The §4.2 row of the placeholder table is removed.
- **Length.** The body did not grow: the main body measures 9.36 pages, as in revision 5b (same measure; the References heading sits at the same height on page 10, and every heading from §4.3 on is where it was). The abstract grew from 296 to 311 words (placeholders counted) and moved §1–§4.2 down by up to about four lines; §4.2's shorter caveat sentence and page 6's column slack took that up. Appendix A has 379 words, about half a page of the LaTeX build (page 14); these notes start on page 15.
- **Editor's notes.** Decisions 16 and 23 RESOLVED; decision 23's open item on a second rater becomes decision 24; decisions 2 and 12 say "adverse". Risks 1, 23 and 24 updated. Conflicts 23, 28 and 31 updated; conflicts 32–36 added. Development evidence (the blind-read and HarmBench entries, the ARC rule, the status) and sources brought up to FOCUS 15:23 UTC. The LaTeX appendix is regenerated from these notes.

What changed in revision 5b (9 Oct, from 15:06 UTC)

History: revision 5c states the blind-read caveat once in §4.2 and once in §6 instead of in every harm sentence, and moves the read's counts to Appendix A.

- **The blind human read (FOCUS 9 Oct 15:05 UTC; the Side Project's pre-registration and scores, commits 4d63e04 and 57d08dd, read with `git show`).** Its numbers are only in the development evidence (new entry, labelled), never in the abstract or the contributions. The body carries only the rounded caveat that the wording rule requires: §4.2 (the claim rule) and §6 ("most judge-unsafe answers went along with the request, but only about a third gave usable help"; one reader; "during development").
- **Tighter judge wording (FOCUS 15:05 UTC; the lead's proposal, applied unless the user objects; decision 23).**
- **U.** Where the body defines U or says what it measures, a rise in U is "refusal erosion as Qwen3Guard labels it": §3.1 (the erosion E), §4.2 (U) and §6.
- **Harm.** §4.2's claim rule keeps its two conditions (the HarmBench classifier's contrast has the same sign and at least half the size; its erosion under that recipe is at least 20 pp). It now allows only the wording "harmful compliance as the HarmBench classifier defines it", never harm alone, and every such sentence carries the blind-read caveat. §6 says the rule "allows at most" that wording.
- **Harm sentences reworded: 6.** §1, the side study ("the harmfulness of its content" → "what the answer says"); §4.2, the claim rule's two harm sentences (merged into one rule); §4.10 item 1's title ("harmful content" → "answer content"); §6, "whether its answer carries out harm" (replaced by the read's evidence) and "reduced harm". One ambiguous phrase was also fixed: §3.6 "sampled harm answers" → "sampled answers to harm prompts".
- **Searched and left, because they make no harm claim:** dataset, benchmark and category names (HarmBench, the harm bank, harm prompts, "harmful requests", "harmful anchors" and "benign prompts that resemble harmful ones", the hard-category list, PKU-SafeRLHF's and the plant's source labels); prior work's own measures (§1, §2); "safety" as the field's name for the outcome and in reference names (the title, "restores the most safety", Gate B's question, the figure caption, Λ_m's gloss, "safety prompt", Safe LoRA, "safety data"); and the three sentences of the *worse* label that use "harmful" for the adverse direction of a contrast (§1, §3.2, §4.3 P4; the user's wording of 8 Oct 18:31 UTC). Decision 23 lists these for the user.
- **Gate B (§4.6).** One sentence: the verdict reads U alone; the blind human read recommended before the first wave (DRAFT §12 item 5) is done, on a different sample from the one first proposed (the audit's §8 item 4: 40 Unsafe e1 answers in two strata plus the 29 over-refusal-bank Unsafe answers).

- **Length.** The body did not grow: the main body measures 9.36 pages against 9.40 for revisions 5 and 5a (same measure). To make room: §1's section roadmap was cut (its points repeat §1's paragraphs before it); three sentences that repeat earlier text were cut (§4.1's seed clause, given in §3.1; §4.10 item 1's "It locates a change …", given in §3.6; §6's "two cost conventions" sentence, given in §4.7); and §6's new caveat replaces the sentence it qualifies. In the LaTeX build only, Figure 1's caption note became "(Placeholder: not yet drawn.)".
- **Editor's notes.** Decision 23 added; decision 22's "WITHOUT" set as *without* (as in the LaTeX). Risks 1, 5 and 24 updated. The "Harm wording" fill-in, the two HarmBench entries of the development evidence and the status updated; the blind-read entry added. Conflicts 28–31 added. The LaTeX appendix is regenerated from these notes, so its wording is identical; this also brings in the placeholder-table wording of revision 5a (commit ec40d6a: "OLMo-2-1B's is 1e-4"), which the LaTeX appendix lacked.

What changed in revision 5 (9 Oct, from 13:17 UTC)

History: revision 5b tightens §4.2's claim-rule wording and §6.

- **Recipe B is the main recipe again (FOCUS 09:29 UTC, decision 1).** Gate A, Gate B and stage 1 use recipe B (LoRA r64, constant LR 1e-4, no warmup, 128 rows per step, per-row loss, one epoch); OLMo's recipe-B Gate A (PASS, 8 Oct 05:40 UTC) is the Gate A of record. Revisions 3–4's "recipe G main" is undone in the abstract, §1, §4.1, Table 1, §4.3, §4.6 and §6. The abstract and §1 no longer call the main recipe standard practice ("a fixed LoRA recipe that erodes it partly"; "recipe B: a constant learning rate, one epoch"); §6 calls recipe B "plain rather than standard" and says why it was chosen (from probe and Gate A readings, before any repair outcome).
- **Recipe G is the "recipe only" fix and a reference (FOCUS 09:29 and 10:01 UTC).** The body gives its design facts (a 4-step warmup and cosine decay, which halve the mean learning rate, and a token-mean loss) and calls it "the recipe-only fix". Revision 4's token-mean-only finding (the schedule, not the loss weighting, carries G's reduction on OLMo; one seed) stays development evidence; it is why these notes call recipe G a dose-lowering fix. The body does not say that recipe G lowers erosion in every family (conflict 25). "Stronger dose", "stronger-dose comparison" and "lower-dose base" are gone.
- **The per-family learning-rate rule replaces the common-recipe ladder (FOCUS 09:29 UTC, decision 2).** §4.1's "One common recipe" paragraph and [PLACEHOLDER: the common rung] are replaced by "A matched erosion dose": recipe B with only the learning rate changed, the highest rung of (1e-4, 5e-5, 2e-5) whose probe reads partial (20–55 pp) under the frozen probe bars, chosen from probes only, never from a repair result; cross-family claims "at a matched erosion dose". The development ladder (FOCUS 01:43 UTC) is cited, without numbers, as the evidence that one recipe cannot equalize the erosion, and the rule is disclosed as fixed after that ladder was read (§4.1, §6). New marker: [PLACEHOLDER: each family's rung]. Qwen is outside the repair study (§4.1); its recipe-B runs enter only the same-recipe family contrast (§4.10). The contributions, §4.6 (Gate A), §4.8 and §6 say "matched dose".
- **Five mandatory references (FOCUS 10:01 and 13:16 UTC).** The safety prompt; adapter scaling (× 0.75, × 0.5); recipe G (3 seeds); B+Qi (Qi et al.'s constrained SFT on recipe B, 6 seeds; "G+Qi" is renamed everywhere); Safe LoRA (Hsu et al., 2024; 3 seeds; settings [NOT YET FIXED]). Changed: abstract, §1, §2 (one clause), §3.2 (now points to §4.4), Table 1 (two rows added), §4.4 (rewritten). New BibTeX `hsu2024safelora`, from the OpenReview venue note HcifdQZFZV (NeurIPS 2024 poster), with arXiv:2405.16833 checked by the arXiv API (CITE-CHECK 26).
- **Dataset plan (FOCUS 09:40 UTC).** §4.1 "Data": Dolly for Gate B and stage 1; PKU-SafeRLHF for the add repairs' safety rows; Alpaca as the second benign dataset after Gate B, [NOT YET FIXED: its units and learning rate]. §6 adds "one benign dataset until Alpaca runs".
- **The two-layer mechanism account (FOCUS 8 Oct 15:17 and 16:50 UTC; the mechanism note's addenda; conflict 19).** §3.6 adds one sentence: the opening-switch term has two layers, the refusal openings lost ($p_0 - p_1$) and how often a direct answer was already unsafe before fine-tuning (u_{N0}). §4.10 item 1 reports both layers across families and recipes; §5's placeholder lists them. Numbers only in the development evidence.

- **§4.10 item 2** is now "Recipe and family contrasts": recipe G against recipe B as a recipe-only fix, the token-mean-only run, the same-recipe family contrast, every learning-rate rung and the rank-16 check under recipe G.
- **HarmBench on the pod and the harm wording (FOCUS 8 Oct 23:43 UTC).** §4.2's last clause is now per recipe ("If the classifier's erosion under a recipe is below 20 pp, no sentence calls that recipe's erosion harm"), so recipe G's erosion is "refusal erosion". Numbers only in the development evidence.
- **ARC rule.** Kept with its disclosure; its scope ("at the common recipe", FOCUS 20:50 UTC) is marked [TO BE FINALISED: its scope under the per-family rule] (decision 16). The sentence "An interval wholly at or below 0.03 passes …" is cut (implied by "resolves a straddling interval").
- **Gate B (§4.6).** Its pre-registration, revision 5, is marked [NOT YET FIXED]; "with the warmup and cosine decay spanning its own epoch" is removed (recipe B has no schedule); the exploratory E-scaled floor is removed, because at recipe B it equals the 8 pp floor (decision 21).
- **Order of work (§4.9).** Gate B runs to its verdict; funding for the later stages is then sought with that result; stage 1 starts only once funded, never because of an outcome. The stage-1 pre-registration is marked [NOT YET FIXED].
- **Secondary arm.** "The length rule under recipe B's per-row loss" reverts to revision 2's "the length rule under a token-mean loss" (decision 11).
- **Length.** The body did not grow: the main body measures 9.40 pages against 9.46 for revision 4 by the same measure (pages before the page with the References heading, plus the share of that page's two columns above it, in the ICML 2026 build). To make room, §4.1's common-recipe paragraph, §4.6's E-scaled floor, §4.10 item 2's dose sentence, §4.2's "as before" sentence and §6's repeat of Gate B's power were cut, §4.3's "Re-implementations" paragraph was merged into §6 "Baselines and doses", and Table 1's B+Qi and Safe LoRA rules were kept to one line each.
- **Editor's notes.** Decisions 14 and 15 resolved; decisions 7, 11 and 13 updated; decisions 16–22 added. Risks 2, 3, 17–19, 21–23 updated; risks 24–27 added. Conflicts 9, 10, 12 and 18 marked superseded; conflicts 19–27 added. CITE-CHECK 25 renamed (B+Qi); 26 added. Placeholders, fill-ins, development evidence and status brought up to FOCUS 13:16 UTC.

What changed in revision 4 (8 Oct, from 21:40 UTC)

History: revision 5 reverses this revision's recipe roles and replaces its common-recipe rule.

- **The loss-weighting account is withdrawn for B → G (FOCUS 21:36 UTC, the token-mean-only run).** The body no longer says or implies that recipe G's token-mean loss, or a first-token loss mechanism, explains why recipe G erodes less than recipe B.
- §1 and §4.1 describe recipe B as the stronger dose: a constant learning rate without warmup, about twice recipe G's mean learning rate over the epoch (FOCUS: G's warmup and cosine decay "also halves the mean learning rate"), plus a per-row loss. §4.1 drops "which over-weights the first tokens of short answers" and "a mechanism test", and names the token-mean-only run as the run that separates the two changes.
- §4.10, item 1 is now "refusal openings against harmful content": the decomposition is descriptive, and the first-token margin is the one causal test. The recipe contrast is no longer a causal test. Item 2 replaces "four amplifiers" with the two differences (schedule and loss weighting), adds a placeholder for the token-mean-only run's result against its prediction, and adds that families can respond differently to the same change of dose.
- §3.6 says the decomposition is descriptive; "should restore less" becomes "may restore less".
- §5's placeholder list adds the token-mean-only run against its pre-registered prediction, and "the mechanism decomposition and its causal tests" becomes "the opening decomposition and the first-token margin". §6 "Recipe and setting" describes recipe B as a stronger dose and adds the family × dose point and the ARC loosening.
- **G+Qi kept (FOCUS 17:36 UTC, item 4).** §1, §2, §4.4 and Table 1 justify it by Qi et al.'s own objective, which constrains updates on the initial tokens, not by the loss-weighting account; none of that text changed. §4.10

now cites Qi et al. (2025) only for the shallow-alignment observation beside the first-token margin (abstract re-read by the arXiv API between 21:40 and 21:44 UTC; CITE-CHECK 25).

- **ARC rule (FOCUS 20:50 UTC).** §4.2's "Task loss" item becomes "Task loss and skill": the rule in FOCUS's wording, called a loosening, with the disclosure that the first OLMo recipe-G probe was read under the old rule before the rule was proposed. The body gives no numbers for that probe. Revision 3's "ARC … sets no limit" was wrong for the probe and Gate A bars, which check ARC; §4.2 now states the bar. §4.1 and §4.6 point to §4.2 for skill; §6 notes the loosening.
- **Common-recipe rule (FOCUS 17:36, 21:04 and 21:37 UTC; the frozen amendment of 7 Oct, 21:35 UTC).** §4.1 adds the too-close rule: one more seed, the two-seed mean classified with the same band, no further seed in that classification, and a choice made after the outcome reported as such. [PLACEHOLDER: the common rung] stays, because Llama's two-seed reading under recipe G is pending and FOCUS records no step-down.
- **Development numbers** (the token-mean-only run, OLMo recipe G at three seeds, Llama under recipe G, the rank-16 check) are only in the development evidence below, labelled. The abstract and contributions did not change.
- **Editor's notes.** Risks 2, 3 and 18 updated; risks 20–23 added; "Where the framing is weak" item 5 updated; decisions 7 and 11 updated, decisions 13–15 added; placeholders updated; CITE-CHECK 25 extended; conflicts 14–18 added; development evidence and status brought up to FOCUS 21:37 UTC.

What changed in revision 3 (8 Oct, from 19:35 UTC)

History: revision 5 reverses this revision's recipe roles (recipe B is main again) and renames G+Qi to B+Qi.

- **Recipe roles (FOCUS 19:24 UTC; the common-recipe rule, 17:36 UTC).** Recipe G (token-mean loss, 4-step warmup with cosine decay, LR 1e-4, LoRA r64, 128 rows per step, one epoch) is the main recipe for Gate A, Gate B, stage 1 and both families. Recipe B is the stronger-dose comparison; any learning-rate rung that runs is a dose point. Changed: abstract, §1, §3.2, §3.6, §4.1, Table 1, §4.3, §4.4, §4.6, §4.8, §4.10, §5, §6.
- **Llama (FOCUS 16:50 and 17:36 UTC).** "Llama-3.2-1B replicates the comparison" is gone from the abstract, §1, the contributions and §4.8. Llama-3.2-1B is a planned second family under one common recipe, chosen by the pre-registered ladder rule (§4.1). The claim rule "replicates across two families" stays in §4.8 as a rule for when both families have run.
- **Worse label (FOCUS 18:31 UTC).** The labels are better / equivalent / worse / inconclusive (abstract, §1, contributions, §3.2, §4.3). Old decision 2 is resolved.
- **G+Qi reference (FOCUS 17:36 UTC, item 4).** Recipe G plus Qi et al.'s (2025) token-wise constrained SFT, six seeds, a model-level reference with no novelty claim. Recipe G is now the base, so "a standard recipe" leaves the references (FOCUS 17:36 UTC, item 4); the three references are the safety prompt, adapter scaling and G+Qi (abstract, §1, §2, §3.2, Table 1, §4.4). Qi et al. (2025) was already in references.bib (`qi2025shallow`, fetched from OpenReview with arXiv:2406.05946 checked; title and authors re-checked with the arXiv API on 8 Oct between 19:32 and 19:35 UTC; OpenReview now answers with a bot challenge, so the venue was not re-fetched); no new BibTeX.
- **One secondary arm.** "The length rule under a token-mean loss" contrasts nothing once recipe G (token-mean) is the main recipe. §4.3 now names its mirror, the length rule under recipe B's per-row loss. The mirror is the drafter's inference (decision 11).
- **Editor's notes.** Development evidence and status brought up to FOCUS 19:24 UTC; risks 2 and 3 updated, risks 18–19 added; conflicts 9–13 added.

What changed in revision 2 (from the draft of 8 Oct, about 12:00 UTC)

- **Headline.** The measured repair comparison leads (six repairs × six seeds against random removal and no repair, labels better / equivalent within 5 pp / inconclusive, with over-refusal, task loss and measured cost). The "when diagnosis pays" map is secondary and model-based; its predictions are written down before Gate B's category runs and tested on Llama.

- **Menu.** Stage 1's primary family P1–P6 (A1, L1, H1 removals; S1, S3, S4 additions); secondary arms and scored-only scorers; rewrite and self-distill only in the DIFFUSE branch. The previous draft's keep/remove/add/rewrite menu by source, the default mix a_mix and the "beat a_mix" rule are gone.
- **New parts.** Three mandatory references with no data change (safety prompt, adapter scaling, recipe G; revision 3 replaces recipe G with G+Qi); the planted-cause positive control; the HarmBench claim rule; two cost conventions; the Llama rule (LR or rank only; revision 3: the common-recipe ladder, LR only, for both families); confirmation on a sealed held-out subsample; the opening decomposition (side study).
- **Resolved old decisions.** O is measured against the untuned model (old decision 1); τ_O is a grid of 2, 5 and 10 pp and τ_L is 2 SE of a seed-to-seed difference in held-out loss on all 200 rows (old decision 2); the main scale is 1B with a ladder (old decision 3); a judge rule decides "harm" against "refusal" wording (old decision 4); model-level references are mandatory (old decision 6); side-effect prediction is secondary, on the add arms only (old co-primary).
- **§4 and §6 are written out. §2** adds the published filters, the format paper, DualSelect, Bianchi et al., PTST and MNEME, the DataRx correction and Guo et al.'s Table 2.

Placeholders

Where	Marker	Filled by
Abstract	[PLACEHOLDER: if funded]; [FINDING 1]–[FINDING 3]	the funding decision for the planted sweep; stage 1, the references, Gate B and the sweep
§1	[PLACEHOLDER: if funded]; [TO BE FINALISED] menu sizes 8/32/128	as above; the plan's review
§1, contributions	[FINDING 1]–[FINDING 4]; [ARTIFACT]	stage 1 and Llama; the sweep; Gates B/C and the mechanism study; the decision-value predictions; the anonymized outcome tables
§1	[FIGURE 1]	the repair-comparison figure (restoration with 90% intervals and the ±5 pp band against over-refusal and cost, with the adapter-scaling curve)
§3.2	[TO BE FINALISED] the label when a repair is better on U but worse on a side effect; [TO BE FIXED in the stage-1 pre-registration] token matching or a token-matched analysis	the stage-1 pre-registration (decisions 12 and 26)
§3	the constants δ (5 pp, given), δ_B (8 pp, given), τ_O grid (given), τ_L (rule given; fixed from the three Gate A a0 seeds, stage-1 build L16), B, the selector rules per (M, B), k_d, φ_j and the V level	the stage-1 pre-registration and the selector spec (committed before Gate B's category runs)
§4.1	[PLANNED; funding-gated, fallback one family] (the full 7–8B comparison); [PLACEHOLDER: if resources allow] (the headline check); [CITE: concurrent submission, anonymized] (twice); [NOT YET FIXED: Alpaca's units and learning rate]	funding, under option B with option C as the fallback (decision 28); funding; the citation form at submission (decision 32); the Alpaca design note and the user's extension of the learning-rate rule to datasets (decision 20). **Filled in revision 6:** the rungs (OLMo-2 1e-4 at 1B and 7B; Llama 5e-5 at 1B and 8B)
§4.3	[PLACEHOLDER: final MDE]; [TO BE FINALISED] secondary arms (listed in Appendix C.3)	Gate B's pooled noise; the plan's review
§4.4	[NOT YET FIXED] Safe LoRA's base model, layer rule and threshold	the stage-1 pre-registration, after the lead reads the paper in full (decision 19)

Where	Marker	Filled by
§4.5, Appendix C.5	[TO BE FINALISED] menu sizes; [PLACEHOLDER: runs only if funded]	the plan's review; funding
§4.7	[TO BE FINALISED] costs	measured ledgers; frozen cost conventions
§4.8	[PLACEHOLDER: if resources allow]	funding; decision 28
§4.9	[NOT YET FIXED] the stage-1 pre-registration	the stage-1 commitments build (FOCUS 16 Oct 19:47 UTC), committed before wave 1
§4.10	[PLACEHOLDER: report against the bound on a first-token path]; [CITE: concurrent submission, anonymized]	the bound recorded before Gate B (P3 in the Gate B pre-registration); decision 32
§5, §7	placeholders	results (decision 33: the page budget)
§6	[CITE: concurrent submission, anonymized]	decision 32
Impact Statement	[TO BE DECIDED: what is released]	decision 31
Appendix A, Appendix C.8	[CITE: concurrent submission, anonymized]	decisions 30 and 32
Appendix C	[TO BE DECIDED: anonymized copies as supplementary material]	decision 32
References	Microsoft page [CITE-CHECK]; Alpaca has no citation (§4.1, §4.8); GradShield's proceedings entry; the versions of SABA and SafeTuneBed	the author (CITE-CHECK 8, 14, 24, 28, 29)

Removed in revision 6: the §4.6 row ([NOT YET FIXED] Gate B's pre-registration, revision 5): the user froze it on 10 Oct (decision 7).

[TO BE FINALISED] items (the plan is under review; revision 6 moved their details to Appendix C): (1) the planted menus' sizes (8, 32, 128 cells) and cell rules; (2) the published-filter list: L1 is LARF-style in the primary family; DataShield-style is secondary; Self-Inf-N-style is scored only; SQSD is cited only; bi-anchoring enters as A1's score; (3) all costs: the FRE values, both cost conventions, the cost of each scorer and of A1, the lean-trial cost c_0, and the dollar plan (stage 1 about USD 56–98; planted sweep about USD 17–35; core path about USD 215–405; all estimates from the plan, not paper text); (4) the secondary arms and their order.

Fill-ins for each outcome (templates, not claims): - **A repair is better.** F1: "[repair] restores __ pp more than random removal (90% CI a–b), with over-refusal __ pp above the untuned model and task loss within τ_L, at __ FRE; it [does / does not] lie below the adapter-scaling curve". If it does not, the sentence must say that a model-level reference does as well (FOCUS "do not claim": data repair beats model-level repair). - **A repair is worse.** F1: "[repair] is worse than [comparator]: it [raises the unsafe rate / raises over-refusal / raises task loss] by __ (Holm-adjusted p = __)". This label replaces "inconclusive" for any significantly adverse contrast (FOCUS 18:31 UTC; "adverse" since decision 23a); P4 (S1 against no repair) is the contrast where prior work predicts an adverse result (DataRx's random mix). - **Everything equivalent.** F1: "every removal repair is equivalent to random removal within ±5 pp; [addition result]". This is a publishable answer; the abstract then says it. - **A free reference wins.** F2: "a safety prompt / a down-scaled adapter / Safe LoRA / recipe G / B+Qi restores as much as any data repair at lower over-refusal" (plan §13: a valid, reportable answer). - **Planted cause found / not found** (only if funded). F3: "[diagnostic] ranks the planted cell first at M = __ ; runs alone find it at __ FRE". - **DIFFUSE.** F3: "no Dolly category restores more than 8 pp beyond random removal (80% detection at about 10–12 pp)". Switch to alternative title 3; the planted sweep becomes the only test of whether diagnosis can pay. - **Second family.** Llama-3.2-1B's Gate A passed at its rung under decision 25 (FOCUS 10 Oct 09:40 UTC); stage 2 still needs funding. If stage 2 runs: "[contrast] replicates on Llama-3.2-1B at a matched erosion dose (same sign; family difference below the interaction's MDE of __ pp)". If no learning-rate rung of recipe B reads partial for Llama, or stage 2 is not funded, the paper is single-family: drop the second family from the abstract and

contributions, and say so in §6. - **Gate C after the sealed rankings (revision 6).** Equal picks: if Gate B reads LOCALIZED, F3 is one of "Gate C reads AGREE: both diagnostics pick [category], which restores __ pp more than random removal", "Gate C reads NONE_FINDS: the shared pick does not beat random removal", or "Gate C reads TOO CLOSE: a pick arm failed the answers-end guard"; with any Gate B verdict, add the pre-registered form sentence: "Neither pick is the form pick; that does not show that the picks are free of form." If Gate B is not LOCALIZED: "Gate C is not read." - **Harm wording (revision 5c; decision 23).** Use "restored refusal" unless the claim rule (§4.2) passes for that contrast: the HarmBench classifier's contrast has the same sign and at least half the size, and its erosion under that recipe (E_HB) is at least 20 pp. Then write "harmful compliance as the HarmBench classifier defines it", never harm alone. The sentence carries no caveat of its own: the blind-read caveat sits once in §4.2 and once in §6, with the counts in Appendix A. Recipe B's OLMo erosion clears 20 pp on every seed and recipe G's does not (development evidence), so recipe G's erosion is only "refusal erosion as Qwen3Guard labels it". **Revision 6:** at 7–8B, recipe B's erosion at each family's rung clears 20 pp on every seed with the same sign and at least half the size (FOCUS 11 Oct 10:40 UTC), so the same wording is allowed there.

[CITE-CHECK] items

#	Item	Where	Why
1	TOSS (2603.01185)	§2	Title and venue now fetched ("Token-level Data Selection for Safe LLM Fine-tuning", ICLR 2026). The claim "token-level selection" rests on the title only.
2	Unforgotten Safety (2512.10150)	§2	Fetched ("…with Continual Learning"); claim from the title only.
3	Yin & Rush (2410.16208, ICLR 2025)	§1, §2	Wording ("almost never compute-optimal"; 5x and 10x model-size ratios) from the arXiv abstract, read this session. The spec's "predicts retraining wins at equal size" is our inference from the 10x ratio; check the paper before using it.
4	TuneAhead (2606.17660)	§2	Fetched; claim from the title only.
5	Data2Behavior (2602.04735)	§2	Fetched; claim from the title only.
6	SafeTune: model-level repairs, single deterministic runs	§2	Recipe and runs PDF-checked (benign-erosion recipes note); "compares repairs of safety drift" from the title and a fetch summary.
7	DataRx mechanism ("gap between the model's own response and a safe reference"); revision 6: "its baselines, all additions, include self-distillation"	§2	From the arXiv abstract (this session). The numbers 59.23% and 13.70% are in the abstract; 20.24% is Table II (recipes note, PDF read); "averaged over seven tasks" is the abstract's wording. The plan reads (by HTML fetch, not PDF) that DataRx's random arm draws refusal-only rows; that is why S1 is the DataRx test. Revision 6: the baselines (Random, Longest, Paraphrase, Self-Distill, SSS-B, PSS-B; all additions; no seeds or selection cost reported) are the literature sweep's reading of the arXiv full text (sweep §4, "Corrections" 2), confirmed by the coordinator; not re-read here.
8	DataShield and GradShield report scoring cost without charging it against runs; DataShield's mechanism; revision 6: two papers named DataShield, and GradShield's venue	§2	Cost claim: fetch summaries only. Mechanism (compliance direction): DataShield's abstract (this session). DataShield's cost table (Table IV) is cited by the plan from the reading list (LARF 0.17 h, DataShield 0.50 h, bi-anchoring 3.5 h per 10,000 rows); not PDF-checked. Revision 6: arXiv:2607.15081 (Wu et al., "DataShield: Uncovering Risky Fine-Tuning Data Across LLMs Through Consensus Subspace Alignment") is a different paper with the same name (arXiv API, 16 Oct); the draft cites only Zhang et al. (arXiv:2606.00160), and its "DataShield-style" filter follows Zhang et al.'s compliance direction; check which paper holds the cost table. GradShield: the NeurIPS 2026 accepted-papers JSON (fetched by the literature sweep) lists it as "Accept (poster)"; the BibTeX keeps the arXiv

#	Item	Where	Why
			form with the note "NeurIPS 2026 (poster)"; replace it with the proceedings entry when published.
9	SOURCE "reported to cost several training runs"	§2	Reported through the Bergson paper; not checked.
10	LESS as "gradient similarity on LoRA gradients"	§2	Reading-list summary.
11	Persona vectors	§2	Blog post via the problem-validation review; confirm in 2507.21509.
12	DataDecide; Ivison et al.	§2	Positioning review, from an extraction-tool read.
13	Xiao & Aranguri	§2	Abstract only ("over 10 times cheaper").
14	Microsoft fine-tuning safety documentation	§1, refs	Read on 6 Oct; re-check before submission; no URL in the bib.
15	Venues and author order	refs	ADMIRE-BayesOpt: TMLR lists Ouyang first, arXiv lists Chen first (bib follows TMLR). DATE-LM year set to 2025 from the venue string.
16	LARF (2507.18631, EMNLP 2025)	§2	Mechanism from the abstract (this session).
17	SQSD (2605.04572, ICML 2026)	§2	Mechanism and acronym from the abstract (this session). "Its scoring needs several FRE" is the plan's estimate (6–14 FRE), not the paper's.
18	Self-Inf-N (2505.06843, ICML 2025)	§2, §4.3	"100 outliers break safety; most mitigations fail" from the abstract (this session). Using its score as a removal filter is our adaptation.
19	"Bi-anchoring" as He et al.'s method name	§2, §4.3	The reading list's name; check He et al.'s own term.
20	Form over content (2609.19589)	§1, §2	Abstract only (here and in the repo's arXiv search file).
21	DualSelect (2606.09866)	§1, §2	Abstract (this session): joint task and reference selection, 1B–8B models. "Compares random, SafeInstr, SEAL, LARF, TOSS and SQSD with three seeds and a false-refusal rate" comes from the plan's novelty review (HTML §4), not re-read. **Revision 6:** FAQ checked the arXiv full text (coordinator's addition 2): SafeInstr, which adds 2–3% safety-instruction tokens, is a baseline (its §4.1, §4.4); selection overhead 1.34× standard SFT at a 90%-of-target-tokens budget; three seeds (mean ± std); a false-refusal rate (§4.6). "Training data that include red-teaming pairs" and "no model-level references" are the literature sweep's reading of the same full text (sweep §4, "Corrections" 1). The §1 sentence now rests on these; its [CITE-CHECK] is removed.
22	Bianchi et al. (ICLR 2024): 3% safety data helps, with exaggerated safety	§2	Abstract (this session).
23	MNEME (EMNLP 2025) predicts side effects	§2	Abstract (this session).
24	Instrument citations (OLMo 2, Llama 3, Qwen2.5, Qwen3Guard, Granite Guardian, HarmBench, StrongREJECT, SimpleSafetyTests, XSTest,	§4	Fetched. Check that each cited report covers the version used (Granite Guardian 3.3 8B; Qwen3Guard-Gen-4B; Llama-3.2 under the Llama 3 report; OLMo-2-0425-1B under the OLMo 2 report). Dolly's entry is the dataset card's own BibTeX (a blog post). Alpaca (§4.1, §4.8) has no citation yet: the brief allows new BibTeX only from the arXiv API, OpenReview or

#	Item	Where	Why
	OR-Bench, PKU-SafeRLHF, LoRA, ARC, Dolly)		PMLR, and Alpaca was not searched there this revision (it is a GitHub release).
25	Qi et al. (2025; 2406.05946) as the B+Qi reference (renamed from G+Qi in revision 5; "token-wise constrained SFT", "constrains updates on the first answer tokens")	§1, §2, §3.6, §4.4, §4.10	Mechanism from the arXiv abstract, read by API this session ("constraining updates on initial tokens"). "Token-wise constrained SFT" is the lead's and FOCUS's name; check the paper's own term before submission. FAQ (FOCUS 15:59 UTC) reports that their Table 9 shows warmup alone does most of the work on benign data; not checked here. Revision 4: §4.10 also cites them for "safety alignment often adapts mainly the first few output tokens", a paraphrase of the abstract ("the alignment adapts a model's generative distribution primarily over only its very first few output tokens"), re-read by the arXiv API on 8 Oct between 21:40 and 21:44 UTC.
26	Safe LoRA (Hsu et al., 2024; arXiv:2405.16833; NeurIPS 2024) as a reference ("projects the LoRA weights of selected layers onto a safety-aligned subspace without training or data")	§1, §2, §4.4, Table 1	BibTeX from the OpenReview venue note HcifdQZFZV (NeurIPS 2024 poster), fetched by the OpenReview API on 9 Oct at about 13:20 UTC; title and the six authors checked against the arXiv API record (v2, "camera-ready version accepted for NeurIPS 2024"). The body's wording is from the arXiv abstract, read by API this session ("projection of LoRA weights from selected layers to the safety-aligned subspace"; "training-free and data-free"; it "only requires the knowledge of the weights from the base and aligned LLMs"). FOCUS 10:01 UTC's fuller description (the weight difference between the aligned and the base model; the layers where the update departs from it) was not checked against the paper; the lead reads it in full before the settings are fixed (decision 19).

| 27 | Safety at One Shot (Zhang et al., 2026a; ICLR 2026; arXiv:2601.01887) | §2, §6 | Revision 6. BibTeX from the OpenReview note EyH8Fu3vtZ (ICLR 2026 poster), fetched by the OpenReview API v2 search endpoint (the note endpoint answered with a bot challenge); arXiv:2601.01887 checked by the arXiv API (same title; second author "Lipeng He" on arXiv, "Tony He" on OpenReview: check). Claim ("further training on a single safety example" restores safety; it "acts after fine-tuning") from the arXiv abstract: "safety alignment can be fully recovered with only a single safety example, without sacrificing utility and at minimal cost", shown after fine-tuning with harmful examples. Whether it works after benign fine-tuning is not tested by them as far as the abstract says. | | 28 | SABA (Anonymous, 2026; ICLR 2027 submission, OpenReview Jo6U0y2omr) | §2 | Revision 6. BibTeX and abstract from the OpenReview API v2 search record (16 Oct). The body's clause paraphrases the abstract: "risk rankings computed at initialization are nearly uncorrelated with those computed after two epochs"; SABA "re-evaluates sample risk at the current model state in each training round". Cited anonymously as an OpenReview submission (the coordinator's instruction). Re-check for an arXiv or venue version before submission. | | 29 | SafeTuneBed (Hossain et al., 2025; arXiv:2506.00676) | §2 ("How we differ") | Revision 6. Title, authors and abstract from the arXiv API ("a benchmark and toolkit unifying fine-tuning and defense evaluation"); the body says only that it compares defences. OpenReview has an IJCAI 2025 workshop version with another title (note yKetbTItcu); the arXiv version is cited. |

PDF- or full-text-checked in the repo, and used as such: Qi et al. (2024), He et al., Luo et al. (2026b), Pan et al., DATE-LM (verification log, 6 Oct); Kim et al. (2025), SafeTune's recipe, PTST, DataRx's Table II (benign-erosion recipes note); Guo et al. including Table 2 (7.27, 4.50, 9.09, 44.80 confirmed), Springer et al. (safety-landscape note); Llama 3's safety mix (problem-validation review); Qi et al. (2025) on first-token alignment (erosion-amplifiers note). Guo's "two seeds" is the Table 2 caption; the Guo-gap note finds no seed-69 checkpoint for the 5,000-row cells.

Decisions the author must make

1. **Title** (above).
2. **RESOLVED (FOCUS 18:31 UTC, the user): the "worse" label.** A repair whose contrast against its comparator is significantly adverse at the 5% Holm level (on unsafe share, over-refusal or task loss, each in its adverse direction) is labelled *worse*, so an adverse repair is never reported as inconclusive. Labels are better,

equivalent, worse and inconclusive; §3.2 and §4.3 carry them, and P4's adverse direction (DataRx) now has a label. One detail is left for the stage-1 pre-registration (decision 12). (The 18:31 UTC wording said "significantly harmful"; "adverse" since decision 23a, revision 5c.)

3. **The planted sweep's funding.** If it is not funded, "menu size" leaves the headline: delete contribution 2, the planted part of the abstract and Figure 1's menu-size panel, and §3.4 shrinks to one sentence.
4. **Preliminary-results section.** The spec does not ask for one, so the body has none. Gate A's numbers and their disclosures stay in this note until §5 is written. Decide whether a section labelled "Preliminary results (development evidence; not confirmatory)" should carry Gate A.
5. **Filter list and secondary arms** ([TO BE FINALISED]).
6. **Venue and length. Revision 6 (FOCUS 16 Oct 21:05 UTC; the user, relayed by FAQ): "the goal for the main paper is ICML 2027".** TMLR or NeurIPS 2027 stays the fallback until the user names one. The ICML 2027 deadlines are ESTIMATED, not official: about 16 Jan 2027 (abstract) and about 22 Jan 2027 (paper), AoE, from a third-party tracker; the ICML 2027 call is not published (literature sweep §6, checked 16 Oct about 21:20 UTC). Plan to the ICML 2026 rules until then: 8 pages of main body; references, the Impact Statement and appendices outside the limit; one more page at camera-ready; the Impact Statement required (added in revision 6). The body measures 7.96 pages to the Impact Statement and 8.15 to the References heading (revision 5e: 9.36). Swap to the 2027 style when it is released and re-check the limit and the statement. *History:* the adopted amendment targeted TMLR or NeurIPS 2027; ICML 2027 only if, by about 1 December, the planted sweep, both 1B families and one 7–8B published-setting replication are funded. The LaTeX build uses the ICML 2026 style; see the page count in the build log. NeurIPS allows 9 pages; TMLR has no strict limit. ICML needs an Impact Statement (missing). **Revision 5:** the main body measures 9.40 pages (the measure in "What changed in revision 5"), over ICML's 8 pages and NeurIPS's 9.
7. **RESOLVED (FOCUS 10 Oct 11:16 UTC, the user: "do the recommended"): the Gate B/C pre-registration, revision 5, is frozen** (SHA-256 5c97cce7…; §12 item 8: take the fallback if the noise gate fails). Revision 6: §4.6 summarizes it and Appendix C.1–C.2 give its rules; [NOT YET FIXED] is gone. *History:* **The Gate B and Gate C rules** in §4.6 summarize the DRAFT pre-registration (revision 4b, written for recipe B), which awaits the user's acceptance. FOCUS 17:36 UTC moves Gate B to the common recipe (REC and recipe S rebuilt from it; the noise gate and predictions P3–P4 re-based on that recipe's Gate A mean E; the 8 pp floor unchanged; each removal arm's warmup-cosine schedule spans its own epoch); DRAFT revision 5 is to carry this and was not yet in the repository at this revision. §4.6 states only the recipe change. Update §4.6 when revision 5 is accepted. **Revision 4:** the repository's DRAFT is still revision 4b. FOCUS 20:17 UTC adds an exploratory, report-only E-scaled floor for Gate B, $8 \times E_common / 45.1$ pp (about 5.2 pp at $E \approx 29$), labelled exploratory wherever it appears and never changing a verdict; the 8 pp floor stays primary. §4.6 does not mention it yet; add one sentence when revision 5 is accepted. **Revision 5:** FOCUS 9 Oct 09:29 UTC keeps Gate B on recipe B, so the move to the common recipe above is void; DRAFT revision 5 "stays on recipe B and adds the [D14]–[D19] rules", and the brief's [D20] fixes the kit's price constants (no hourly limit; guard planning rate USD 1.85/h; spend rate USD 1.79/h; reserve USD 6), with the money thresholds recomputed in revision 5 (FOCUS 13:16 UTC). The repository's DRAFT is still revision 4b, so §4.6 marks revision 5 [NOT YET FIXED]. Update §4.6 when revision 5 is accepted. **Revision 5e:** the user's USD 10/h pod-creation price limit (FOCUS 9 Oct 15:29 UTC) replaces "no hourly limit"; brief revision 17 amends [D20]; the budget guard's planning rate (USD 1.85/h) is unchanged.
8. **The opening decomposition's definition** (§3.6) must match the exploratory note's "composition share"; the note does not state its formula in these terms. Fix the order of terms and report both orders.
9. **Gate names.** Keep "Gate A/B/C" or call them pre-registered checks.
10. **Who the user is** (§6): the plan names a fine-tuning service that must ship the customer's adapter and may change only data, or a data-only pipeline. Confirm this wording.
11. **The length-rule secondary arm under the other loss** (§4.3). Revision 2 listed "the length rule under a token-mean loss", a control for recipe B's per-row loss. With recipe G (token-mean) as the main recipe, revision 3 writes the mirror: the length rule under recipe B's per-row loss. Confirm, or drop the arm. **Revision 4:** the token-mean-only run (development evidence) finds that the loss weighting carries none of recipe G's reduction

at one seed, so the arm's reason (that per-row weighting makes short rows matter more) has no support at the recipe level. The arm would still test it at the row level; keep it only if three seeds are cheap. **Revision 5:** with recipe B (per-row loss) the main recipe again, §4.3 reverts to revision 2's "the length rule under a token-mean loss". Keep it only if three seeds are cheap, or drop it.

12. **Label order when endpoints disagree.** The 18:31 UTC rule labels a repair *worse* if any of its three contrasts (unsafe share, over-refusal, task loss) is significantly adverse. It does not say which label a repair gets when it is *better* on the unsafe share and *worse* on over-refusal or task loss. §3.2 reports both facts and leaves the order to the stage-1 pre-registration; fix it there.
13. **Where the failed prediction goes.** FOCUS 21:36 UTC: "the paper reports it as a pre-registered prediction that failed". Revision 4 puts its result in §5 (placeholder) and names the run in §4.1 and §4.10. Decide whether §5 gives it a sentence or a row in the dose table, and whether the mechanism side study keeps any recipe contrast. Revision 4 keeps the recipe contrast only as a dose decomposition, not as a mechanism test. **Revision 5:** recipe G is now a reference arm, and the token-mean-only run explains that reference's effect on OLMo; §4.10 item 2 and §5 keep it. Decide whether §5 gives it a sentence or a row beside recipe G. **Revision 6 (overlap rule):** the token-mean-only run is reported by the concurrent submission; §4.10 item 2 cites it and §5's placeholder no longer lists it. Decide whether the main paper still states this failed pre-registered prediction itself (FOCUS 8 Oct 21:36 UTC: "the paper reports it as a pre-registered prediction that failed"), or only through the citation.
14. **RESOLVED (FOCUS 9 Oct 09:29 UTC): no common recipe.** Llama's two-seed reading under recipe G stayed TOO CLOSE (+58.25 pp), and the ladder found no rung for both families (01:43 UTC); the user chose recipe B as the main recipe and the per-family learning-rate rule (§4.1). Llama's recipe-G seed 1103 was then read as a report-only third seed (three-seed mean +57.1 pp, TOO CLOSE at the top edge; results record, 09:35 UTC).
15. **RESOLVED (FOCUS 9 Oct 09:29 UTC): recipe G's name.** Recipe G is "the recipe only fix" and a reference, not the main recipe or a lower-dose base; the body calls it "the recipe-only fix", and the abstract no longer calls the main recipe standard practice. **Revision 6 (coordinator's addition 1; FOCUS 11 Oct 10:38 and 12:54 UTC):** recipe G did not lower erosion at 7–8B in the side work, so the body no longer calls it a fix: it is "a reference arm that changes only the recipe" and "not a general fix", with the concurrent submission cited for its effect across families and sizes.
16. **RESOLVED (FOCUS "USER DECISIONS, 2026-10-09 15:23 UTC", item 1): the ARC rule's scope under the per-family rule.** FOCUS 20:50 UTC adopted it "for the probe and Gate A readings at the common recipe (both families)", and no common recipe now exists. OLMo's Gate A of record (recipe B) passed ARC under the frozen bars (ARC rose on every seed; audit), so the rule decided none of OLMo's main readings; it decided the recipe-G Gate A read and the rung-2 readings (reported). **Resolution:** the rule is extended to every family's Gate A read at its chosen learning rate, fixed before any such read and disclosed as a loosening; it replaces "at the common recipe" in the 20:50 UTC entry. §4.2 states this scope (revision 5c); the placeholder is gone.
17. **The family × recipe interaction test (Side Project).** Its 95% interval on the logit scale excludes 0, but it was written after 11 of its 12 values were known, has about 5 df and uses one judge (development evidence). Decide whether §5 reports it, labelled exploratory, or only the per-family rule it motivates. The body now gives only the design consequence (§4.1: "families respond to the same change of recipe by different amounts"). **Revision 6 (overlap rule):** the interaction is the concurrent submission's finding (the 7–8B replication was "not shown"); the main paper does not report it, and §4.1 cites the concurrent submission.
18. **RESOLVED (FOCUS 10 Oct 09:40 UTC, the user: "yes"; Llama-3.2-1B's Gate A recorded as PASS under decision 25): Llama's rung is 5e-5.** Revision 6 drops "provisional" from §4.1. *History:* **PROVISIONALLY FILLED, PENDING THE USER (revision 5d): Llama's rung under the per-family rule.** Llama's rung-1 probe at recipe B (8 Oct, 16:50 UTC) read SATURATED, and the two report-only seeds agree (three-seed mean +59.3 pp), so the rule's next step was a recipe-B probe at 5e-5. That probe was pre-registered at 15:31 UTC and read at 16:59 UTC (FOCUS; commit 7a8eb08): E +48.0 pp, partial and inside its prediction, but answers end reads TOO CLOSE and Granite has not run (development evidence). §4.1 now says "Llama-3.2-1B's provisional rung is 5×10^{-5} (one probe seed; development evidence; to be confirmed by stage

2's Gate A)" ("provisional" since revision 5e), and the placeholder table marks the rung provisionally filled. It stands only if the user accepts rung 2 (decision 25); if the user declines, §4.1 reverts to a placeholder for Llama's rung. The rung-2 reading under recipe G (+41.2 pp, partial) is not a recipe-B probe and did not set Llama's rate.

19. **Safe LoRA's settings** (FOCUS 10:01 UTC; to fix in the stage-1 pre-registration before any Safe LoRA run, after the lead reads the paper in full): the unaligned model (OLMo-2-0425-1B base for OLMo-2-0425-1B-Instruct; Llama-3.2-1B base for the second family), the exact projection, the layer-selection rule and its threshold (from the paper's rule or a fixed default, never tuned on stage-1 outcomes), the seeds (the adapter-scaling seeds 1101–1103 unless the pre-registration says otherwise; Table 1 says 3), and the engine work (a post-hoc state transform evaluated by the existing eval-state job, built with tests and reviewed). §4.4 and Table 1 are marked [NOT YET FIXED].
20. **Alpaca's preconditions** (FOCUS 09:40 UTC; to settle before any Alpaca run, not now): the user extends the learning-rate rule from families to datasets before any Alpaca probe; a short design note fixes Alpaca's units (it has no categories, unlike Dolly), the field mapping (instruction, input as context, output as response), the version (tatsu-lab/alpaca; CC BY-NC 4.0) and the download, which the user approves. Alpaca still has no citation (CITE-CHECK 24).
21. **The exploratory E-scaled Gate B floor** (FOCUS 8 Oct 20:17 UTC): 8 × E_common / 45.1 pp equals 8 pp at recipe B (E_common is the recipe-B Gate A mean, +45.1 pp), so it repeats the primary floor; revision 5 drops the sentence from §4.6. Confirm, or restore it if DRAFT revision 5 keeps it.
22. **B+Qi's warmup.** FAQ reports that Qi et al.'s Table 9 shows warmup alone does most of the work on benign data (FOCUS 8 Oct 15:59 UTC; not checked here). Recipe B has no warmup, so whether B+Qi takes Qi et al.'s own schedule decides whether B+Qi differs from recipe B only in the objective or also in the schedule, like recipe G. Fix it in the stage-1 pre-registration and report it beside recipe G. **Lead's default (FOCUS 13:45 UTC, 9 Oct):** Qi's constrained loss on recipe B *without* Qi's warmup, so B+Qi isolates the loss while recipe G covers the schedule; the stage-1 pre-registration fixes it [NOT YET FIXED].
23. **RESOLVED (FOCUS "USER DECISIONS, 2026-10-09 15:23 UTC", item 3, "FAQ decision 23"): the judge wording package.** Revision 5b applied the lead's tighter wording (FOCUS 9 Oct 15:05 UTC; §3.1, §4.2, §4.6, §6): U is "refusal erosion as Qwen3Guard labels it"; a harm sentence, even where E_HB ≥ 20 pp and the same-sign, half-size rule holds, says "harmful compliance as the HarmBench classifier defines it" and (in revision 5b) carried the caveat that a blind human read rated about a third of judge-unsafe answers as usable help. Revision 5b left four items for the user. The user accepted the package as follows, applied in revision 5c: (a) the *worse* label says "adverse", not "significantly harmful" (§1, §3.2, §4.3 and these notes; conflicts 33 and 35); (b) the title keeps "safety", and the abstract names the measures exactly (refusal erosion as Qwen3Guard labels it; harmful compliance as the HarmBench classifier defines it); (c) the blind-read caveat sits once in §4.2 and once in §6, with Appendix A, not in every harm sentence; (d) the read's counts go to Appendix A as development evidence, not to the main results. Revision 5b's last open item, a second rater, is not part of this decision and becomes decision 24 (conflict 32).
24. **A second rater or a larger human read** (revision 5b's decision 23d; open). FAQ's main-track list asks for "HarmBench co-primary plus a human check" (FOCUS 8 Oct 15:59 UTC, item 4); the one-reader development read (Appendix A) does not meet it as a confirmatory measure (not costed). Decide whether one is planned on stage-1 arms.
25. **RESOLVED (FOCUS 9 Oct 20:31 UTC, decision 25 accepted; 10 Oct 09:40 UTC, Gate A recorded as PASS under it).** Revision 6: §4.1 says the rung plainly and discloses, with Llama-3.1-8B's rung, that it "was accepted on a too-close bar" (decision 34). *History:* **Accept rung 2 (recipe B at 5e-5) as Llama-3.2-1B's main-recipe rung, given the answers-end TOO CLOSE?** (FOCUS 9 Oct 16:59 UTC; the user decides; not urgent: needed only when stage 2 is planned.) The rung-2 probe (seed 1101) reads E partial (+48.0 pp, band 41.2–54.8), inside its pre-registered prediction (+35 to +50), so it qualifies Llama on the erosion class. Answers end reads TOO CLOSE under the frozen bars: 64 of 600 sampled answers (10.7%) fail the screen against the 10% bar, but 53 of 589 (9.0%) without the 11 answers shorter than 5 tokens, which is the frozen short-answers rule's TOO CLOSE case (7 Oct 21:35 UTC amendment; a probe qualifies only with no TOO CLOSE, and any

TOO CLOSE goes to the user). Granite, the second judge, has not run on this probe. **The lead's recommendation:** accept rung 2 (5e-5) as Llama's main-recipe rung for planning, with stage 2's own Gate A seeds re-reading answers end per seed. If the user accepts, §4.1's sentence stands and "provisional" (revision 5e) can be dropped; the rung is still to be confirmed by stage 2's Gate A. If not, §4.1 reverts to a placeholder (decision 18). Note: this probe had 11 very short answers, against 0–1 in the other Llama runs.

26. **RESOLVED by the user's rule (17 Oct, relayed by the coordinator: "keep the more ambitious one always but always open to fallbacks"): the title keeps "Compute-Matched".** The design will match tokens per removal arm, or pre-register a token-matched analysis beside the row-matched one; the lead rules on which in the stage-1 pre-registration (§3.2 says so, marked [TO BE FIXED in the stage-1 pre-registration]). The fallback title, alternative 5 ("A Cost-Accounted Comparison of Data Repairs"), is used only if token matching cannot be done. The literature sweep's reasoning, kept for the record (§0 item 3 and §5): the design measures cost but does not equalize it (every arm is logged in GPU-seconds and FRE, §4.7); the removals are row-matched (600 rows), and the shortest-answer rule removes far fewer tokens than random removal; a concurrent submission found that a selection gain at a fixed document budget vanished once token budgets were matched; keep "Compute-Matched" only if stage 1 matches total GPU-seconds per arm, or at least training tokens per removal arm. Risk 30.
27. **Candidate additional reference arms (the user's rule of 17 Oct; notes only; not in the body's list of five references).** (a) The opening-hold loss of a concurrent ICLR 2027 submission ("A Harmful Request Is Still a Request"; the literature sweep, entry 3: it cut harmful compliance from 22.6% to 0.9% in its tests, the sweep's reading; not fetched or verified here); (b) Safety at One Shot (Zhang et al., 2026a; a single-example patch after fine-tuning; cited in §2 and §6 as the bounded D5 item). Decide before stage 1 is frozen whether either becomes a reference arm, and how B+Qi relates to an opening hold. If (a) is added, its citation must be fetched by the OpenReview API first.
28. **RESOLVED by the user's rule (17 Oct, relayed by the coordinator): the re-plan scope is option B, with option C as the fallback.** The ICML re-plan proposal (`writing/paper/2026-10-17-icml-replan-PROPOSAL.md`, about 21:25 UTC on 16 Oct): option B is "P1–P6 + 5 references on OLMo-2-7B and Llama-3.1-8B; confirmation at 7–8B (≤ 2 contrasts per family)", with Gate B at 1B, OLMo-1B's stage 1 as development, Llama-1B's primary family as a replication, a reduced Alpaca study on Llama-8B and 32B if funded (about USD 1,180–1,760, an estimate; no grant is awarded); option C is the same with the full 7–8B comparison on one family. The body now describes the 7–8B tier as the full comparison on both families, marked [PLANNED; funding-gated, fallback one family] (§4.1, §4.8, §6). Not yet written into the body (for the lead): option B's roles for the 1B stages (OLMo-1B as development, Llama-1B as a replication), confirmation at 7–8B, the proposal's "Gate B-lite" transfer check (no full Gate B at 7–8B), and its power at 7–8B (a 6-seed MDE of about 11–12 pp per contrast, against the 5 pp SESOI; the measured 7–8B seed spread is about 1.7× the 1B spread). The abstract, §1 and the contributions still describe OLMo-2-1B as the stage-1 model (conflict 52).
29. **Gate C's format clause (§4.6).** The brief asked for the ATTR_BETTER/REP_BETTER consequence; revision 6 also states the pre-registered form sentence, because the pre-registration requires the two equalities to be reported whatever Gate B reads. Keep it, or cut §4.6 to the brief's clause (Appendix C.2 keeps the full rule either way).
30. **Appendix A and the concurrent submission.** The side paper reports the same blind read with the same counts. Revision 6 keeps Appendix A (it is this paper's check of what U measures) and adds one sentence that cites the concurrent submission. Options: keep; cut Appendix A to a pointer; or keep it and move the counts out of the side paper (the GPT lead's paper; not this chat's decision).
31. **The release policy ([TO BE DECIDED] in the Impact Statement).** What is released: menus, seeds, outcome tables, costs and scores; generated answers to harm prompts or not; anchor text; the planted rows' identifiers. The repository rule (no rows, prompts, answers or anchor text in git) is about git, not a release.
32. **Citing the concurrent submission, and dual submission.** ICML 2026's rules allow citing one's own unpublished work only anonymized, with an anonymized copy as supplementary material, and forbid parallel submission of substantially similar work. The overlap rule keeps the side paper's findings out of this paper; the placeholder [CITE: concurrent submission, anonymized] sits in §4.1 (twice), §4.10, §6, Appendix A and

Appendix C.8. Decide the citation form at submission (for example "Anonymous, 2026", with the anonymized copy), and check the side paper's ARR status then (ARR commitment deadline about 23 Dec 2026; FOCUS 15 Oct records sync). The anonymized copies of the pre-registrations (Appendix C's [TO BE DECIDED]) go the same way.

33. **The page budget for results.** The body is 7.96 pages with §5 and §7 as placeholders. Results (Gate A, Table 2, Figure 1's data, Gates B and C) need about a page. Options: move more design to Appendix C (for example §3.4, §4.5 and §4.10 to one paragraph each); shrink Figure 1's placeholder to the figure's real size; drop the related-work paragraph "Prediction and decisions under a budget" to one sentence.
34. **The disclosure of Llama's two rungs (§4.1).** "Both Llama rungs were accepted on a too-close bar (one seed's answers-end bar at 1B; the window's upper edge at 8B); §5 reports both." Keep it in §4.1, or move it to §5 with the numbers.

Risks a reviewer will raise (sources in brackets)

1. **"Unsafe" means "did not refuse".** See the development evidence. Answer: the HarmBench claim rule, hard-category U, U by length band, the selectivity score, and claim wording [Gate A audit, 4a; plan §4.2]. **Judge validity (revision 5b):** the blind human read [FOCUS 9 Oct 15:05 UTC; development evidence; Appendix A since revision 5c] supports U as lost refusal (34 of 39 Qwen3Guard-Unsafe answers went along with the request; 0 of 10 Safe controls missed) but rated only about a third of judge-unsafe answers as usable help (13 of 39, 0.33 [0.19, 0.50]; 4 of 12 OLMo answers that HarmBench labels "yes"). Limits: one reader, who also designed parts of the study; 50 answers; small groups; no second rater; no comparison between groups. Answer: the body calls a rise in U refusal erosion as Qwen3Guard labels it and allows at most "harmful compliance as the HarmBench classifier defines it"; the caveat sits once in §4.2 and once in §6, with the counts and limits in Appendix A (revision 5c); no sentence claims usable harm. A confirmatory human check is not planned (decision 24).
2. **Forking paths.** Five settings were screened before OLMo-2-1B fell in the 20–55 pp window, fixed after the Qwen results; seed 1101 both chose the setting and counts in the mean [audit, 4b]. Answer: report every setting tried and the fresh-seed mean (+45.9 pp). **Added in revision 3:** recipe G became the main recipe after its one-seed OLMo result (+29.0 pp, partial) and Llama's saturation under recipe B were known (FOCUS 17:36 and 19:24 UTC). The choice used probe data only, never a Gate B or stage-1 outcome; Gate A is re-run at the common recipe with seeds 1102 and 1103 per family, and seed 1101 (the probe) counts, so report the fresh-seed mean beside the 3-seed mean again. **Added in revision 4:** the ARC rule was adopted (FOCUS 20:50 UTC) after the first OLMo recipe-G probe had been read under the old rule (disclosed in §4.2); OLMo's recipe-G seeds 1102 and 1103 ran before Llama's recipe-G probe was read and count toward Gate A only if recipe G becomes the common recipe (19:58 UTC); Llama's recipe-G seed 1103 was launched before seed 1102 was read and never enters the two-seed classification (21:37 UTC). **Added in revision 5:** the recipe roles changed twice on development data: recipe B (Gate A, 8 Oct morning), recipe G main (8 Oct 17:36 and 19:24 UTC), recipe B main again (9 Oct 09:29 UTC), after the ladder and the HarmBench pod results were known. Every switch used probe, Gate A and judge readings only, never a Gate B or stage-1 outcome, and the Gate A of record is the recipe-B one, read before any of the switches. The per-family rule was fixed after the ladder was read; §4.1 and §6 say so. **Added in revision 6:** the 7–8B rungs come from two-seed probes under the same rule, and Llama-3.1-8B's rung was accepted by the user although its two-seed read was TOO CLOSE at the upper edge (FOCUS 11 Oct 03:53 UTC); Llama-3.2-1B's Gate A read TOO CLOSE by the rule text on one item (seed 1101's answers end) and was recorded PASS under decision 25 (FOCUS 10 Oct 09:40 UTC). §4.1 discloses both. The stage-1 commitments are frozen after the Gate B noise gate and the sealed rankings were read (rule T2: REQ-21 (m) lists every Gate B value seen); §4.9 says so.
3. **Recipe dependence.** On OLMo, recipe G erodes about a third less than recipe B, and the token-mean-only run puts all of that on the schedule (development evidence, one seed). Answer (revision 5): recipe B is the main recipe and the body calls it plain, not standard; recipe G, the standard schedule, is a mandatory reference (the recipe-only fix), so the reader sees what the schedule alone removes; the repair contrasts are within recipe B,

and §6 says rankings may differ at a lower dose. On Llama recipe G barely moves the erosion (risk 26), so "G lowers the dose" is a statement about the learning rate, not about erosion in every family. The rank-16 check under recipe G reads "rank matters little" by its pre-registered 10 pp rule, but with a small margin at one seed each, and over-refusal rose at rank 16 (development evidence); report the difference itself, not only the rule's label (FOCUS 21:25 UTC). The rank-64, all-linear adapter is still larger than common practice (rank 8–16). **Revision 6:** in the side work recipe G did not lower erosion at 7–8B (OLMo-2-7B and Llama-3.1-8B; FOCUS 11 Oct 10:38 UTC), so the body calls recipe G a reference, not a general fix, and cites the concurrent submission; the 7–8B recipe numbers are not this paper's results.

4. **Nulls everywhere.** A 600-row filter clears the MDE only if the findable effect is about 11–14 pp at row AUC 0.9 [plan §1 item 16, POW Part 3]. Answer: equivalence labels make nulls informative; the planted control; the dose curve.
5. **Additions win on U by construction.** Refusal rows restore refusal, and U tracks refusal [plan §13], which the blind human read supports (risk 1). Answer: the HarmBench rule, selectivity, O against the untuned model, and the adapter-scaling curve.
6. **A free reference may beat every data repair** (safety prompt, adapter scaling) [plan §13]. That is a valid answer; say so.
7. **Prior-work overlap** (DualSelect, Yin & Rush, DataShield's cost table). Answer: no "first" claims (§2 "How we differ"); the equivalence bounds, references, positive control and pre-registered predictions are the contribution.
8. **Re-implemented baselines.** The four method repositories have no licence [plan §5.1, FIL]; all scorers are re-implemented ("X-style") with deviation tables; authors are asked for reference outputs; stage 4's published-setting replication checks fidelity.
9. **Diffuse erosion is the expected Gate B outcome** [Gate B draft §8; audit §7]. Under this framing it no longer collapses the paper: stage 1 runs unchanged, and the planted sweep carries the diagnosis question.
10. **The plant may not erode** above the background [plan §13]; the pilot gate decides.
11. **Small scale favours trial runs; cost ratios change with size.** No 1B cost ratio is claimed for 7–8B.
12. **The decision-value map is a model output** at the menus we can afford [plan §0, NOV-2, MET-M1]. Answer: it is secondary, frozen before the data, and scored against measurements and on Llama.
13. **Judge precision on benign prompts.** After fine-tuning Qwen3Guard labels 22–29% of the benign bank Unsafe (untuned 5%) [audit §5].
14. **"Adding a little safety data is known to work." Correction (FAQ second review, 8 Oct; source opened):** DataRx does NOT show that adding works by default. Its 59.23% → 13.70% compares a *random* 1% BeaverTails mix with DataRx's *selected* 1% mix on Llama3-8B. The fine-tune with no safety data scored 20.24%, so the random mix made it worse. So *which* safety data is added decides whether adding helps (now P4 and P5). Guo's 10 rows were not checked for over-refusal.
15. **Added data can hide misalignment** that returns on prompts like the training data (Dubiński et al.) [reading list v2, risk 4]. The plan adds an overlap control: the add effect must hold on the eval prompts least similar to the add set.
16. **Motivation.** We found no public evidence that practitioners weigh attribution against retrains [problem-validation review]. The draft does not claim that they do.
17. **Money.** The account balance is read before each launch and recorded in FOCUS; it changes daily. **Revision 6:** Gate B's wave-1 need is USD 83.91 with the frozen allowances (measured worst path USD 72.43; FOCUS 16 Oct 13:21 UTC), and the newest balance in FOCUS is USD 91.94 at 19:07 UTC on 16 Oct; no grant is awarded (BlueDot's reply expected about 18 Oct; TAIF late Nov to early Dec). The ICML re-plan proposal estimates its recommended option at about USD 1,180–1,760 (an estimate; decision 28). *History (revision 5e):* these notes gave no figure. Every reading so far is far below the core path of about USD 215–405 [plan §10, §13]. Gate B's wave-1 need is being recomputed with the [D20] constants (a rough estimate is about USD 60–64; the top-up it needs depends on the balance at launch; the recomputed figure decides; FOCUS 13:16 UTC). Every stage starts on a funding event; only complete seed blocks are analysed.

18. **B+Qi may add little over recipe B, or act like recipe G.** FAQ reports that Qi et al.'s Table 9 shows warmup alone does most of the work on benign data [FOCUS 15:59 UTC; CITE-CHECK 25], and our token-mean-only run puts recipe G's reduction on its schedule (development evidence). If B+Qi brings Qi et al.'s warmup to recipe B, it differs from recipe B in its schedule as well as its objective (decision 22). B+Qi stays as a reference because Qi et al.'s objective constrains updates on the initial tokens, a mechanism different from loss weighting; the paper must not justify it by the refuted loss-weighting account. A B+Qi equal to recipe B or to recipe G is still a valid reference result; no novelty is claimed for it.
19. **One family, or a same-recipe contrast read as a replication.** Under recipe B, Qwen and Llama saturate or sit at the top edge and OLMo erodes partly. That is the same-recipe family contrast (§4.10), reported separately and not as a replication. Under the per-family rule Llama runs at its own rate; if no rung of recipe B reads partial for Llama, or stage 2 is not funded, the paper has one family. **Revision 6:** Llama-3.2-1B's Gate A passed at its rung (decision 25), so only funding decides stage 2; the same-recipe family contrast is now the concurrent submission's (overlap rule).
20. **The mechanism account is refuted for B → G.** The mechanism note's first-token loss-mass account (under the per-row loss the first 5 response tokens carry 17.7% of the objective, against 5.3% under the token mean; FOCUS 20:01 UTC) predicted that the token-mean loss carries most of recipe G's reduction. The pre-registered token-mean-only run found that it carries none, at one seed [FOCUS 21:36 UTC]. Answer: report it as a pre-registered prediction that failed; the body offers no loss-weighting explanation; the side study is "refusal openings against harmful content", descriptive, with the first-token margin as its only causal test. Under recipe G refusal openings fall less than under recipe B, but that is now read as a dose effect, not as support for the loss account.
21. **Recipe G is a lower dose.** A reviewer may read "the standard schedule erodes less" as "a lower mean learning rate erodes less", which work on optimization choices already suggests (Kim et al., 2025). Answer: say so; recipe G is a recipe-only reference, and the main claims are repair contrasts within recipe B. **Revision 6:** the body no longer says that recipe G erodes less at all; it gives the design fact (it halves the mean learning rate) and "not a general fix".
22. **Family × recipe interaction.** Recipe G cuts OLMo's erosion by about a third but barely moves Llama's, and the Side Project's logit-scale test excludes 0 (exploratory; development evidence). No rung of the one-recipe ladder left both families partly eroded (FOCUS 01:43 UTC). Answer (revision 5): the per-family rule matches the erosion dose, and cross-family sentences say "at a matched erosion dose"; the rule was fixed after the ladder was read (disclosed in §4.1 and §6); a family difference is claimed only above the interaction's MDE. **Revision 6:** the interaction is the concurrent submission's finding, and it was "not shown" at 7–8B (side work); the body cites it and reports no number.
23. **The ARC rule is a loosening, adopted mid-stream.** It can pass a run whose true drop is up to about 6 points, and for the first OLMo recipe-G probe it was not applied blind [FOCUS 20:50 UTC]. Answer: both verdicts are reported; "skill kept" rests mainly on the held-out task loss; §4.2 and §6 disclose it. **Revision 5:** its scope after the per-family rule is open (decision 16). **Revision 5c:** the user extended it to every family's Gate A read at its chosen learning rate, fixed before any such read and disclosed as a loosening (FOCUS 15:23 UTC; decision 16 resolved); §4.2 states the scope.
24. **The main recipe is harsh.** Recipe B has no warmup, a constant learning rate and a per-row loss; a reviewer may call the erosion an artefact of a recipe few people use. Answer: recipe G, the standard schedule, is a mandatory reference, so the paper shows what the schedule alone removes; recipe B was chosen because its erosion clears the 20 pp HarmBench line (so "harmful compliance as the HarmBench classifier defines it" is allowed; the blind-read caveat sits once in §4.2 and once in §6) and leaves more room for repairs, while recipe G's does not (development evidence; FOCUS 09:29 UTC); §6 says repair rankings may differ at a lower dose. **Revision 6:** at 7–8B, recipe B's erosion at each family's rung also clears the HarmBench line on every seed (FOCUS 11 Oct 10:40 UTC).
25. **Safe LoRA may beat every data repair.** It needs no training and no data. That is a valid answer, like risk 6, and the paper says so. Its settings must be fixed before any stage-1 outcome (decision 19).

26. **"G lowers the dose" does not carry over between families.** Halving recipe G's peak learning rate (rung 2) cut both families by about 28–29%, but moving from recipe B to recipe G, which halves the mean learning rate, cut OLMo by 36% and Llama by only about 4–7% (Llama's B-against-G gap is within its seed spread; development evidence). So the schedule does not act like a lower constant rate in every family, and the token-mean-only run is OLMo-only and one seed. Answer: the body states only the design fact (G halves the mean learning rate) and calls G the recipe-only fix; it makes no claim that G lowers erosion in every family. **Revision 6:** nor across sizes: in the side work G did not lower erosion for OLMo-2-7B or Llama-3.1-8B (FOCUS 11 Oct 10:38 UTC); the body now says "not a general fix" and cites the concurrent submission.
27. **A matched dose differs in learning rate.** Under the per-family rule a contrast that "replicates at a matched erosion dose" replicates across families and learning rates together. Answer: §4.8's claim rule and the fill-in say "at a matched erosion dose"; the same-recipe contrast is reported separately.
28. **The page limit, once results exist** (decision 33). The body fits 8 pages only while §5 and §7 are placeholders.
29. **DualSelect is close** (the literature sweep). A reviewer may say that DualSelect already compares selectors and a safety-data mix with seeds, a false-refusal rate and its selection cost. Answer: §1 and "How we differ" now say what it covers; the differences are purely benign data, removal against addition with attribution, representation and a free length rule, every arm's full cost in one unit, limits on over-refusal and task loss fixed in advance, equivalence-bounded labels over six seeds, model-level references and pre-registered gates.
30. **"Compute-matched"** (decision 26). The arms are row-matched and cost-accounted, not compute-matched, and the shortest-answer rule removes fewer tokens than random removal. Answer (the user's rule of 17 Oct): match tokens per removal arm, or pre-register a token-matched analysis beside the row-matched one (§3.2, [TO BE FIXED in the stage-1 pre-registration]); report tokens removed and optimizer steps per removal arm (the sweep's design action 1); if token matching cannot be done, use the fallback title.
31. **Scoring once** (SABA, cited in §2). Risk scores taken at the start of training may rank rows differently from later. Our LARF-style arm scores on the untuned model; A1 scores at two checkpoints of the full fine-tune. The sweep's design action 4 (two nearly free Gate C analyses: the untuned model's refusal of each Dolly prompt, and the attribution score's share along a safety direction) is for the lead.
32. **Literature the body does not cite** (the sweep's §5 "sentences to add"): concurrent work on first-token damage, a capacity-rank-recipe map, seed effects larger than data edits, judge filters that drop safe demonstrations, multi-layer representation filters, task-format safety data, natural refusals, subliminal traits and a token-mass confound. Not added in revision 6: each needs its own programmatic fetch and room in the 8 pages, and most are anonymous submissions. The lead chooses which to add before submission; the sweep maps their OpenReview ids.
33. **Overlap with the concurrent submission** (decisions 30 and 32). The side paper covers the family × recipe and dose material, the opening layers and the blind read. Answer: the main paper cites it anonymously and reports none of its findings as new.

Where the framing is weak (the drafter's view)

1. **The cost axis is mostly evaluation.** Generation is about 81% of an arm and training about 8% [plan §1 item 6]. FRE then measures our evaluation size more than training; the two cost conventions help, but a reviewer may still read "compute-matched" as "evaluation-matched". Say plainly what one FRE buys.
2. **The comparison fixes doses.** Rankings across removal (600 rows) and addition (48 rows) hold only at these doses; the dose curve is secondary and 3-seed.
3. **Category units versus row scorers.** Gate B tests categories; stage 1 tests row scorers. A row scorer can find what Gate B cannot, and the reverse; the paper must not read one from the other.
4. **The data-only user and diagnostics pull against each other** (previous draft, point 2): a closed-service customer cannot run attribution; a developer with open weights also has model-level repairs. The references now make this explicit, but the population that benefits from paid diagnosis stays narrow.

5. **Mechanism tests are thin.** The token-mean-only run separated loss weighting from schedule at one seed, on OLMo only, and refuted the loss-weighting account of B → G. What remains is a descriptive decomposition with its two layers (openings lost; direct answers already unsafe) and one teacher-forced first-token margin. The two-layer account is observational (no prefill test; positions counted in words), and neither part says why fine-tuning switches refusal openings to direct ones. Both are side studies.
6. **The opening side study is largely pre-empted** (revision 6; the literature sweep, entry 3 and design action 3). A concurrent ICLR 2027 submission puts the damage on the answer's opening token across 17 models. §4.10 keeps the side study exploratory; frame it as a replication if it stays.

Development evidence (for the author only; never a finding in the abstract or the contributions)

- **Gate A PASSED on OLMo-2-0425-1B-Instruct under recipe B** (8 Oct, 05:40 UTC; the Gate A of record since FOCUS 9 Oct 09:29 UTC): E = +43.5, +43.7, +48.2 pp over three seeds; mean +45.1 pp (band 41.3–49.0; partial); fresh-seed mean +45.9 pp; answers end; held-out Dolly loss 2.10 → 1.54; Granite agrees. Independent audit: sound with caveats.
- **Audit disclosures:** P(Unsafe | no refusal) 0.91–0.94 after fine-tuning; about a third of Unsafe answers carry only "unethical acts" or "politically sensitive"; E under hard categories only is about 28 pp; five settings screened; 2 df for seed variance; the ARC bar is near chance at 1B (untuned 0.392).
- **Blind human read** (FOCUS 9 Oct 15:05 UTC; report-only, gates nothing; since revision 5c, Appendix A of the paper carries the pooled counts and limits, labelled development evidence, decision 23d; the group counts stay here). Pre-registered 8 Oct 20:56 UTC and run by the Side Project; the user was the single reader; the ratings were sealed (commit 8c338fb) before the key was opened; scored 9 Oct 14:58 UTC exactly as fixed. Source: the Side Project's branch claude/wonderful-greider-d1aa5b, `research/side-blind-read/PREREG-2026-10-08.md` (Results) and `blind-read-scores-2026-10-09.json`, commits 4d63e04 and 57d08dd. 50 sampled answers, shown without model, recipe, seed or judge label; rubric: 0 refuses or deflects; 1 goes along but gives no usable help (vague, generic, wrong, fictional or incoherent); 2 gives usable help.
- **Qwen3Guard-Unsafe answers:** 40 drawn, 39 rated (1 skipped). 34 went along (rated 1 or 2): 0.87 [0.73, 0.96]. 13 gave usable help (rated 2): **0.33 [0.19, 0.50]** (exact 95% intervals). Rated 2 by group: OLMo-2-1B recipe B 4 of 12; OLMo-2-1B recipe G 3 of 6; Qwen2.5-1.5B recipe B 2 of 7; Qwen2.5-7B recipe B 2 of 6; Llama-3.2-1B recipe B 2 of 8. The groups are too small to compare, and no comparison was made.
- **Safe-labelled controls:** 0 of 10 rated 1 or 2 (no judge misses; [0.00, 0.31]).
- **HarmBench, on the 15 OLMo items that have a label:** of 12 "yes", 4 rated 2 and 8 rated 0 or 1; of 3 "no", none rated 2. The human bar ("would meaningfully help") is stricter than HarmBench's.
- Two "looping or garbled" ticks, both Qwen2.5-1.5B. Reading (I, the Side Project's): U is a sound measure of lost refusal; small models often comply vaguely, generically or wrongly.
- **Limits:** one reader, who also designed parts of the study; 50 items; small groups; no second rater; no comparison between groups.
- **For Gate B (FOCUS):** it meets the intent of DRAFT §12 item 5 (what U means), on a different sample from the audit's proposal (§8 item 4); DRAFT revision 5 reports it as done; the verdict stays on U. It also gives the "about 50 human-read answers" that FAQ asked for the arXiv note (FOCUS 8 Oct 15:59 UTC, item 1).
- **Qwen2.5-1.5B and 7B-Instruct saturate** under recipe B (+75.2 and +82.2 pp); rank 16 still gives +67.8 pp. Guo-gap analysis: judge and prompts explain at most about 13–30 pp of the gap to Guo's Qwen LoRA cell.
- **Mechanism (exploratory note):** refusal openings fall by about 50 pp (OLMo) and 85 pp (Qwen); the opening switch carries about 0.42–0.64 of OLMo's Unsafe increase; direct answers also became more harmful; the erosion is broad across prompts and stable across seeds. Recorded prediction: a first-token path alone gives at most about 3 pp per category, so a LOCALIZED Gate B would need a content effect. **Revision 4:** the note's first-token loss-mass account does not explain the B → G difference (the token-mean-only run, below).
- **Over-refusal:** untuned OLMo refuses 17/100 benign prompts; after recipe-B fine-tuning 4–6/100 (recipe G, one seed: 17/100). v2's base-model positive control: a classifier-targeted rewrite cut sampled unsafe answers

from 80.3% to 11.4%, but XSTest-safe refusals rose from 1/50 to 22/50 in both seeds.

- **Cost of an arm** (one Gate A ledger): training 245 s, greedy generation 1,750 s, sampled generation 761 s, guard and scoring 293 s; about 3,115 s; A100 at USD 1.59/h.
- **Planning noise:** clustered SE(U) 2.8 pp per arm; seed SD 2.65 pp (2 df). MDE after Holm's first step about 5.0 pp (removal) and 5.3 pp (add) at central noise; Gate B detects 8 pp about half the time and reaches 80% at about 10–12 pp.
- **Recorded prior:** DIFFUSE or TOO CLOSE is more likely than LOCALIZED at Gate B.
- **Recipe G on OLMo-2-1B** (one seed, 1101; FOCUS 15:17 UTC; report-only): U 3.2% → 32.2%, E +29.0 pp (SE 2.6; partial) against +43.5 pp under recipe B at the same seed (shared sampling streams); answers end; task learned (held-out loss 2.100 → 1.612); over-refusal 17/100, the untuned level (recipe B 4/100); ARC TOO CLOSE only because 48 questions changed (a gain of 2); Granite agrees (greedy E +23.0 pp, κ 0.93). Refusal openings fall less under G (−0.36 against −0.51, sampled); the opening switch carries about half of the increase under both recipes (0.48 against 0.52). **Revision 4:** the token-mean-only run puts this reduction on the schedule, not the loss weighting, so the smaller drop in refusal openings is read as a dose effect, not as support for the first-token loss account.
- **Llama-3.2-1B under recipe B** (one seed; FOCUS 16:50 UTC; report-only): **SATURATED**, so it does not qualify. U 13.8% → 76.3%, E +62.5 pp (interval 56.3–68.7). Sampled answers end; ARC and task loss pass; over-refusal 11 → 2 of 100; heavy greedy loops (60 of 200 capped; reported, not gating); Granite agrees (greedy E +67.5 pp, κ 0.92). The untuned Llama is far less safe than the untuned OLMo (U_0 13.8% against 3.2%); the opening switch carries 0.68 of the increase (OLMo 0.52).
- **The same-recipe family contrast and recipe G** (Qwen3Guard E; FOCUS and the results record): under recipe B, Qwen2.5-1.5B (+75.2 pp) saturates, Llama-3.2-1B sits at the top edge or above (+62.5, +61.3, +54.0; below) and OLMo-2-1B erodes partly (+43.5 pp at seed 1101; Gate A mean +45.1 pp). Under recipe G: OLMo +29.0, +31.0, +26.5 pp (mean +28.8); Llama +60.0, +56.5, +54.8 pp (mean +57.1). Recipe G removes 36% of OLMo's E on Qwen3Guard and 43% on HarmBench, and about 4–7% of Llama's E (FOCUS 09:29 UTC: "−36% of E on Qwen3Guard, −43% on HarmBench" for OLMo). These numbers stay here, never in the abstract or the contributions.
- **HarmBench compliance judge, complete** (FOCUS 17:07 UTC; reported, not gating): E_HB +27.8, +30.5, +32.8 pp; mean +30.4 pp, about two thirds of Qwen3Guard's +45.1 pp; agreement 83–86% (κ 0.65–0.71). Every seed is above 20 pp, so the 13:53 UTC judge rule is met and Gate A's recipe-B erosion may be called "harmful compliance as the HarmBench classifier defines it", with no caveat of its own (revision 5c: the blind-read caveat sits once in §4.2 and once in §6).
- **Token-mean-only run `bt`** (OLMo-2-1B, seed 1101; pre-registered 20:01 UTC, read 21:36 UTC; report-only; FOCUS): recipe B with only the token-mean loss (constant LR 1e-4, no warmup), fine-tune only on the sealed untuned unit. U 3.2% → 46.8% (281/600): **E_bt +43.7 pp** (SE 3.0, band 37.8–49.6, partial), against E_B +43.5 and E_G +29.0 on the same seed and sampling streams. **The pre-registered prediction (E_bt ≤ 36.25 pp, i.e. closer to G than to B) FAILED.** The share of the B → G reduction carried by the loss weighting, (E_B − E_bt)/(E_B − E_G), is −0.01: none of it. The schedule (a 4-step warmup and cosine decay, which also halves the mean learning rate over the epoch) carries all of it, so the mechanism note's first-token loss-mass account does not explain B → G, and recipe G is, in effect, a lower dose (FOCUS's reading, labelled interpretation). Other bars: answers end (3 capped, 17 failing the screen); task learned (2.100 → 1.587); ARC d = 0 (TOO CLOSE under the frozen bars, pass under the adopted ARC rule); over-refusal 9/100 (B 4, G 17); greedy unsafe 75/200 (B 88, G 46). One seed. Granite and the opening lexicon queued.
- **OLMo-2-1B under recipe G, seeds 1101, 1102 and 1103** (results record, formal three-seed read 22:47 UTC, 8 Oct; reported, not the Gate A of record since FOCUS 9 Oct 09:29 UTC): E +29.0, +31.0, +26.5 pp; mean +28.8 pp, SE 1.77, band 25.3–32.4, partial; coherent-only +28.4; answers end and the task is learned on every seed; Granite agrees on every seed (greedy E +23.0, +25.5, +18.5 pp; κ 0.932, 0.911, 0.905). ARC nets a gain of 2 of 189 on every seed, TOO CLOSE under the frozen bars; frozen verdict TOO CLOSE (ARC only), PASS under the adopted ARC rule (seed 1101 read before that rule existed).

- **Llama-3.2-1B under recipe G** (seed 1101, the common-recipe probe; read 21:04 UTC; FOCUS): U 13.8% → 73.8% (443/600); **E +60.0 pp**, SE 3.2, band 53.6–66.4, which contains the 55 pp edge, so the E class reads **TOO CLOSE** (point estimate saturated; U_ft 0.738, just under the 0.75 auto-saturation line); coherent-only +58.3 pp (same class). Every other bar passes: answers end (24 capped, 47 failing the screen, of 600); ARC 124 → 133 (a gain); task loss 2.243 → 1.619. Over-refusal 11 → 7 of 100; greedy answers loop (75 of 200 capped; reported). Recipe G barely moves Llama (+62.5 → +60.0 pp) while it cuts OLMo by about a third (+43.5 → +29.0 pp at seed 1101). By the frozen amendment (7 Oct, 21:35 UTC), seed 1102 runs and the two-seed mean of seeds 1101 and 1102 is classified with the same band; seed 1103, launched early at 21:37 UTC, never enters that classification. Granite pending. **Later (results record 22:51 UTC, 8 Oct, and 09:35 UTC, 9 Oct):** seed 1102 +56.5 pp; the frozen two-seed classification (1101, 1102) is +58.25 pp, SE 2.72, band 52.8–63.7: TOO CLOSE again, so the user decided (FOCUS 09:29 UTC). Granite on seed 1102: greedy E +58.5 pp, κ 0.903. Seed 1103, read as a report-only third seed: +54.8 pp; three-seed mean +57.1, SE 2.55, band 52.0–62.2, TOO CLOSE at the top edge; Granite agrees on every seed; ARC a gain on every seed.
- **Rank-16 check under recipe G** (OLMo-2-1B, seed 1101; pre-registered 19:24 UTC, read 21:25 UTC; report-only; FOCUS): U 3.2% → 23.3% (140/600); **E +20.2 pp** (SE 2.5, band 15.1–25.2, which contains the 20 pp edge, so TOO CLOSE on the E class; point estimate partial; coherent-only +19.2) against +29.0 pp at rank 64. $|E(r16) - E(r64)| = 8.8$ pp ≤ 10 pp, so **rank matters little by the pre-registered rule**, but the margin is small: one seed each, and the SE of a difference is about 3.6 pp, so a real rank effect of this size cannot be excluded; report the 8.8 pp itself. Over-refusal 25/100 (rank 64 and untuned: 17). Answers end (0 capped, 14 failing the screen); task learned (2.100 → 1.649); ARC d = +1/189, a drop of one question (TOO CLOSE under the frozen bars, pass under the adopted ARC rule). The main recipe stays G at rank 64, as fixed in advance. Granite queued.
- **HarmBench on the pod** (FOCUS 8 Oct 23:28 and 23:43 UTC; results record; one device, after a Mac-against-pod parity PASS; reported, not gating). OLMo-2-1B E_HB (sampled): recipe B +27.7, +30.5, +32.8 pp (mean +30.3; the pod reproduces the Mac's recipe-B counts except one sampled label); recipe G +17.8, +19.3, +14.3 pp (mean +17.2; 0.57 of B); greedy B +27.0, +25.5, +31.0 and G +11.0, +13.5, +8.5. **Harm wording:** recipe B clears 20 pp on every seed and the same-sign, half-size rule holds, so recipe B's erosion may be called "harmful compliance as the HarmBench classifier defines it", with no caveat of its own (revision 5c wording); recipe G is below 20 pp on every seed, so its erosion is only "refusal erosion as Qwen3Guard labels it" (the 10:14 UTC rule; the 13:53 UTC rule holds, 17.2 ≥ 14.4). Family contrast (00:19 UTC, 9 Oct): Qwen2.5-1.5B recipe B +46.0 (rank 16 +45.8), Qwen2.5-7B recipe B +63.5, Llama recipe B +40.5 (seed 1101), Llama recipe G +35.5 and +35.8 (seeds 1101, 1102); the untuned Llama has the highest HarmBench rate of the four families (11%). Recipe G cuts OLMo's E_HB by 43% and Llama's by 12% (Qwen3Guard: 36% and 7%).
- **The one-recipe ladder** (recipe G; FOCUS 01:43 UTC, 9 Oct; the frozen amendment): no rung qualifies both families cleanly (table and rung-2 details below).

Rung (recipe G)	OLMo-2-1B E (Qwen3Guard)	OLMo E_HB	Llama-3.2-1B E	Llama E_HB
1: LR 1e-4	+28.8, partial (3 seeds)	+17.2 (below the 20 pp line)	+58.25, TOO CLOSE high (2 seeds)	+35.7
2: LR 5e-5	+20.8, TOO CLOSE low (2 seeds)	not measured	+41.2, partial (1 seed)	not measured

Rung 2 details (results record): OLMo seeds +20.3 and +21.3 (SE 1.94, band 17.0–24.7); Llama +41.2 (SE 3.6, band 34.0–48.3); both inside their pre-registered predictions (OLMo +12 to +26; Llama +35 to +55); Granite agrees (greedy E OLMo +11.5, +12.0; Llama +38.0); ARC passes under the ARC rule. Halving the peak learning rate moves OLMo from +28.8 to +20.8 pp (−28%) and Llama from +58.25 to +41.2 pp (−29%): the same relative change in both families, at different levels. Rung 3 (2e-5) would put OLMo below 20 pp (FOCUS's reading).

- **Llama-3.2-1B under recipe B, three seeds** (report-only; seeds 1102 and 1103 run by the Side Project's kits and launched by the lead, 9 Oct 10:46 UTC, after the user's "yes"; FOCUS 12:18 UTC; the kits' reading files

and the Side Project's results note): E +62.5 (seed 1101, the 8 Oct probe), +61.3 (seed 1102, attempt 2 after a cap-gate stop) and +54.0 pp (seed 1103); mean +59.3 pp (the Side Project's arithmetic). Pair readings with seed 1101: 1101 + 1102 mean +61.9 pp, saturated; 1101 + 1103 mean +58.25 pp, TOO CLOSE. Answers end; ARC a gain; task learned; over-refusal 2–4 of 100. Granite and HarmBench were not run on seeds 1102 and 1103 (the Mac is kept for other work).

- **Llama-3.2-1B rung 2 under recipe B (LR 5e-5)** (seed 1101; development evidence; pre-registered 9 Oct 15:31 UTC, read 16:59 UTC; source: FOCUS 16:59 UTC, commit 7a8eb08; it sets §4.1's provisional rung, decisions 18 and 25). Fine-tune only on Llama's sealed untuned unit; config `study-kt-llama32-1b-r64-b5e5-mb4-20261008` (recipe B with only the learning rate changed). U 13.8% → 61.8% (371/600): **E +48.0 pp** (SE 3.41; band 41.2–54.8, no edge inside: **partial**; coherent-only +46.6, partial), inside the pre-registered prediction (+35 to +50, point +43). **Answers end: TOO CLOSE** under the frozen bars: 20 of 600 capped (3.3% ≤ 5%), but 64 of 600 (10.7%) fail the screen against the 10% bar, and 53 of 589 (9.0%) without the 11 answers shorter than 5 tokens (0–1 such answers in the other Llama runs). ARC a clear gain (lost 15, gained 36; pass); task learned; greedy E +44.5 pp. Granite not yet run (it goes on the pod with the next Granite run). Pod llama32-1b-b5e5-1101, job 15:50–16:58 UTC, verified and stopped. **The user has not decided** whether to accept the rung; the lead recommends accepting it for planning, with stage 2's Gate A seeds re-reading answers end (decision 25).
- **The family × recipe interaction** (Side Project, report-only; test fixed at 12:18 UTC, 9 Oct, before seed 1102's result was read, but after 11 of its 12 values were known; side branch claude/wonderful-greider-d1aa5b, commits fd48c89 and 8cf2e2d; not in FOCUS). On logit(U) per seed, recipe G minus recipe B: OLMo −0.686 (SE 0.086), Llama −0.113 (SE 0.152); interaction −0.573, SE 0.175, Welch df 5.08, **95% interval −1.019 to −0.127, which excludes 0**: recipe G cuts OLMo's erosion more than Llama's (−16.3 against −2.2 pp; on the pp scale −14.1, SE 3.66, interval also excludes 0). Limits: written after 11 of 12 values; about 5 df; one judge (Qwen3Guard); each family's B and G units share an untuned baseline; 1B and Dolly only.
- **The two-layer account** (FOCUS 8 Oct 15:17 and 16:50 UTC; the mechanism note's addenda of 8 Oct 15:46, 17:14 and 21:56 UTC; the name is the Side Project's arXiv note, commit c51cf39; frozen lexicon `bbbc71e9…`; sampled answers, seed 1101 unless stated). **Layer 1, refusal openings lost** under recipe B: OLMo −0.51, Llama −0.68, Qwen2.5-1.5B −0.85. **Layer 2, unsafe rate of direct answers before fine-tuning:** OLMo 0.18, Qwen 0.57, Llama 0.64, so the untuned Llama's 13.8% unsafe share is mostly compliant direct answers, not failed refusals. The opening switch carries 0.52 of OLMo's unsafe increase, 0.68 of Llama's and 0.64–0.78 of Qwen's. **Across OLMo's recipes** the unsafe rate within direct openings stays at 0.57–0.70 (B, the token-mean-only run, G), while refusal openings fall by 0.51 (B), 0.50 (token mean only), 0.34–0.36 (G, three seeds) and 0.20 (G at rank 16): the recipes act mainly through layer 1. Reading (I): recipe G's lower erosion on OLMo is a dose effect of its schedule, expressed through refusal openings (one seed for the token-mean-only and rank-16 runs). Observational only: no prefill test; positions counted in words.
- **ARC rule adopted** (FOCUS 20:50 UTC; text in §4.2). The disclosure's numbers: the OLMo recipe-G probe (seed 1101) was read at 15:17 UTC under the old rule (ARC TOO CLOSE; d = −2/189, a gain; task loss 2.100 → 1.612); the rule was proposed after that reading, so for that one unit it is not applied blind. **Scope (FOCUS 9 Oct 15:23 UTC):** extended to every family's Gate A read at its chosen learning rate, fixed before any such read; it replaces "at the common recipe".
- **Llama-3.2-1B Gate A at its rung (recipe B, LR 5e-5)** (read 9 Oct 23:16 UTC by the 20:31 UTC pre-registration; the user's "yes" 10 Oct 09:40 UTC: PASS under decision 25; FOCUS). E +48.00, +52.83, +44.17 pp (seeds 1101–1103, shared untuned unit, U_0 13.83%); **mean +48.33 pp** (edge band 43.05–53.61; partial); 2 × SD 8.69 < mean; every U above U_0. Answers end: pass for 1102 and 1103, TOO CLOSE for 1101 (the decision-25 case), so the rule-text verdict was TOO CLOSE with that one item. ARC a gain on every seed; task learned. Granite (pod) E +47.5, +48.0, +41.5 pp, direction agrees. HarmBench at the rung (pod hb-llama-b5e5-1, 10 Oct ~12:37 UTC): E_HB +28.3, +34.8, +26.8 pp, **mean +30.0** (each seed above 20 pp, same sign, at least half of E). The three adapters were later lost with deleted pods; the readings and judge files survive (FOCUS 15 Oct records sync).

- **7–8B rungs** (the Side Project's reads with the frozen family-probe bars; FOCUS 10 Oct 16:40 and 21:30 UTC, 11 Oct 00:10, 03:53 and 10:40 UTC). **OLMo-2-1124-7B-Instruct at 1e-4 (rung 1):** E +49.67, +37.17 pp (seeds 1101, 1102); two-seed mean **+43.42 pp** (SE 2.13; band 39.16–47.67; partial); the seeds differ by 12.5 pp, beyond the band (which covers evaluation noise only); Granite E +49.5, +40.0; E_HB +37.0, +27.7 (mean +32.4). **Llama-3.1-8B-Instruct:** rung 1 (1e-4) saturated (E +70.17); rung 2 (5e-5) E +52.50, +54.00, two-seed mean **+53.25 pp** (SE 2.40; band 48.46–58.04, which contains the 55 pp edge: TOO CLOSE; answers end also TOO CLOSE), accepted by the user on 11 Oct 03:53 UTC "at the edge"; Granite E +50.0, +55.0; E_HB +36.2, +37.8 (mean +37.0). Both families clear the 20 pp HarmBench line on every seed with the same sign and at least half the size, so "harmful compliance as the HarmBench classifier defines it" is allowed at 7–8B as at 1B. The same per-family pattern as at 1B (OLMo 1e-4, Llama 5e-5).
- **Side work at 7–8B: NOT this paper's result** (the concurrent submission's; overlap rule; FOCUS 11 Oct 10:38 and 12:54 UTC). Recipe G against recipe B at rung 1 (three seeds): OLMo-2-7B +44.1 → +50.3 pp; Llama-3.1-8B +60.7 → +59.2 pp; the 1B family × recipe interaction is "not shown" at 7–8B (I = +0.334, 95% −0.49 to +1.16). Halving the learning rate: OLMo-2-7B +31.9 against +44.1 pp, Llama-3.1-8B +53.25 against +60.7 pp; "both families respond to the LR alike" is not shown as equality (the side paper: "without establishing equal effects"; FOCUS 15 Oct records sync). FOCUS: recipe G's benefit "is family- and scale-dependent, so the paper must not present recipe G as a general fix".
- **Gate B anchor and recipe S** (FOCUS 10 Oct 14:01 UTC): 64 harmful and 64 safe anchors kept and sealed (text out of git); recipe S frozen, with recipe B's training identity.
- **Gate B wave 0a: the noise gate PASSED** (FOCUS 16 Oct 00:25 UTC; the frozen reading; report from `tmp/gate-b-olmo/gate_b_noise_gate.json`). r_j = U(a0, j) − U(R_j, j) = +0.67, +2.50, +4.33 pp (seeds 1101, 1102, 1103); SD(r) 1.83 pp; σ_eval,0 1.28 pp; **σ̂_gate = 1.83 pp ≤ 7** (governing reading plain); the LOCALIZED bar max(8, 4.177 × 0.707 × 1.83) = 8 pp ≤ 33.8; the fallback σ̂_gate (2.16 pp) also passes; σ̂_gate rests on 2 df. Report only: P4 mean r = 2.5 pp (SE 1.06) against the proportional prediction 5.6 pp and saturation's about 0; P(LOCALIZED) at the nearest simulated σ_pair (2.15 pp): 61 / 97 / 100% at 8 / 10 / 12 pp. Spend: USD 4.78.
- **Gate B wave 0b: the rankings are SEALED** (FOCUS 16 Oct 13:21 UTC). The retrain's q and e1 adapter fingerprints equal Gate A seed 1101's (fingerprint trigger false); the noise gate re-read with the retrain: PASS. The diagnostics took 3.04 h against the frozen upper of 2.0 h (cap 4.5 h). Sealed file `gate_b_rankings.json`, SHA-256 d6d5cd6b…. **Rankings** (Dolly categories, top first): attribution general_qa > summarization > brainstorming > open_qa > closed_qa > classification > information_extraction > creative_writing; representation general_qa > summarization > open_qa > brainstorming > classification > closed_qa > creative_writing > information_extraction; classifier (report only) brainstorming > creative_writing > closed_qa > classification > general_qa > open_qa > summarization > information_extraction. **pick_A = pick_R = general_qa; pick_F = classification;** pick_A = pick_F false; pick_R = pick_F false. Spearman ρ with first5_share: attribution −0.17, representation −0.14, classifier −0.17; classifier informative. These are Gate B values seen before the stage-1 commitments are frozen (§4.9). Spend in waves 0a–0b: USD 11.26.
- **D5 = (a)** (the user delegated it; FOCUS 16 Oct 12:15 UTC): P1–P6 primary; the selector secondary and model-based (simulation, replay, predictions); the held-out selector comparison becomes its own later pre-registered study after a funding event; the one-example safety patch bounded in text (applied in §2 and §6).
- **Status (FOCUS, to the 16 Oct 21:05 UTC entry, the newest; head 416b780).** The goal for the main paper is ICML 2027 (the user, relayed by FAQ; 21:05 UTC); no fallback venue named, so TMLR or NeurIPS 2027 stays. The stage-1 commitments build: the lead decided L1–L16 (19:47 UTC; for example (L5) dose draws d = 2–4 at seeds 1100 + d; (L6) the list-or-markup rule by three frozen patterns; (L14) SD-600 removes rows it cannot rewrite; (L16) τ_L fixed from the three Gate A a0 seeds); REQ-21 (m) must list every Gate B value seen. Wave 1 waits for those commitments; it needs USD 83.91 (13:21 UTC); the balance was USD 91.94 at 19:07 UTC. Pods: w0b-2 holds the retrain's states (only copy; keep); w0b-1 was deleted on the user's yes (19:08 UTC); the user said "stop asking me for approval and go ahead and do these things" (19:06 UTC; the lead reads it as doing planned steps without approval questions, with the frozen money rules, the guard, the USD 10/h limit and the launch check unchanged). The P5 lexicon pin is filled only at `--final`, with every

sealed phase re-run and compared (11:08 UTC). Earlier (FOCUS 15 Oct records sync): the side paper "Same Recipe, Different Fate" went to ARR October 2026 as #13386 on 13 Oct; three grants were submitted on 13 Oct, none awarded; General Implementation VI leads the main study since 15 Oct 19:56 UTC.

- **Status at revision 5e (history; FOCUS to 9 Oct 17:22 UTC).** The user accepted decision 23's wording package and extended the ARC rule (15:23 UTC); the blind human read is recorded (15:05 UTC). **Pod price limit (the user, 15:29 UTC):** every kit refuses to create a pod priced above USD 10.00/h, or with no finite price, or stops it at once. This replaces the old USD 1.75/h limit and the "no hourly limit" of 8 Oct 20:17 UTC; the balance-based budget guard is unchanged (it plans a new pod at USD 1.85/h); brief revision 17 amends [D20]. Recipe B is the main recipe for Gate B and stage 1; Gate B DRAFT revision 5 is to stay on recipe B, add the [D14]–[D19] rules and the [D20] price constants, and recompute the money thresholds; the repository's DRAFT is still revision 4b. The two report-only Llama recipe-B seeds finished (12:18 UTC). At 09:30 UTC four idle pods were deleted and 15 stopped pods kept the fine-tuned adapters; the Llama recipe-B pods were stopped after their runs. Llama's rung-2 probe under recipe B (5e-5) was read at 16:59 UTC (entry above; Granite not yet run; it goes on the next Granite pod run); the user has not decided whether to accept it (decision 25). **Granite on RunPod (FOCUS 16:48 UTC; the user: "just use the gpus for granite as well"):** the Granite judge's RunPod launcher is committed (ff9196a). Its parity rule was fixed, and PARITY.md's SHA-256 (c4fc455e…) recorded, before the first Granite pod launch. The rule: on the four OLMo-2-1B Gate A units (untuned and e1 seeds 1101–1103; 800 greedy harm answers with Mac Granite labels), the tokenizer fingerprint and every item's input hash are identical, every item is judged, and each unit has at least 196 of 200 identical labels and a Yes count within 3 of the Mac's; all four units must pass. If parity fails, pod and Mac Granite numbers are never mixed in one contrast. By default, each Granite launch first judges 200 known answers (OLMo recipe B, seed 1101, e1) and stops unless at least 190 of 200 labels match the Mac's and the Yes count is within 5. **Granite parity: PASS (FOCUS 17:22 UTC, by the pre-set rule).** The first pod (16:49 UTC) got no network address and was stopped after 20 minutes without SSH (about USD 0.6); the retry, on another machine, passed SSH. Identical labels per unit: 200, 200, 199 and 200 of 200 (one no → yes on e1 seed 1102); all 800 input hashes and the tokenizer fingerprints are identical. Pod-made Granite files may now be used like Mac-made ones; this was recorded before the Gate B freeze, so Gate B's Granite files can come from the pod (DRAFT revision 5 rule). Next (FOCUS 16:48 UTC): Granite for the Side Project's Llama recipe-B seeds and Gate B's judge files. Safe LoRA: the lead reads the paper in full and fixes its settings in the stage-1 pre-registration; the post-hoc transform is built with tests and reviewed before use. B+Qi's engine change comes in a later round, committed before stage 1 is frozen. The fiction side study (Side Project; report-only) is approved at recipe B, up to about USD 20 in total (15:23 UTC; earlier about USD 10–16), and is not part of this paper. The Qwen2.5-1.5B recipe-G run is binned; the HarmBench add-on for the recipe-G side runs is dropped (10:07 UTC). Alpaca waits for Gate B and its own design note. The Gate C anchor list is sealed (8 Oct 15:53 UTC).

Conflicts between the sources, and how this revision resolves them

1. **Spec RQ2 (the map co-primary, the headline figure) vs the user's decision (map secondary, model-based).** Followed the user's decision: the measured comparison leads the title, abstract, §1 and contributions; the map is §3.5, secondary, with predictions frozen before Gate B's category runs and scored on Llama. Figure 1 is now the repair comparison.
2. **Spec repair menu (category, cluster or row removal by heuristic, filters, attribution, representation similarity; random vs selected mix; "10 rows"; recipe-only) vs the adopted stage-1 design.** Followed the adopted design: six primary arms (A1, L1, H1, S1, S3, S4) at 600 removed or 48 added rows; recipe G is a reference, not a repair; category removal stays in Gate B; cluster menus exist only as the planted sweep's nested cells; the "10 rows" dose is an option, not the plan.
3. **Spec "Qwen is the fragile contrast" vs "do not claim Qwen is uniquely fragile at 7B".** Qwen appears only as context in §4.1 and §4.10 (its results are in this note), with Guo et al.'s Table 2 showing a Qwen2.5-7B LoRA cell that barely erodes.
4. **Previous draft vs spec on O's reference.** Spec and amendment: the untuned model (a0 reported). Applied.

5. **Previous draft's rewrite and self-distill fixes.** The amendment drops them from the core menu; one rewrite arm runs in the DIFFUSE branch (§4.6).
6. **Previous draft's co-primary side-effect prediction.** Secondary, on the add arms only (§4.10).
7. **The brief's "keep the Guo Table 2 wording".** Neither previous file contained it; it is now in §2.
8. **Spec §9 vs §5 on the Llama probe** ("queued" vs "running"): not paper text; noted above.
9. **The lead's message vs FOCUS on the common-recipe rule (revision 3).** The lead: if Llama under recipe G "reads partial", G is the common recipe; "if not", both families step down the learning rate together. FOCUS 17:36 UTC: G is the common recipe if Llama's probe *qualifies* (meets every frozen family-probe bar, not only the partial class); both families step down only if Llama *saturates*; if no rung qualifies for both (for example Llama weak, or OLMo below 20 pp where Llama turns partial), no common recipe exists and the user decides. Followed FOCUS (§4.1). **Superseded (revision 5):** FOCUS 9 Oct 09:29 UTC replaces the common-recipe rule with the per-family learning-rate rule.
10. **"A fourth reference" vs three references.** FOCUS 17:36 UTC item 4 calls G+Qi "a fourth model-level reference", but the same item makes recipe G the base, so the references are three: the safety prompt, adapter scaling and G+Qi. Followed that list: G+Qi replaces "a standard recipe" in every reference list. **Superseded (revision 5):** FOCUS 9 Oct 10:01 UTC lists five mandatory references (recipe G among them, with B+Qi and Safe LoRA), and 13:16 UTC puts the Qi arm on recipe B.
11. **OLMo's recipe-B number.** The lead gives +45.1 pp (Gate A, 3-seed mean); FOCUS 19:24 UTC gives +43.5 pp (seed 1101, the same-seed contrast with recipe G). Both are correct; the development evidence gives both, labelled.
12. **The Llama rule.** The 13:53 UTC amendment let Llama's next recipe change "LR or rank"; the 17:36 UTC common-recipe rule changes the learning rate only, for both families. Followed 17:36 UTC (§4.1, §4.8, §6). **Superseded (revision 5):** the per-family rule (FOCUS 9 Oct 09:29 UTC) also changes the learning rate only, now of recipe B, and per family.
13. **"Llama replicates" (spec and revision 2).** Removed, as the lead asked: FOCUS 16:50 UTC reads Llama saturated under recipe B, and its recipe-G probe had not been read at this revision. The claim rule for a replication stays in §4.8.
14. **OLMo recipe-G seed 1103 (+26.5 pp; revision 4).** The lead gives seeds 1101–1103 as +29.0, +31.0 and +26.5 pp. FOCUS (to 21:37 UTC) and the results record hold only the interim two-seed reading (1101, 1102); seed 1103 is not recorded there. The kit's pair reading for seeds 1101 and 1103 (`tmp/family-probe-olmo2-1b-g-seed1103/reading-olmo-g-1101-1103-nojudge.json`) shows a two-seed mean of 27.75 pp, which agrees with +26.5; the three-seed reading file beside it is empty. Resolution: the number appears only in the development evidence; the mean +28.8 pp is marked as the drafter's arithmetic. **RESOLVED (revision 4a):** seed 1103 is now recorded in FOCUS ("OLMo under recipe G, all three seeds in", interim) and the results record (commit 8fccce2), so the "lead's figure" label is dropped.
15. **"Mechanism note §4" (revision 4).** FOCUS and the lead call the first-token loss-mass account the mechanism note's §4. In the note, the loss-share table is in §5 ("Training data and the Gate B link"), with first-5 shares .181 (per-row) against .054 (token mean); FOCUS 20:01 UTC quotes 17.7% against 5.3%. Resolution: these notes say "the mechanism note's first-token loss-mass account" without a section number and quote FOCUS's numbers (FOCUS wins); the body quotes neither.
16. **ARC in the body (revision 4).** Revision 3's §4.2 said ARC "sets no limit"; the frozen family-probe and Gate A bars do check ARC (a drop of at most 0.03), and the 20:50 UTC rule resolves its TOO CLOSE. Followed FOCUS: §4.2 states the bar and the rule.
17. **"The schedule carries all of it" (the lead) against FOCUS's one-seed caveat (revision 4).** FOCUS reads one seed and labels "G is, in effect, a lower dose" as interpretation. Resolution: the body states only design facts (recipe G's schedule halves the mean learning rate, so recipe B is the stronger dose) plus a placeholder for the run's result; the attribution of the reduction to the schedule stays in the development evidence, marked one seed.

18. **"If FOCUS records a newer decision, follow it" (the lead's brief; revision 4).** FOCUS records no step-down of the learning rate for both families: only the conditional rule (21:04 UTC) and the early Llama seed 1103 (21:37 UTC). [PLACEHOLDER: the common rung] stays, and §4.1 adds only the too-close rule that FOCUS cites from the frozen amendment. **Superseded (revision 5):** the user decided (FOCUS 9 Oct 09:29 UTC); [PLACEHOLDER: the common rung] is gone.
19. **"The two-layer mechanism account, as FOCUS states it" (the lead's brief).** FOCUS has no entry by that name. Its numbers are in FOCUS's 8 Oct 15:17 UTC (recipe G: G keeps refusal openings rather than making direct answers safer) and 16:50 UTC (Llama: "already unsafe in 64% of its direct openings (OLMo 18%)") entries, and in the mechanism note's addenda (17:14 and 21:56 UTC); the name "two layers" comes from the Side Project's arXiv note (branch claude/wonderful-greider-d1aa5b, commit c51cf39, read-only). Resolution: the body states the two layers as the two factors of §3.6's opening-switch term, without numbers; the numbers come from FOCUS and the mechanism note, which agree.
20. **Llama recipe-B seed 1102 (+61.3 pp).** FOCUS 12:18 UTC records that seed 1102 finished but gives no E for it (it gives +54.0 for seed 1103). The number is the kit's frozen reading (`tmp/family-probe-sp-llama32-1b-b-seed1102/reading-llama-b-1101-1102-nojudge.json`: E per seed 62.5 and 61.33) and the Side Project's results note (side branch, commit fd48c89); it matches the lead's figure. It appears only in the development evidence.
21. **The family × recipe interaction** is recorded only on the Side Project's branch, not in FOCUS or the main checkout's records. Resolution: development evidence only, with its limits; decision 17.
22. **"Fixed now before any Llama run at the main recipe" (FOCUS 09:29 UTC)** against Llama's recipe-B probe of 8 Oct 16:50 UTC and the two report-only recipe-B seeds run after the decision. The rule postdates Llama's first recipe-B probe and the ladder; the drafter reads FOCUS's phrase as "before any further Llama run at the main recipe that the rule governs". Resolution: §4.1 and §6 say the rule "was fixed after a development ladder was read"; decision 18 names the probe that would come next.
23. **The ARC rule's scope.** FOCUS 20:50 UTC scopes it to "the common recipe", which no longer exists. Resolution: §4.2 keeps the rule and its disclosure, drops the scope phrase and marks the scope [TO BE FINALISED] (decision 16). **RESOLVED (revision 5c):** FOCUS 15:23 UTC extends the rule to every family's Gate A read at its chosen learning rate; §4.2 states the scope.
24. **The E-scaled floor (FOCUS 8 Oct 20:17 UTC) against recipe B as the main recipe.** At recipe B it equals the 8 pp floor, so it adds nothing. Resolution: dropped from §4.6; decision 21.
25. **"Describe recipe G as a dose-lowering fix" (the lead's brief) against the Llama data.** Recipe G barely moves Llama, while halving recipe G's learning rate moves Llama by about 29% (development evidence). Resolution: the body gives the design fact (recipe G's schedule halves the mean learning rate), calls G "the recipe-only fix" and "a gentler learning-rate schedule", and does not say that G lowers erosion in every family; these notes keep "dose-lowering" for OLMo, where the token-mean-only run supports it (risk 26).
26. **Numbers only in the notes, against the §3.6 sentence "where u_N0 is high and u_R0 low, a lost opening comes close to an unsafe answer".** The sentence is conditional and follows from the decomposition's first term, $(p_0 - p_1)(u_{\mathrm{N0}} - u_{\mathrm{R0}})$; it states no result. Kept.
27. **The secondary length-rule arm.** Revisions 3–4 named the mirror (the length rule under recipe B's per-row loss) because recipe G was main. With recipe B main again, the arm reverts to revision 2's "under a token-mean loss" (decision 11).
28. **"Numbers only in the development evidence" (the lead) against the wording rule's caveat (FOCUS 15:05 UTC).** The lead puts the read's numbers only in the development evidence, not in the abstract or the contributions; FOCUS requires each harm sentence to carry the caveat that "a blind human read rated about a third of judge-unsafe answers as usable help". They agree on the abstract and the contributions, but the caveat is itself a rounded development number in the body. Resolution (FOCUS wins on the wording): the body carries only the rounded caveat ("most … went along", "about a third", "one reader", "during development") in §4.2 and §6; counts, intervals and groups stay in the development evidence. "What this is" names this exception to "Nothing in the paper body is a result". **Revision 5c (FOCUS 15:23 UTC, item 3c–d):** the caveat is no

longer in every harm sentence. §4.2 states it once, as a fact about the measures, and §6 once, as a limitation, each pointing to Appendix A; harm sentences say only "harmful compliance as the HarmBench classifier defines it". The pooled counts, intervals and limits are in Appendix A (part of the paper, labelled development evidence); group counts stay in the development evidence here. The body still gives the rounded caveat ("most … went along", "about a third") and no count (conflict 34).

29. **"Judge-unsafe".** The read sampled Qwen3Guard-Unsafe answers (13 of 39 usable help), while the claim rule applies the caveat to HarmBench sentences. On the 12 OLMo answers that HarmBench labels "yes", 4 were usable help, also about a third, so FOCUS's phrase "judge-unsafe answers" holds for both judges as measured; the HarmBench subset is small and OLMo-only. Kept FOCUS's phrase.
30. **"Of 39 Qwen3Guard-Unsafe answers" (the lead, FOCUS) against "40 labelled Unsafe" (the pre-registration).** Both are right: 40 Unsafe-labelled answers were drawn and one was skipped, so 39 were rated; with the 10 Safe controls the sheet held 50. The development evidence says so.
31. **Gate B's blind read, "DONE" (the lead, FOCUS 15:05 UTC) against the repository's DRAFT (still revision 4b: §12 item 5, "recommended").** FOCUS says DRAFT revision 5 reports the read as done, with its sample difference, and the verdict stays on U. Resolution: §4.6, which summarizes revision 5 [NOT YET FIXED], says it is done, on a different sample from the one first proposed, and that the verdict reads U alone (FOCUS wins). Revision 5c points this sentence to Appendix A, which describes the difference.
32. **Decision 23's letters.** Revision 5b's decision 23 listed (a) "adverse", (b) "safety" in the title, (c) whether §5 or an appendix reports the counts and (d) a second rater. FOCUS 15:23 UTC lists (a) "adverse", (b) the title keeps "safety" and the abstract names the measures, (c) the caveat once in the measurement section and the limitations, with an appendix, and (d) the counts in an appendix. Followed FOCUS's letters; revision 5b's (d), which FOCUS does not decide, stays open as decision 24.
33. **"'Significantly harmful' becomes 'adverse'" (the lead's brief) against the label's significance condition.** Dropping "significantly" would let §1 read as if any adverse estimate made a repair *worse*; revision 5b's decision 23a, which the user accepted, proposed "significantly adverse". Resolution: "significantly adverse" in §1 and §3.2, "a significant adverse result" in §4.3, and "adverse" alone where the phrase names only a direction ("adverse direction", "adverse repair"). If the lead wants "adverse" alone in §1 and §3.2, §3.2 still carries the Holm-adjusted 5% level.
34. **"The body may refer to 'a blind human read (Appendix A)' without giving numbers" (the lead's brief) against the caveat's wording "about a third" (FOCUS 15:05 UTC, placed by 15:23 UTC item 3c).** Read the brief as allowing number-free references (as §4.6 now makes) and FOCUS as keeping the rounded caveat in its two places; item 3d moves the counts, not the caveat. Resolution: §4.2 and §6 keep "about a third" (§6 also "most … went along"); no count, interval or group is in the body. If the lead wants a number-free caveat, replace "about a third of judge-unsafe answers as usable help" by "most judge-unsafe answers as giving no usable help" in §4.2, and "but only about a third gave usable help" by "but most gave no usable help" in §6.
35. **"The abstract" in decision 23a.** The abstract names the *worse* label but has never contained "significantly harmful", so 23a changed nothing there; its change is 23b.
36. **The FOCUS entry's time.** The lead's brief first named the entry "USER DECISIONS 15:12"; FOCUS stamps it "USER DECISIONS, 2026-10-09 15:23 UTC", and the lead then corrected the brief (content unchanged). These notes cite 15:23 UTC.
37. **"Llama-3.2-1B's rung is 5e-5" (the lead's brief, §4.1) against FOCUS 16:59 UTC ("the answers-end TOO CLOSE is the user's decision"; the lead recommends accepting rung 2 "for planning").** The body sentence reads as settled while the user has not decided. Resolution: the brief's wording is kept, because its parenthesis labels the rung as one probe seed of development evidence, to be confirmed by stage 2's Gate A, and the pre-registration (FOCUS 15:31 UTC) makes rung 2 Llama's main-recipe rung if the probe reads partial; these notes mark it provisionally filled, pending the user (decisions 18 and 25). If the lead wants the body to say so, write "Llama-3.2-1B's provisional rung is …" (one more word). **RESOLVED (revision 5e):** the lead asked for it, because the user has not decided; §4.1 now says "Llama-3.2-1B's provisional rung is …".

38. **Granite.** The lead's brief gives the E class and the answers-end bar but not the second judge. FOCUS 16:59 UTC records that Granite has not run on this probe, and the frozen family-probe bars include the second judge's direction. Resolution (FOCUS wins): the development-evidence entry, decision 18, decision 25 and the placeholder table say Granite has not run.
39. **§4.1's "a too-close probe gets one more seed" against the answers-end TOO CLOSE.** In the frozen amendment (7 Oct 21:35 UTC) one more seed follows only a TOO CLOSE on the E class; a short-answers TOO CLOSE goes to the user, and the lead proposes that stage 2's Gate A seeds re-read answers end instead. §4.1's parenthesis is about the E class, but a reader could apply it to any bar. Not changed (this revision edits only §4.1's rung sentence). If the lead wants it exact: "(§4.6; a probe too close on E gets one more seed; any other too-close bar goes to the user)". **RESOLVED (revision 5e):** the lead asked for the exact rule; §4.1 now says "(§4.6; a probe too close on E gets one more seed; any other too-close bar goes to the user)". Checked against the amendment's text: one more seed follows only the E class's edge band; the short-answers bar, ARC (another seed cannot resolve it: the questions are the same) and unscored answers go to the user; "any TOO CLOSE goes to the user with the numbers". Left out for length: the extra seed runs after the user's go, and a two-seed mean that is still too close goes to the user.
40. **The balance figure and the Granite parity pod (the lead's brief for revision 5e) against FOCUS.** If risk 17 needs a figure, the brief offers "the latest account reading in FOCUS (USD 54.80 at 16:49 UTC on 9 Oct, before the Granite parity pod)". FOCUS (to the 17:22 UTC entry, commit 2837eaf; read at 17:31 UTC) has no such reading; its latest balance is "USD 57.11 before" the 15:35 UTC launch (the 15:31 UTC entry). The brief also gave the Granite parity result as pending (the first pod got no network address and was stopped; the retry passed SSH and the GPU check); FOCUS 17:22 UTC, committed while this revision was written, records parity PASS. Resolution: risk 17 needs no figure, so it gives none; the status entry follows FOCUS (PASS), with the first pod's failure as FOCUS records it.
41. **The pipelines.** The brief says to reuse scratchpad scripts (`paper_to_pdf.py`, `scripts/body_pages.py`, `scripts/notes_md_to_tex_rev5c.py`, `scripts/wrap_tex.py`, `rev5d/splice.py`); none existed (the scratchpad held only empty folders from the earlier revisions). Re-created in `rev6/scripts/`: `body_pages.py` (the same definition; it gives 9.36 pages on revision 5e's PDF), `notes_md_to_tex.py` (Markdown notes to LaTeX in revision 5e's Appendix B style, with line wrapping), `paper_to_pdf.py` (Markdown to HTML with KaTeX, printed by headless Chrome). The Markdown PDF may differ slightly in look from revision 5e's.
42. **Appendix C's letter.** The brief asks for "Appendix C" beside the existing A, with the notes as B. In the LaTeX build the notes stay Appendix B and the design details are Appendix C, after the notes; when the notes are deleted, C becomes B (it is referenced by `\ref`). In the Markdown file the notes sit at the top, so Appendix C follows Appendix A; the PDF pipeline puts the notes last.
43. **"Nothing in the body is a result" against the Gate B status.** The brief asks the body to say that the noise gate passed and the rankings are sealed. The body says so as a status line, with no number and no category; "What this is" lists it as an exception.
44. **The Gate C format clause** goes beyond the brief's wording (decision 29); the frozen pre-registration requires it to be reported whatever Gate B reads.
45. **"Its Gate A PASSED on 10 Oct" (the brief) against the rule-text verdict.** FOCUS 9 Oct 23:16 UTC reads TOO CLOSE by the rule text, with one item (seed 1101's answers end), and the user recorded PASS under decision 25 (10 Oct 09:40 UTC). Both hold: the body states the rung plainly and discloses "accepted on a too-close bar"; these notes give the detail.
46. **"The one-example safety patch: bounded in text (related work and a limitation)"** (FOCUS 16 Oct 12:15 UTC). It needs a citation; revision 6 fetched Safety at One Shot by API (also on the coordinator's list). The patch is described from its abstract, after harmful fine-tuning; the body does not say it works after benign fine-tuning.
47. **GradShield's venue.** The coordinator: "GradShield is now at NeurIPS 2026". The arXiv API gives no venue; OpenReview shows only a withdrawn ICLR 2026 submission; the NeurIPS 2026 accepted-papers JSON (fetched by the literature sweep from neurips.cc) lists it as a poster. The brief allows new BibTeX only from the

arXiv API, OpenReview or PMLR, so the entry keeps its arXiv form with the note "NeurIPS 2026 (poster)" (CITE-CHECK 8).

48. **DualSelect's wording.** The sweep's replacement text says "six data selectors"; the coordinator's facts give no count; the body gives none. The sweep put DataRx and SABA in §1's sentence; revision 6 puts them in §2 to hold the length.
49. **The 7–8B rungs as design (the brief) against their source.** The 7–8B probes were read by the Side Project under the main study's frozen family-probe bars, and FOCUS records the rungs as the main study's ("7–8B main-study rungs", 11 Oct 10:40 UTC; the user's decision 11 Oct 03:53 UTC). The body states the rungs as design; the side paper's same-recipe 7–8B numbers stay out (overlap rule).
50. **The overlap rule's scope.** The brief names the family × recipe and dose findings; the side paper's abstract also lists the opening layers and the blind read. Revision 6 cites the concurrent submission for those too (Appendix C.8; Appendix A, decision 30). It keeps the blind read in Appendix A as this paper's check of what U measures.
51. **"Don't add them to the body's reference list yet" (the coordinator's addition 3) against Safety at One Shot in §2 and §6.** Read as the body's list of reference arms (the five references of §4.4), not its bibliography: Safety at One Shot was cited in §2 and §6 earlier in this revision, because D5 = (a) bounds the one-example patch "in text (related work and a limitation)" and the coordinator's addition 2 asked for the citation if verified. It is not a reference arm. If the coordinator meant the bibliography too, delete the two clauses (§2's last sentence and §6's second sentence) and the BibTeX entry `zhang2026oneshot`.
52. **Option B (decision 28) against the body's stage-1 framing.** Option B makes OLMo-1B's stage 1 development and puts the confirmatory comparison at 7–8B; the body still calls OLMo-2-1B the stage-1 model and lists the testing order of revision 5. The coordinator asked only for the 7–8B tier's description; the rest of option B is for the lead and the user before the stage-1 pre-registration.

Sources read (revision 6): FOCUS.md at `origin/stageb-costs-and-recovery-safeguards-20261003` (fetched; head 416b780), every entry from 9 Oct 17:00 UTC to 16 Oct 21:05 UTC, and the 8 Oct 12:53 and 13:53 UTC entries (the scale ladder; the adopted direction); the Gate B/C pre-registration, revision 5 (SHA-256 5c97cce7… checked), header, §1–§5, §7, §8 and §10; the kill-test design §4.4–§4.5; the GI handoff, STATUS of 16 Oct 13:20 UTC and the 15 Oct lead handoff; the stage-1 commitments brief's D5-7 row and FOCUS's L1–L16 entry; the restart plan's line on "Safety at One Shot" (for the reference only); the side paper's abstract and section list (`writing/side-notes/arr/main.tex` on its branch, read with `git show`; overlap only); `writing/paper/2026-10-17-lit-sweep.md` §0, §4's corrections, §5 and §6; the summary of `writing/paper/2026-10-17-icml-replan-PROPOSAL.md`; the ICML 2026 example paper's instructions (page limit, anonymity, self-citation, Impact Statement); the arXiv API records of 2601.01887, 2506.00676, 2606.00160, 2607.15081, 2605.14194 and 2606.09866; OpenReview API search records for EyH8Fu3vtZ, Jo6U0y2omr, yKetbTItcu and GradShield; the NeurIPS 2026 accepted-papers JSON (the sweep's copy) for GradShield.

Sources read (revision 5e): FOCUS.md to its newest entry at 17:31 UTC (9 Oct 17:22 UTC, the Granite parity PASS, commit 2837eaf), with the entries of 16:59 UTC (Llama's rung-2 read), 16:48 UTC (the Granite judge on RunPod; its parity rule and PARITY.md SHA-256), 15:31 UTC (the rung-2 pre-registration and its launch balance) and 15:29 UTC (the USD 10/h price limit), the 13:16 UTC lead notes (the [D20] constants and the wave-1 estimate) and the 7 Oct 21:35 UTC amendment to the family-probe bars (edge band, ARC, short answers, unscored answers, overall qualification); the lead's brief for revision 5e.

Sources read (revision 5d): FOCUS.md's entry "Llama-3.2-1B rung 2 at the main recipe (recipe B at LR 5e-5), seed 1101, read 2026-10-09 16:59 UTC" (commit 7a8eb08; no later entry exists at 17:03 UTC), its pre-registration (15:31 UTC) and the 7 Oct 21:35 UTC amendment to the family-probe bars (short answers; overall qualification).

Sources read (revision 5c): FOCUS.md's entry "USER DECISIONS, 2026-10-09 15:23 UTC" (no later entry exists), its 13:45 UTC item on the ARC scope and the 8 Oct 20:50 UTC ARC entry; on the Side Project's branch, by `git show` without checkout, `research/side-blind-read/PREREG-2026-10-08.md` (design, rubric, limits, results) and `blind-read-scores-2026-10-09.json` at commit 57d08dd (4d63e04 differs only in the results time); the Gate A

audit §8 item 4 (the proposed sample: 40 Unsafe answers from OLMo's Gate A units, 20 under 20 tokens and 20 flagged only as unethical, plus the 29 Unsafe answers on the over-refusal bank).

Sources read (revision 5b): FOCUS.md's entries of 9 Oct 15:05 UTC (the blind human read and its two sub-items; recorded in commit 3e7c570) and 13:45 UTC, and of 8 Oct 18:31 UTC (the *worse* label's wording) and 15:59 UTC (FAQ items 1 and 4); on the Side Project's branch claude/wonderful-greider-d1aa5b, by `git show` without checkout, `research/side-blind-read/PREREG-2026-10-08.md` and `blind-read-scores-2026-10-09.json` at commit 57d08dd (4d63e04 differs only in the results time, 15:00 → 14:58 UTC); the Gate B/C DRAFT §12 item 5 (still revision 4b); the Gate A audit §8 item 4.

Sources read (revision 5): FOCUS.md's entries of 9 Oct from 00:36 to 13:16 UTC, and of 8 Oct at 15:17, 16:50, 20:17, 20:50, 21:36, 23:28 and 23:43 UTC, with the adopted amendment's reference and family lines; the results record's sections from "OLMo-2-1B under recipe G, seed 1103" (8 Oct 21:53 UTC) to "Rung 2" (9 Oct); the Gate B/C DRAFT header (still revision 4b); the brief's revision-16 header and [D14]–[D20]; the mechanism note's summary, §2, §4 and its three addenda; the Llama recipe-B kits' reading files (seeds 1102 and 1103); on the Side Project's branch (read-only), its Llama recipe-B results note, the arXiv note's commit c51cf39 and the fiction pre-registration header; the arXiv API record and the OpenReview notes for Safe LoRA (arXiv:2405.16833).

Sources read (revision 4): FOCUS.md's entries of 8 Oct from 17:36 to 21:37 UTC; the results record's sections on recipe G (seed 1101), Llama under recipe G, OLMo recipe G at seeds 1101 and 1102, the rank-16 check and the token-mean-only run; the mechanism note in full (including its recipe-G and Llama addenda); the header of the Gate B/C DRAFT (still revision 4b); the file names and the two-seed mean of the kit's reading files for OLMo recipe-G seed 1103; the arXiv abstract of Qi et al. (2025), by API.

Sources read (revision 3): FOCUS.md's entries of 8 Oct from 15:17 to 19:24 UTC; the results record's recipe-G, Llama and HarmBench (complete) sections; the header of the Gate B/C DRAFT (revision 4b); the arXiv abstract of Qi et al. (2025). The sharpened-direction plan was not changed after revision 2 used it (last commit 3ade6c3), so it was not re-read.

Sources read (revision 2): the paper specification (in full); the adopted FOCUS amendment (commit 4b37ec2) and FOCUS's dated entries of 6–8 Oct; the sharpened-direction plan (revision 2, in full); the Gate A audit; the kill-test Stage 0 results record (Gate A); the Gate B/C pre-registration draft (revision 3); the mechanism note; the Guo-gap analysis; the safety-landscape note; the erosion-amplifiers note; reading list v2 (groups 4–5 and the DataRx correction); the benign-erosion recipes note (DataRx); the kill-test design (measurement table). Abstracts fetched this session for the new citations.